Supporting material for Orthopolity: a general law of resource allocation

Accounting proofs, physical derivations, observation models and complete comparisons

Prepared 10 October 2026; revised 22 October 2026 as a companion to the main article. This document preserves the substantive technical material of the article at source commit `f137d4efb050ccfcb5c5f814d32da651c645421c`, supplemented by its existing study reports. This is an editorial reorganization, with the Dunaliella treatment/control and log-change interpretations corrected explicitly. No frozen input, protocol, forecast, registry entry or numerical output is changed. The general-law claim and the ambition to establish it are retained; each theorem, physical realization, simulation and empirical comparison keeps its stated scope.

The comparison measure is part of every allocation statement. Except where a linear, mode-counting or word-type measure is explicitly specified, the size applications concern resource per logarithmic size, **log-orthopolic allocation**. Natural-system observations, engineered measurements and simulations are identified separately. A local Git freeze records execution order; it does not establish external preregistration or erase prior exposure.

Contents:

- S1. Accounting, composites and identification proofs
- S2. Measures, ensembles and counterexamples
- S3. Exchange dynamics and physical derivations
- S4. Exploratory sources, methods and results
- S5. Frozen forecasts, full comparisons and eligibility
- References

Each study's linked report remains the route to its exact source files, configuration, archived algorithms and retained results. Figures and tables below use independent S labels; they reuse retained presentation artifacts.

S1. Accounting, composites and identification proofs

The physical resource measure, the class comparison measure, and the object partition have distinct roles. The following definitions and proofs state those roles before inverse inference.

S1.1. Objects, resources, finite capacity and complete hierarchies

Let k be a positive size coordinate on a finite domain $D = [a, b]$, with $0 < a < b < \infty$. Choose a reference measure μ, with $0 < \mu(D) < \infty$, for the classes being compared. Let $n_\mu(k) = dN/d\mu$

denote object abundance and $\bar{q}(k) = E[q \mid k] > 0$ the conditional arithmetic mean of an additive resource per object. Then

$$\mathcal{O}_\mu(k) = \frac{dR}{d\mu} = \bar{q}(k) n_\mu(k). \tag{S1}$$

For an observed class, the corresponding identity is simply $\sum_i q_i = N_{\text{class}} \bar{q}_{\text{class}}$. Resources may be a standing stock, such as biomass, or an accumulated event quantity over a declared observation period. The latter is not a stock resident in a size class, and neither is a flux through size classes. Additivity is needed for (S1); a dynamical conservation law is not.

Define the **neutral allocation form of orthopolity relative to** μ, denoted O_μ, by

$$\mathcal{O}_\mu(k) = C > 0 \quad \text{for } \mu\text{-almost every } k \in D. \tag{S2}$$

This is an assumption about the neutral allocation, rather than a claim that every constrained system has this profile. It gives $n_\mu = C/\bar{q}$. Equivalently, if an object is sampled uniformly from the population, its size density with respect to μ is

$$p_N(k) = \frac{1}{Z\bar{q}(k)}, \qquad Z = \int_D \frac{d\mu(k)}{\bar{q}(k)} < \infty. \tag{S3}$$

Sampling instead in proportion to the resource held gives $p_R(k) = \bar{q}(k) p_N(k)/E_N[q] = 1/\mu(D)$. Thus uniform resource-weighted size and inverse-cost object abundance are equivalent statements, under the same reference measure. This is an accounting equivalence, not a derivation of the allocation assumption. The count, resource total, and constants also satisfy $N = CZ$ and $R = C\mu(D)$. For a finite observed population, the identity is exact for class totals; a smooth density is a population model or approximation.

Finite capacity and complete hierarchies. The box intuition has an exact formulation that precedes the neutral-allocation postulate. Let ν be a nonnegative additive resource measure on a system Ω, with $B = \nu(\Omega) \in (0, \infty)$. This physical resource measure differs from the class comparison measure μ. For disjoint units $A_i \subseteq \Omega$ with requirements $q_i = \nu(A_i)$, finite additivity gives the capacity bound

$$N_{\geq t} = \#\{i : q_i \geq t\} \leq B/t, \qquad t > 0.$$

Indeed, $tN_{\geq t} \leq \sum_{q_i \geq t} q_i \leq B$. Equal units of resource q therefore obey $Nq \leq B$, with equality when they exhaust the budget. In a cube of side L, cubes of side $\ell = L/m$, for integer $m \geq 1$, tile it with $N = m^3$ and $N\ell^3 = L^3$. For other side lengths an aligned grid accommodates $\lfloor L/\ell \rfloor^3$ cubes; the volume bound remains $N\ell^3 \leq L^3$. The grid construction is not a claim about optimal packing under arbitrary shape or orientation constraints.

Hierarchy theorem. Suppose each level j consists of a nonempty finite collection of positive-resource sets A_{ji}, and define the multiplicity $m_j(x) = \sum_i \mathbf{1}_{A_{ji}}(x)$. If N_j is their number and $\bar{q}_j = N_j^{-1} \sum_i \nu(A_{ji})$, then

$$N_j \bar{q}_j = \int_\Omega m_j(x)\, d\nu(x) = B\gamma_j, \qquad \gamma_j = B^{-1} \int_\Omega m_j\, d\nu.$$

This follows by integrating the finite sum of indicator functions. In particular, if every level completely partitions the same resource, $m_j = 1$ almost everywhere and

$$N_j \bar{q}_j = B, \qquad \frac{N_j}{N_h} = \frac{\bar{q}_h}{\bar{q}_j}.$$

No equal-size assumption is needed for the mean-resource relation. Nesting between levels is allowed: a unit cube, its eight half-side subcubes, and its sixty-four quarter-side subcubes each account for one unit of volume at their respective levels. Within-level disjointness permits the same matter to be represented at several resolutions. Summing J complete levels counts that underlying resource J times; it does not create J independent physical budgets. Hollow containers likewise require a choice between enclosed volume and wall material.

For incomplete disjoint levels, γ_j is the occupied resource fraction; for overlapping descriptions it is the mean multiplicity and may exceed one. The exact ratio becomes $N_j/N_h = (\gamma_j/\gamma_h)(\bar{q}_h/\bar{q}_j)$. Thus independently measured coverage supplies a quantitative correction to the inverse relation. A bound $m_j \le M$ also gives $N_{j,\ge t} \le MB/t$. Arbitrary overlap with no multiplicity bound has no finite capacity bound based on B alone. Complete partitions fix the comparison structurally: one full accounting per level. Equal steps in logarithmic size arise when each refinement uses the same scale factor.

This theorem establishes an exact inverse count–mean-resource relation across complete descriptions. A mixed population's size classes usually account for different parts of one budget. Writing their fractions as f_j gives $N_j\bar{q}_j = Bf_j$, with $\sum_j f_j \le 1$ for disjoint classes. Equal allocation across equally weighted classes then means equal f_j; finite capacity alone leaves those fractions undetermined. Sections S2.2 and S3.1 supply additional symmetry and dynamical conditions for that allocation statement. Additivity and multiplicity accounting are established mathematics (Tao, 2011); inverse-scale covering counts also have a developed geometric theory (Bishop and Peres, 2017). The capacity and hierarchy derivation gives proofs, examples, and the implications for composite resources.

S1.2. Effective resources with coupled requirements

An effective resource can combine several constituent requirements. For an object i, let $q_{i\ell}$ be its requirement or assigned amount of constituent ℓ, and let $s_\ell > 0$ be a fixed reference quantity in the same units. A general representation is

$$Q_i = Q_0 F\left(\frac{q_{i1}}{s_1}, \ldots, \frac{q_{im}}{s_m}; z_i, \theta\right) > 0, \qquad \bar{q}_{\text{eff}}(k) = E[Q_i \mid k].$$

Here Q_0 supplies effective-resource units, z_i records relevant environmental or historical conditions, and θ specifies the combination rule. The reference scales are declared independently of the observed abundance. Nonlinearity can represent complementarity, substitution, saturation, or a bottleneck. The arithmetic class mean is taken after combining the constituent resources. In general, $E[F(X,Y) \mid k] \ne F(E[X \mid k], E[Y \mid k])$; for $F(X,Y) = XY$, the difference is the conditional covariance. Marginal resource means alone may therefore miss a dependence that changes the effective resource.

The quantity $R_{\text{eff}} = \sum_i Q_i$ is additive over the declared object partition, so equation (S1) still holds with $\bar{q} = \bar{q}_{\text{eff}}$. Nonlinearity within an object does not prevent this additivity across objects. It does, however, make the physical partition important: $\sum_i F(\mathbf{q}_i)$ need not equal $F(\sum_i \mathbf{q}_i)$. Splitting or merging objects can change the effective total. Nor is R_{eff} automatically conserved merely because some constituent resources are conserved. The proposed combination must be interpreted as a physical stock, an accumulated requirement, an opportunity cost, or another explicitly defined quantity, with its appropriate balance law. Collective interaction terms must be included or assigned to objects through a physical accounting convention without double counting.

The hierarchy theorem yields a further restriction when an effective resource is required to describe the same underlying constituents at different resolutions. Let F be homogeneous of

degree h, and split a parent constituent vector into r identical proportional parts. Their effective total is

$$\sum_{i=1}^{r} Q_0 F(\mathbf{x}/r) = r^{1-h} Q_0 F(\mathbf{x}).$$

Exact level invariance in this proportional subdivision therefore requires $h = 1$ for a positive parent and $r > 1$. For product resources this becomes $\sum_\ell \theta_\ell = 1$. Arbitrary heterogeneous subdivision is more restrictive: invariance requires $F(\mathbf{x} + \mathbf{y}) = F(\mathbf{x}) + F(\mathbf{y})$. If all nonnegative constituent vectors are admissible, F is finite and nonnegative, and $F(0) = 0$, additivity implies $F(\mathbf{x}) = \sum_\ell c_\ell x_\ell$ with $c_\ell \geq 0$. Thus nonlinear composition remains admissible on a physically specified object partition, or on a restricted family of constituent proportions, while invariance under every regrouping imposes a linear form. The supporting derivation proves these statements. They give a consistency test for applying a composite at several resolutions, without imposing homogeneity on resources defined only at one level.

For algae, nutrient stocks per cell and available nutrient budgets differ; light can be an absorbed photon flux or an exposure integrated over a specified interval. Space and water can be capacities or environmental conditions. Time can enter through residence time, growth history, or an additive occupancy such as cell-hours; elapsed time is not automatically a divisible stock. For cities, land, buildings and people are stocks, while food, water, energy and money can involve stocks, flows and exchanges over the chosen observation horizon. The combination rule must respect those distinctions and avoid counting a resource twice along a supply chain.

Multiple physical budgets remain explicit. For fixed class requirements, feasibility can require $\sum_j N_j q_{\ell j} \leq B_\ell$ for each constituent ℓ. A scalar Q does not generally replace all these inequalities. A specified optimization model can produce a local effective cost $Q_i = \sum_\ell \lambda_\ell q_{i\ell}$, with dimensionally compatible scarcity prices λ_ℓ, but those prices and their range of validity belong to that model (Boyd and Vandenberghe, 2004). A bottleneck gives another illustration: for a separately supported population of identical objects with fixed requirements, $N_{\max} = \min_\ell B_\ell / q_\ell$ depends nonlinearly on the budgets. It does not prescribe the allocation among competing types. These are ways to derive candidate effective quantities; neither scalarization nor feasibility by itself proves neutrality.

A simple nonlinear candidate is a product of positive normalized inputs, $Q = Q_0 \prod_\ell (q_\ell / s_\ell)^{\theta_\ell}$. If constituent requirements are deterministic functions $q_\ell(k) \propto k^{d_\ell}$, then $Q(k) \propto k^{d_{\text{eff}}}$ with $d_{\text{eff}} = \sum_\ell \theta_\ell d_\ell$. Under logarithmic neutrality the abundance density exponent is $1 + d_{\text{eff}}$. For differentiable F, the more general local elasticity is

$$\frac{d \ln Q}{d \ln k} = \sum_\ell \frac{\partial \ln F}{\partial \ln(q_\ell / s_\ell)} \frac{d \ln q_\ell}{d \ln k},$$

when the environment and parameters are fixed and F has no further explicit size dependence. Varying constituent elasticities or interactions can generate curvature. This chain rule for specified functions is distinct from a claim that independent random resource distributions have additive tail exponents. With heterogeneous objects, the class mean $E[Q \mid k]$ remains the relevant quantity. The effective-resource derivation develops the finite-bin and heterogeneity implications.

S1.3. Inverse identification and illustrative urban and linguistic targets

The law can be used in two directions. Given a resource model and a specified constraint, it predicts abundance. Given an abundance profile, it supplies information about the effective resource required by that allocation. Write a positive constrained profile as

$$\bar{q}_{\text{eff}}(k)n_\mu(k) = Ca(k), \qquad a(k) > 0,$$

where neutrality has $a = 1$. The class-exchange example below derives $a = e^{-V}$ from prescribed kinetics. In other systems a needs its own mechanism or conditional assumption; an arbitrary fitted factor is not an explanation. On a domain with positive population density, inversion gives

$$\frac{\bar{q}_{\text{eff}}(k)}{a(k)} = \frac{C}{n_\mu(k)}, \qquad \frac{\bar{q}_{\text{eff}}(k)/a(k)}{\bar{q}_{\text{eff}}(k_0)/a(k_0)} = \frac{n_\mu(k_0)}{n_\mu(k)}.$$

Thus a distribution identifies the shape of effective resource divided by its constraint factor, conditional on the comparison measure. With the constraint profile and its normalization specified, an independently measured effective-resource total fixes $C = R_{\text{eff}}/\int_D a\, d\mu$. Without further information, the constituents and their combination are not unique; multiplying both $\bar{q}_{\text{eff}}$ and a by the same positive size-dependent function leaves the prediction unchanged. This is a precise statement of the inverse information available, not a reason to discard inverse inference. Measured constituents, restricted combination families, and responses in different environments can resolve parts of that ambiguity. Finite observed counts require a sampling model and bin integration; a zero-count bin does not establish infinite physical cost.

A constructive inverse example. Prespecify $Q = Q_0 X^{\theta_1} Y^{\theta_2}$ for positive normalized resources. In environment s, suppose independently measured class requirements scale as $X_s \propto k^{d_{1s}}$ and $Y_s \propto k^{d_{2s}}$, with no within-class variation, and a known constraint has $V_s = v_s \ln(k/k_0)$. A Pareto count density $dN/dk \propto k^{-(1+\alpha_s)}$ then implies

$$\alpha_s - v_s = \theta_1 d_{1s} + \theta_2 d_{2s}.$$

Two environments with scaling vectors $(1, 1)$ and $(1, 2)$ and corrected exponents $\alpha_s - v_s$ equal to 1.5 and 2 identify $\theta_1 = 1$ and $\theta_2 = 1/2$. The inferred nonlinear resource is $Q = Q_0 X\sqrt{Y}$. A third environment with scaling vector $(2, 1)$ predicts $\alpha_3 = 2.5 + v_3$ without refitting. The two independent scaling directions identify the composition within this family; the third supplies a test. The remaining scale Q_0 requires an independent anchor. These are algebraic illustrative values, not measurements. They show how allocation distributions can reveal a composite resource and generate further predictions.

Identification and transfer. More generally, assume a common product composition $Q = Q_0 \prod_{\ell=1}^{m} X_\ell^{\theta_\ell}$, a fixed logarithmic comparison measure, independently characterized constituent exponents $D_{s\ell}$, and known constraint slopes v_s. Assume deterministic class requirements or write $X_{s\ell} = A_{s\ell}(k/k_0)^{D_{s\ell}} U_{s\ell}$ with a finite positive residual moment $E[\prod_\ell U_{s\ell}^{\theta_\ell} \mid k, s]$ independent of size. Then the corrected abundance slopes satisfy

$$b = D\theta, \qquad b_s = \alpha_s - v_s.$$

For a compatible b and unrestricted real exponents, θ is unique precisely when D has full column rank. With deficient rank, the compatible parameters are $D^+ b + \ker D$, provided b lies in the column space. A new environment with constituent row d_* still has an identified prediction if and only if d_* lies in the row space of D, for unrestricted real exponents. Restrictions such as nonnegative exponents can reduce this ambiguity further; they must be included in the identification problem.

Every vector ℓ satisfying $\ell^\top D = 0$ supplies a parameter-free cross-environment consequence,

$$\ell^\top b = 0.$$

For the three environments above, this gives $b_3 = 3b_1 - b_2$. The third slope is predicted before its abundance data are used. Thus additional environments can test a shared effective resource even when an individual profile supplies only a conditional reconstruction.

Proof and precision. Two parameter vectors give the same calibration slopes exactly when their difference lies in $\ker D$. A target row gives the same prediction for all such vectors exactly when it annihilates that kernel. Multiplication by a left-null vector gives the stated contrast. For full column rank and fixed D, least-squares recovery obeys

$$\|\widehat{\theta} - \theta\|_2 \leq \frac{\|\delta b\|_2}{\sigma_{\min}(D)}.$$

Near-proportional constituent scalings can therefore make a formally unique composition practically unstable. A contrast with slope-error covariance Σ_b has variance $\ell^\top \Sigma_b \ell$, including correlations between environments. Uncertain constituent scalings, constraint corrections, and size-dependent heterogeneity require their own error model; this fixed-design bound does not cover them. These statements apply standard linear inverse-problem results to the allocation model. The supporting identification analysis supplies generalized least squares, partial identification, and reproducible synthetic recovery and transfer checks.

City sizes. Upper city-size tails often admit Pareto descriptions, with Zipf scaling as a prominent special case. The result depends on how cities and the fitted size range are defined. Eeckhout (2004) found a lognormal description across the full US Census Places distribution; Ioannides and Skouras (2013) found a lognormal body and Pareto tail, with an estimated tail exponent of 1.25 for their preferred Census Places model. These findings concern different parts and definitions of an urban system, not a requirement that every city-size observation follow exact Zipf scaling.

Let S denote population per city and $n(S)$ the positive number density of cities per unit population interval. For a Pareto survival exponent α, $N(\geq S) \propto S^{-\alpha}$ and $n(S) \propto S^{-(1+\alpha)}$; ranked size scales as $S(r) \propto r^{-1/\alpha}$. Finite upper cutoffs modify the survival and rank relations near the largest city. Under logarithmic allocation,

$$\frac{dN}{d\ln S} = Sn(S), \qquad \frac{\bar{q}_{\text{eff}}(S)}{a(S)} \propto S^{\alpha}.$$

Exact Zipf scaling has $\alpha = 1$. If $a = 1$ and population is the resource, $q(S) = S$, the relation expresses equal population in equal logarithmic city-size intervals. If the operative resource is a nonlinear combination of urban requirements, the same observation instead constrains that combination to have approximately linear net scaling, after accounting for a. This is the proposed orthopolity interpretation and an inverse target for measurements of land, infrastructure, energy, water and other inputs. The exponent alone does not identify the combination or assign a probability to its causal explanation. Existing proportional-growth mechanisms, including Gabaix's (1999) model with lower-size regularization, provide comparisons that the resource account should explain, connect to, or distinguish. No new city dataset is analyzed here.

Language. A word type is a distinct lexical form and a token is one occurrence. For type w, let f_w be its token frequency and L_w its letter count. The corpus expenditure of letters on that type is $f_w L_w$. Equality on a counting measure over word types would therefore imply $f_w \propto L_w^{-1}$. Zipf's rank-frequency relation $f(r) \propto r^{-\zeta}$ concerns a different coordinate. In a smooth rank approximation, letter expenditure per logarithmic rank is $rf(r)L(r) \propto r^{1-\zeta}L(r)$. For $\zeta = 1$, tokens per log rank are constant, while letters per log rank vary with word length. Finite rank groups require the actual sums, including tied frequencies. Length classes introduce another distinction: if M_L vocabulary types have length L, equal letter expenditure per type gives total token count at that length proportional to M_L/L. The number of available types is part of the comparison measure; a peaked word-length distribution alone does not negate this conditional allocation.

The abbreviation tendency, in which frequent words are shorter, motivates resource candidates without fixing either exponent or comparison measure. Contextual information is also relevant to word length (Piantadosi, Tily and Gibson, 2011). A declared candidate is $Q_w = Q_0 L_w^{\theta_L} P_w^{\theta_P}$, where P_w is a dictionary phoneme count. Under neutral type allocation it predicts $p_\theta(w) \propto L_w^{-\theta_L} P_w^{-\theta_P}$ on a fixed vocabulary. Section S5.5 applies this inverse model across text genres. Letter and phoneme counts are symbolic resource candidates; sonority, duration and acoustic energy need separate measurements and constitutive definitions.

S1.4. Additional proofs and sharpness limits

Layer-cake identity and the finite-domain restriction. In a finite inventory,

$$\sum_i q_i = \int_0^\infty N_{>t}\, dt.$$

Integrate $q_i = \int_0^\infty \mathbf{1}_{\{t<q_i\}}\, dt$ and exchange the finite sum and integral. A continuum approximation $N_{>t} = C/t$ on $[a, b]$ contributes $C \ln(b/a)$ to the budget on that interval alone; it cannot extend over an unbounded logarithmic range with finite resource. The sharper integer capacity bound is $N_{\geq t} \leq \lfloor B/t \rfloor$. It is attainable only when the budget and feasibility restrictions admit that many units.

Boundary occupancy and a mixed-size counterexample. An aligned grid of side-ℓ cubes in a side-L box in d dimensions has occupied fraction

$$\phi(\ell) = \left(\frac{\ell \lfloor L/\ell \rfloor}{L}\right)^d, \qquad N = \phi(\ell) L^d/\ell^d.$$

Complete use of the total volume does not imply equal class budgets. Tile a unit cube with 64 quarter-side cubes and subdivide one into eight eighth-side cubes. The final inventory has 63 units holding 63/64 of the volume and eight units holding 1/64. Both classes coexist, fill the volume without overlap, and have unequal resource totals. The construction specifies the missing allocation premise; it does not contradict the capacity bound.

Define uncovered fraction and excess multiplicity by

$$g_j = \frac{\nu(\{x : m_j(x) = 0\})}{B}, \qquad o_j = \frac{1}{B}\int_\Omega (m_j(x) - 1)_+\, d\nu(x).$$

Because integer-valued $m_j = \mathbf{1}_{\{m_j>0\}} + (m_j - 1)_+$,

$$\gamma_j = 1 - g_j + o_j, \qquad N_j \bar{q}_j = B(1 - g_j + o_j).$$

For two nonempty levels the resulting prediction is

$$\frac{N_j}{N_h} = \frac{\bar{q}_h}{\bar{q}_j}\frac{1 - g_j + o_j}{1 - g_h + o_h}.$$

Incomplete disjoint coverage has $o_j = 0$. Complete overlapping coverage has $g_j = 0$. A pointwise bound $m_j \leq M$ gives the threshold bound $N_{j,\geq t} \leq \lfloor MB/t \rfloor$ by the capacity argument in S1.1. Arbitrary repeated copies with no overlap bound can have arbitrarily large counts while enclosing the same physical volume.

Coverage and overlap can be determined from independently specified geometry. In that setting they supply a quantitative correction rather than a residual named after observing an abundance curve. If they are computed solely from the same count/resource accounting being compared, the relation is a consistency identity; it provides no independent forecast.

Proof of the arbitrary-subdivision restriction. Assume every vector in $\mathbb{R}^m_+$ and every split into two such vectors are admissible, F is finite and nonnegative there, and $F(0) = 0$. Invariance under every split or merge is equivalent to

$$F(\mathbf{x} + \mathbf{y}) = F(\mathbf{x}) + F(\mathbf{y}).$$

It forces

$$F(\mathbf{x}) = \sum_{\ell=1}^{m} c_\ell x_\ell, \qquad c_\ell \geq 0.$$

Proof. Additivity gives $F(\mathbf{x}) = \sum_\ell F(x_\ell \mathbf{e}_\ell)$. For each axis $g_\ell(t) = F(t\mathbf{e}_\ell)$ is additive and nonnegative. It is monotone since $g_\ell(t + s) - g_\ell(t) = g_\ell(s) \geq 0$. Rational arguments give $g_\ell(r) = r g_\ell(1)$; bounding a real argument above and below by rationals gives $g_\ell(t) = t g_\ell(1)$. Set $c_\ell = g_\ell(1)$.

Degree one is necessary but insufficient for arbitrary heterogeneous splits. For $F(x, y) = \sqrt{xy}$, parents with vectors $(9, 1)$ and $(1, 9)$ each contribute 3, while their merged vector $(10, 10)$ contributes 10. If attainable vectors are restricted to common constituent proportions, a nonlinear homogeneous degree-one form can remain invariant on that restricted family. The full-cone assumption must not be omitted.

This result preserves the programme's nonlinear resources while identifying their physical domain. They can describe assembly-dependent requirements at a specified object partition. Applying them across arbitrary partitions needs additional interaction accounting or the additivity property above. The restriction does not apply to the linguistic study merely because its fitted product has a degree different from one: that study fixes word types and makes no claim of invariance under arbitrary splitting and merging.

S1.5. Constructed identification, transfer and uncertainty checks

Prespecify a dimensionless product resource $Q = Q_0 X^{\theta_1} Y^{\theta_2}$ and the logarithmic size measure. Deterministic constituents scale as $X_s \propto k^{D_{s1}}$ and $Y_s \propto k^{D_{s2}}$. The known allocation factor is $a_s(k) \propto k^{-v_s}$. A count density $dN_s/dk \propto k^{-(1+\alpha_s)}$ therefore has corrected slope $b_s = \alpha_s - v_s = D_{s1}\theta_1 + D_{s2}\theta_2$.

Environment	**Constituent exponents**	**Known offset** v_s	**Corrected slope** b_s	**Abundance slope** α_s	**Role**
1	$(1, 1)$	0.15	1.5	1.65	Calibration
2	$(1, 2)$	-0.10	2.0	1.90	Calibration
3	$(2, 1)$	0.20	2.5	2.70	Held-out prediction

Table S1. Fixed environments in the constructed product-resource identification example.

The first two rows recover $\theta = (1, 1/2)$, or $Q = Q_0 X\sqrt{Y}$. Since $D_3 = 3D_1 - D_2$, the prediction fixed without the third abundance slope is

$$b_3 = 3b_1 - b_2, \qquad \alpha_3 = 3(\alpha_1 - v_1) - (\alpha_2 - v_2) + v_3.$$

The scalar Q_0 is not identified by slopes. No resource amplitude or conservation principle is inferred from its omission. The composition is assumed shared across environments, while the known constituent scalings and known constraint offsets change.

Correlated uncertainty and deliberate misspecification

Every estimated slope has Gaussian marginal standard error 0.02. The joint correlation is

$$\mathcal{C} = \begin{pmatrix} 1 & 0.35 & 0.20 \\ 0.35 & 1 & 0.40 \\ 0.20 & 0.40 & 1 \end{pmatrix}, \qquad \Sigma_b = 0.02^2 \mathcal{C}.$$

The design and offsets are exact in this experiment. The calibration estimate uses generalized least squares. Its analytic covariance is

$$\mathrm{Cov}(\widehat{\theta}) = \begin{pmatrix} 0.00144 & -0.00078 \\ -0.00078 & 0.00052 \end{pmatrix}.$$

For the held-out residual, $c = (-3, 1, 1)^\top$ gives $\mathrm{Var}(r) = c^\top \Sigma_b c = 0.0034$ and $\mathrm{SE}(r) = 0.0583095$. Retaining the correlations is essential: the prediction uncertainty and held-out observation uncertainty cannot simply be added as if independent. The reported intervals are centered on the calibration prediction and assessed against the noisy held-out slope; they include both sources of uncertainty. Coverage is unconditional across the specified joint repetitions.

The run generates 20,000 independent joint error vectors with seed 20261006. Each error vector is reused in a paired alternative where only the held-out composition changes to $(1, 0.8)$; its true α_3 is then 3.0 instead of 2.7. The same frozen calibration forecast is used for both scenarios. No parameter is refitted to the third slope.

Quantity	**Analytic prediction**	**Seeded outcome**
Mean recovered θ_1	1	0.999741
Mean recovered θ_2	0.5	0.500149
Shared-model standardized residual mean	0	0.009145
Shared-model standardized residual variance	1	1.011435
Shared-model 95% interval coverage	95%	94.815%
Changed-composition standardized residual mean	5.144958	5.154103
Changed-composition two-sided rejection fraction	99.9276%	99.915%

Table S2. Analytic expectations and all 20,000 seeded repetitions under shared and changed composition.

The Monte Carlo standard error of nominal 95% coverage is 0.1541 percentage points. These results check the inference calculation under its declared Gaussian model. The high rejection fraction is the power for one specified alternative and one selected noise level; it does not establish general sensitivity to changes in effective resources. In empirical work a failed forecast could also arise from an incorrect offset, constituent measurement, sampling model, or heterogeneity correction.

Partial identification and conditioning

With calibration rows $(1, 1)$ and $(2, 2)$, the data identify only $\theta_1 + \theta_2 = 1.5$. For example, $(1, 0.5)$ and $(0.25, 1.25)$ give exactly the same two corrected slopes. The target row $(3, 3)$ nevertheless

predicts 4.5. A target row $(1, 0)$ cannot be determined. The implementation rejects unidentified components and targets instead of interpreting a minimum-norm coefficient vector as a discovered physical composition.

Formal uniqueness also need not imply useful precision. Compare the original calibration design to rows $(1, 1)$ and $(1, 1.0001)$, and perturb the second corrected slope by 0.001 in both cases:

Design	**Condition number**	**Coefficient change**	**Error norm**	**Fixed-design bound**
Original distinct scalings	6.8541	$(-0.001, 0.001)$	0.0014142	0.0026180
Nearly proportional scalings	40002.0	$(-10, 10)$	14.1421	20.0005

Table S3. Precision loss when constituent scaling directions become nearly proportional.

Both matrices have full column rank at the declared relative SVD threshold 10^{-12}. This separates rank from precision. The error bound is $\|\delta b\|_2/\sigma_{\min}(D)$; the tests also verify attainment along the weakest singular direction. The derivation states a separate deterministic bound for perturbed constituent exponents, but this run does not simulate or estimate error in D.

The configuration, retained summary and source report preserve the design and all 20,000 repetitions. Analytic predictions were known during construction; their execution before random sampling is not an empirical holdout.

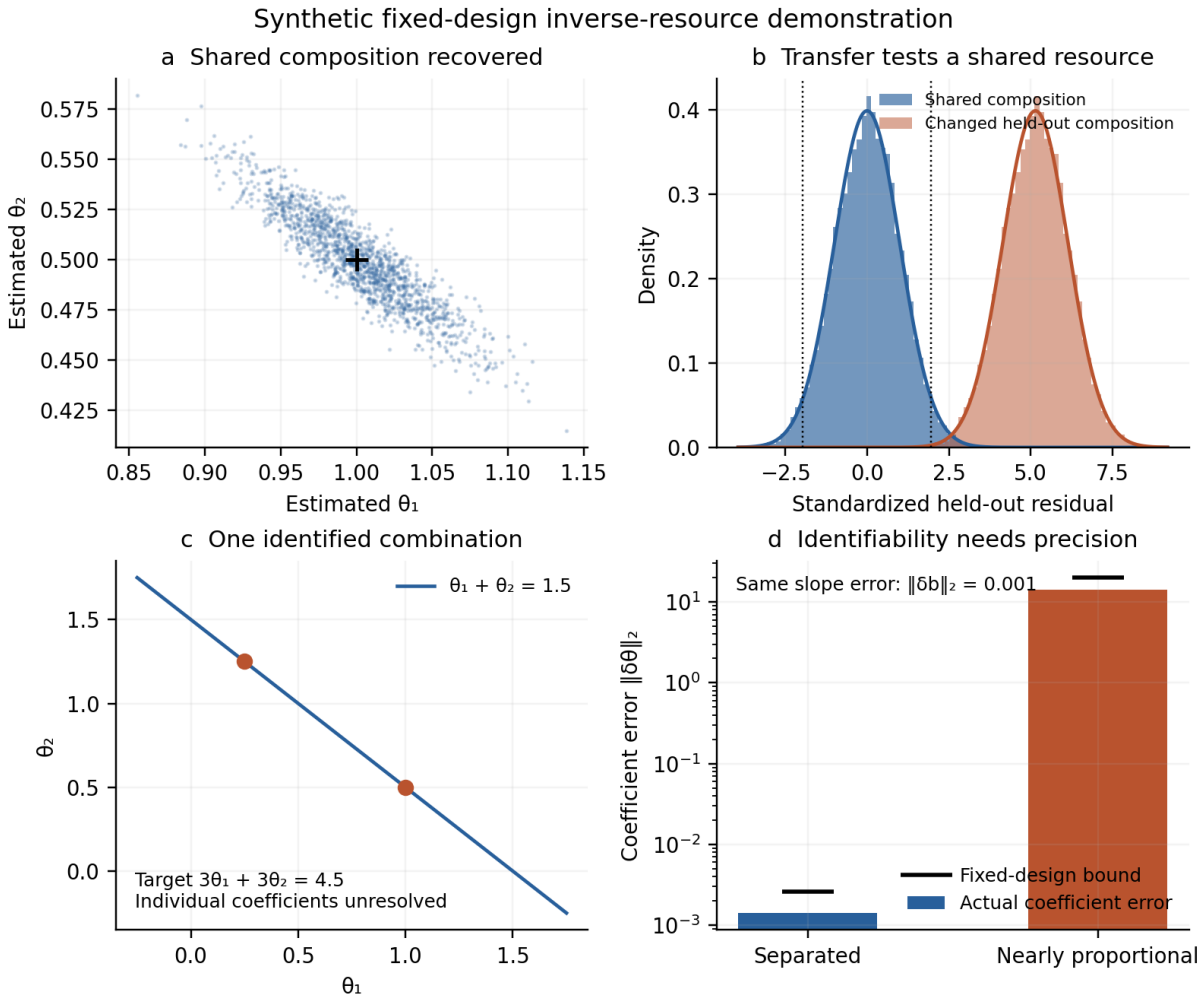


Figure S1. Constructed inverse-resource recovery and transfer. The first panel displays 1,500 of 20,000 replicates; all replicates enter the tabulated checks. These are simulations, with no new natural-system observations.

S2. Measures, ensembles and counterexamples

These distinctions delimit what a distribution can establish without replacing the general-law claim with a single fitted exponent.

S2.1. The comparison measure changes the exponent

Write $u = \ln(k/k_0)$ for a fixed reference size k_0. For $\bar{q}(k) = q_0(k/k_0)^d$, with $d > 0$, two different hypotheses give:

Equal resource with respect to	**Resource spectrum**	**Count density with respect to** dk
Linear size, $d\mu = dk$	$\bar{q}(k)dN/dk = C$	$dN/dk \propto k^{-d}$
Logarithmic size, $d\mu = du = dk/k$	$\bar{q}(k)dN/du = C$	$dN/dk \propto k^{-(d+1)}$

Table S4. Different hypotheses induced by linear and logarithmic size comparison measures.

The second row follows because $dN/du = k\,dN/dk$. For biomass with $q = k = m$, the predictions are consequently density exponents 1 and 2, respectively. The 2017 manuscript's explicit uniform density over a linear wealth interval is not interchangeable with the logarithmic convention used in the empirical analyses below. Adopting logarithmic classes is a substantive specification, motivated by size-spectrum practice.

Logarithmic allocation is unchanged by changing size units or logarithm base, apart from a constant density multiplier. It also survives $z = (k/k_0)^c$ for constant $c > 0$, since $d\ln z = c\,d\ln k$. It is not invariant under arbitrary nonlinear coordinates. In general, $dR/dv = (dR/du)|du/dv|$. Choosing a transformation after observing R can therefore manufacture flatness; the coordinate and measure must be fixed independently. More precisely, changing coordinates while carrying the same measure along preserves the allocation statement. Replacing that measure by a uniform measure in the new coordinate changes the hypothesis. The complete hierarchy in Section S1.1 chooses its levels through the construction of the system; that structural choice is separate from relabeling those levels.

Under $O_{\log}$ and the power-cost assumption, the slope of abundance per log interval is $-d$, the ordinary density exponent is $d + 1$, and the survival function on $[a, b]$ is

$$P(K \geq k) = \frac{k^{-d} - b^{-d}}{a^{-d} - b^{-d}}. \tag{S4}$$

The familiar survival exponent d and rank-size exponent $1/d$ are approximations away from the upper boundary, not exact finite-domain identities. For an unbounded domain with $d > 0$, count can be finite while logarithmically uniform resource has infinite total. Finite boundaries are essential to a finite-budget interpretation.

S2.2. Allocation postulates and accounting counterexamples

Resource symmetry as a candidate law. The proposed rationale is that, absent restrictions favouring particular resource costs, allocation should not distinguish those costs. A natural law can adopt this as a physical postulate without a deeper mechanistic derivation. One precise sufficient formulation uses K declared classes with fixed positive per-object costs q_i, allocations $R_i = q_i N_i$, and fixed total $\sum_i R_i = R > 0$. If the probability law of $(R_1, \ldots, R_K)$ is invariant under exchanging allocations across the fixed classes, then symmetry and the sum give

$$E[R_i] = R/K, \qquad E[N_i] = R/(Kq_i).$$

This postulate selects resource allocations as the symmetric quantities; symmetry of counts would give a different prediction. It is stronger than conservation and is not implied by an unspecified absence of constraints. Indivisible objects may require an approximation where exact exchangeability is infeasible. The class measure also matters: with cost itself as coordinate, equal resource per dq predicts $dE[N]/dq \propto q^{-1}$, whereas equal resource per $d\ln q$ predicts $dE[N]/dq \propto q^{-2}$. Transformation-based treatments of indifference likewise require a specified symmetry (Jaynes, 1968); an inferential rule does not itself establish a physical law.

Equal expectation does not establish equal snapshots or relaxation toward equality. In a divisible-resource idealization, assigning all resource to one uniformly chosen class has the required symmetry and unequal allocation in every realization. Time-average equality needs an additional condition connecting time averages to expectations or its own empirical postulate. A test must therefore specify what "tends to allocate" means and identify eligible systems through conditions independent of their observed allocation. A deeper mechanism is optional for an empirical law; the scope and predictive evidence are not.

Conservation does not imply equal allocation. On any bounded logarithmic domain, $\mathcal{O}(u) = A\exp(\beta u)$ can be normalized to the same resource total for every β. Conservation fixes the integral and leaves its shape undetermined. Likewise, maximum entropy over finitely many equally weighted object classes, constrained by a mean cost, yields $p_j \propto \exp(-\lambda q_j)$. Different reference measures or constraints give different distributions (Jaynes, 1957; Visser, 2013).

Equipartition does not imply a power law without a cost law. If $\bar{q}(k) = q_0\exp(k/k_0)$, then $O_{\log}$ gives $dN/du \propto \exp(-k/k_0)$ and $dN/dk \propto \exp(-k/k_0)/k$. Neither is a power law. Geometric dimension supplies d only in a specified geometric model such as similar objects with volume proportional to length cubed. It does not establish the number of independent resource inputs in an arbitrary system.

Independent inputs do not generally add tail exponents. For example, two independent unit-scale Pareto variables with densities hx^{-h-1}, $x \geq 1$, $h > 0$, have product density $h^2 z^{-h-1}\ln z$, $z \geq 1$. Integration over the possible factorizations introduces a logarithmic factor. This limits the independent-input argument in the 2017 essay.

Forward tests and inverse inference have different roles. For any positive abundance, defining $\bar{q} = C/n_\mu$ makes (S2) hold identically. In Section S1.3 this is the starting inverse relation: it identifies a conditional effective-resource profile and motivates resource hypotheses. Agreement on the same fitted profile is not independent evidence for the law or for the inferred physical composition. A forward test fixes the resource model through definitions, independent measurements, or a separate calibration, then checks a prediction on additional observations. A constraint inferred from the original profile likewise needs additional information or a transferred prediction to establish its explanatory value.

S2.3. Ensemble statements and averaging

Suppose a system has a power-shaped resource spectrum $\mathcal{O}_i(u) = A_i\exp(\beta_i u)$. A proposed ensemble condition $E[\beta_i] = 0$ concerns mean slopes. It does not imply $E[\mathcal{O}_i(u)]$ is constant. With equal $A_i = A$ and $\beta_i \sim N(0, \tau^2)$,

$$E[\mathcal{O}_i(u)] = A\exp(\tau^2 u^2/2), \tag{S5}$$

which is curved whenever $\tau > 0$. Equal-total normalization changes this formula but does not make mean-zero slopes sufficient: averaging normalized spectra proportional to $\exp(cu)$ and

$\exp(-cu)$ on a symmetric domain produces a nonconstant hyperbolic cosine. A median slope is a different estimand again. A claim about average resource occupancy requires spectra, a common domain, and an explicit weighting of systems; slopes alone do not identify it.

More generally, let $H(u) = \sum_{i=1}^{n} w_i A_i e^{\beta_i u}$ on a common log-domain, with fixed positive weights and amplitudes. Then

$$H''(u) = \sum_i w_i A_i \beta_i^2 e^{\beta_i u} \geq 0.$$

If any slope is nonzero, the derivative is strictly positive, so the arithmetic mixture cannot be constant on an open interval. Exact flatness requires every contributing slope to be zero. With $p_i(u) = w_i A_i e^{\beta_i u}/H(u)$, its log-curvature is $d^2 \ln H/du^2 = \mathrm{Var}_{p(u)}(\beta)$. Per-system total normalization preserves these conclusions. They are standard convexity consequences for exact power spectra; arbitrary curved profiles can cancel to a flat mean. They therefore neither exclude a general resource-symmetry law nor convert noisy fitted slopes into a profile test. The scientific assessment gives a quantitative bound for approximate flatness and its weighting restrictions.

Mean and variance also do not require a Gaussian distribution. Gaussianity is an additional maximum-entropy modelling choice, and selecting it by AIC is not a goodness-of-fit test. If $\beta_i \sim N(\mu_\beta, \tau^2)$ and reported slope error is independent $N(0, \sigma_i^2)$, the predicted probability that the *reported slope* lies in $[-c_i, c_i]$ is

$$\Phi_{\mathrm{N}}\left(\frac{c_i - \mu_\beta}{\sqrt{\tau^2 + \sigma_i^2}}\right) - \Phi_{\mathrm{N}}\left(\frac{-c_i - \mu_\beta}{\sqrt{\tau^2 + \sigma_i^2}}\right). \tag{S6}$$

The expression $2\Phi_{\mathrm{N}}(c_i/\tau) - 1$ applies to zero-centred *latent* slopes. Comparing it to observed estimates without their errors is a different calculation. Fitting τ on the same slopes makes either comparison an internal diagnostic, not independent validation.

There is a valid simultaneous-resource restriction for an individual system: $\mathcal{O}_1/\mathcal{O}_2 = \bar{q}_1/\bar{q}_2$. Both spectra can be exactly flat only if their mean resource ratio is size-independent. Across systems, however, $\beta_{1i} - \beta_{2i} = d_{1i} - d_{2i}$ for power cost laws. Equal variances and correlation one follow only if that difference is constant and slope variance is nonzero, whether or not equipartition holds. They are not a general test of ensemble equipartition.

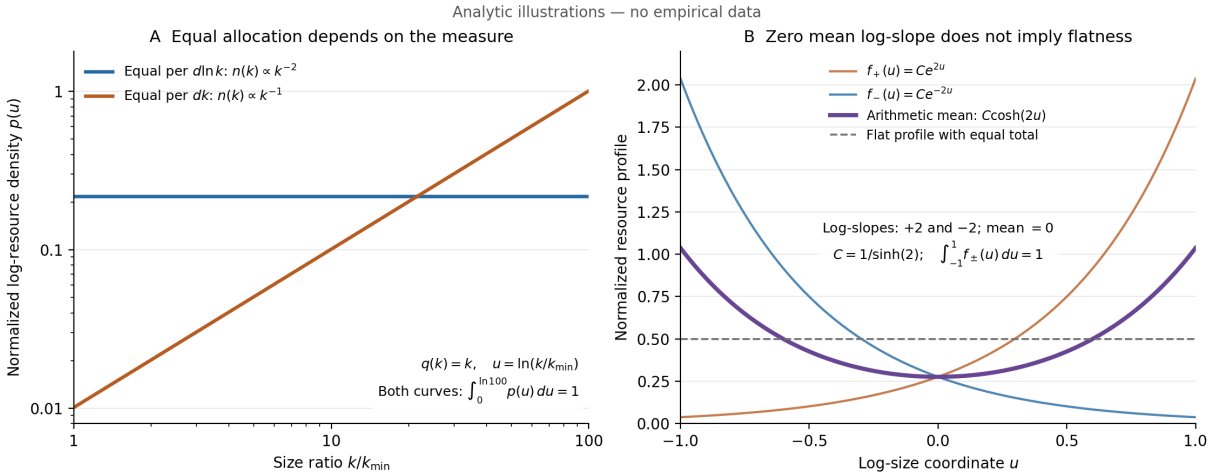


Figure S2. Analytic illustrations, not empirical data. (A) With resource equal to size, linear and logarithmic resource equipartition produce different count exponents. Both curves show unit-total resource density in the logarithmic coordinate. (B) Two unit-total resource profiles on a common domain have opposite log-slopes and therefore zero mean log-slope, but their arithmetic mean is not flat. Generated by run_theory.py; a vector version is available.

S2.4. Constraints and observable distributions

Three levels of description must be kept distinct: an underlying resource-equalizing tendency, the neutral signature (S2), and the outcome of the complete system. The latter depends on the proposed tendency together with resource costs, other processes, forcing, and initial and boundary conditions. A model containing an equalizing contribution can therefore predict a non-neutral stationary state or a transient with no clear neutral signature. Establishing such a contribution requires an explicit model or discriminating observations; equation (S1) alone does not identify it.

Age and height distributions are motivating examples of constrained outcomes. A worked age-distribution account must specify birth history, survival, and migration; a height account must specify developmental processes, biological variation, and environmental conditions, as well as the population sampled. An orthopolity application must additionally identify the resource, its concentrations, and the comparison measure. No completed orthopolity model of either example is supplied here. The relevant task is to derive the combined outcome and explain how its form depends on the constraints.

A failure of the neutral prediction is a finding about that prediction in the stated setting. It does not automatically establish the absence of the broader tendency, and counting distributions without its visible signature does not establish a general absence of evidence in Nature. A constrained model has its own predictions, which can succeed or fail. Specify the constraint through physical reasoning or independent measurements, derive its effect on the profile, and test the predicted response when it changes. Where recovery of neutrality is predicted, specify its conditions and timescale. This permits both neutral and non-neutral outcomes to inform the theory, while preserving the distinction between an observed departure, a proposed cause, and a tested explanation.

S3. Exchange dynamics and physical derivations

The following mechanisms establish allocation relations under their explicit physical and stochastic assumptions. They do not assert a shared microscopic dynamics for every application.

S3.1. Exchange on concentration classes and recovery

Consider K fixed size classes, with positive reference widths w_i, fixed per-object resource costs q_i, nonnegative class stocks R_i, and conserved total $B = \sum_i R_i$. The classes represent populations of resource-bearing objects; the state is their aggregate resource. Equivalent counts $N_i = R_i/q_i$ may be continuous or expected counts, and their total need not be conserved. This effective model does not specify individual-object assembly or demographic kinetics. The full derivation states its interpretation and assumptions.

Let $g_{ij} = g_{ji} \geq 0$ be conductances on a connected undirected graph. Neutral exchange has current $J^0_{ij} = g_{ij}(R_i/w_i - R_j/w_j)$ and dynamics $\dot{R}_i = -\sum_j J^0_{ij}$. Positivity follows from nonnegative jump rates g_{ij}/w_i; antisymmetry of currents conserves total resource. Connectedness makes the stationary resource density constant, giving

$$R^0_i = \frac{Bw_i}{\sum_j w_j}, \qquad N^0_i = \frac{Bw_i}{q_i \sum_j w_j}.$$

This is a mechanism selecting equal resource per declared class measure. The local exchange symmetry and the measure are assumptions of the model; they are not supplied by conservation alone. To model a constraint, prescribe dimensionless class preferences V_i and add downhill transitions:

$$a_{ij} = \frac{g_{ij}}{w_i} \exp(\max\{V_i - V_j, 0\}).$$

The rate a_{ij} is from i to j. Its neutral part g_{ij}/w_i remains, and the additional part is nonnegative. With column generator $A_{ji} = a_{ij}$ for $j \neq i$, $A_{ii} = -\sum_{j \neq i} a_{ij}$, define

$$\pi_i = \frac{w_i e^{-V_i}}{Z}, \qquad Z = \sum_j w_j e^{-V_j}.$$

Then $\pi_i a_{ij} = \pi_j a_{ji}$ on every edge. Detailed balance selects the unique stationary resource $R^V_i = B\pi_i$. This uses standard reversible-chain mathematics (Levin and Peres, 2017), specialized here to fixed concentration classes and an explicit additive constraint mechanism. At that stationary state the neutral edge current is

$$J^0_{ij} = \frac{Bg_{ij}}{Z}(e^{-V_i} - e^{-V_j}), \qquad J^c_{ij} = -J^0_{ij}.$$

Thus a nonuniform stationary distribution can have an active equalizing contribution balanced by the constraint. Setting V to a constant removes the added rates and lets the actual constrained allocation relax toward neutrality.

For $p = R/B$, set $E^2 = \sum_i (p_i - \pi_i)^2/\pi_i$. Writing $z_i = p_i/\pi_i$ gives

$$\frac{dE^2}{dt} = -2\sum_{i<j} \pi_i a_{ij}(z_i - z_j)^2 \leq -2\lambda E^2,$$

where $\lambda > 0$ is the smallest nonzero eigenvalue of the symmetric matrix $-D_\pi^{-1/2} A D_\pi^{1/2}$. Consequently $E(t) \leq E(0)e^{-\lambda t}$ and $\mathrm{TV}(p, \pi) \leq E/2$. The same bound applies after release, using the neutral generator and the state actually present at the switch. If connections are removed, each component retains its own resource total and only component-wise equalization follows.

The simulation fixes 24 equal-log size classes on $[1, 64]$, $q_i = k_i^{1.5}$, and adjacent conductances $1/\Delta \ln k$. A prescribed quadratic preference in $\ln k$, centered at $k = 8$ with width 0.75, produces a curved resource profile. The corresponding continuous count form is $dN/dk \propto k^{-2.5} e^{-V(\ln k)}$, a

truncated lognormal shape for this quadratic choice. Each phase runs for eight relaxation times. The neutral and constrained spectral gaps are 0.569805 and 1.991712 per model time unit. The final relative errors in the neutral, constrained, and released phases are 4.73×10^{-4}, 2.41×10^{-7}, and 5.51×10^{-11}, each below its predicted bound. The opposing stationary drift vectors cancel to Euclidean residual 3.22×10^{-15}.

There are 4,096 equal-resource packets in each of 128 realizations, initially all in the first class. Exact matrix-exponential checkpoint transitions preserve the state at both switches. Packets remain independent, giving class-count law $X(t) \sim \text{Multinomial}(4096, p(t))$ and share covariance $(\text{diag}(p) - pp^T)/4096$. Uneven finite snapshots are therefore predicted even at neutral mean allocation. Rates, costs, preferences, and predictions were fixed before drawing the stochastic trajectories. The computational results and checks are retained with the model report; the simulation supplies a constructive example, not new natural-system observations.

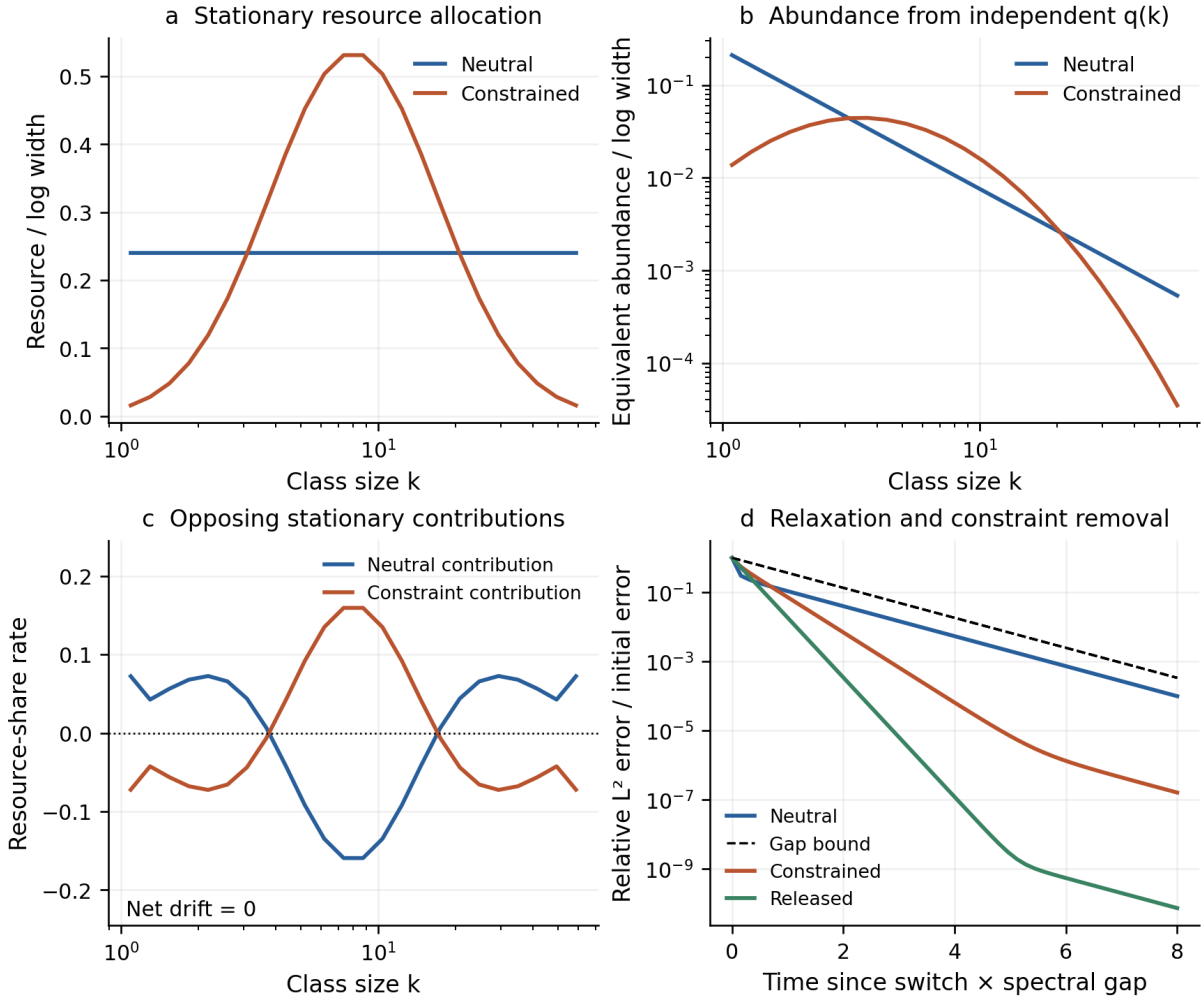


Figure S3. Fixed class-exchange model. Equalizing kinetics select neutral resource shares; an added class preference selects a curved profile and changes equivalent abundance. At constrained equilibrium the neutral and added contributions cancel. Removing the preference preserves the current state and predicts recovery under the neutral dynamics. The displayed convergence curves are deterministic expectations; stochastic packet realizations and their uncertainty are retained in the linked report.

S3.2. Electromagnetic transport, absorption and release

For electromagnetic energy density u and Poynting vector $\mathbf{S}$, Maxwell's equations give $\partial_t u + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}$ (Poynting, 1884; Haus and Melcher, 1989). Define $P(r) = \int_{S_r} \mathbf{S} \cdot \mathbf{n}\, dA$ outside a source

enclosed by an inner sphere r_0 with steady power $P(r_0) = L$. Integrating the balance over a transparent, source-free shell with no net energy accumulation gives $P(r_2) = P(r_1) = L$. Isotropy is a separate condition: it makes the local radial flux density $F = L/(4\pi r^2)$; otherwise this is its sphere average.

Under outgoing radial transport at constant speed c, without scattering or secondary emission in the modeled component, a specified absorber with coefficient $\kappa(r)$ gives

$$\frac{dP}{dr} = -\kappa(r)P, \qquad P(r) = L\exp\left[-\int_{r_0}^{r}\kappa(s)\,ds\right].$$

This is the absorption law specialized to spherical transport (Condon and Ransom, 2016). An absorbing layer with optical depth $\ln 2$ halves downstream power. The missing radiative power is deposited in matter, so the modeled radiation component has an open budget. Its departure follows from the prescribed medium and the same energy balance.

These nested surfaces receive the passage of the same energy; their powers cannot be summed as disjoint resource shares. Disjoint shell stocks do give an additive comparison: in transparent stationary radial transport, $dU = (L/c)dr$. Thus equal radial thicknesses hold equal energy, whereas $dU/d\ln r = Lr/c$ increases with radius. The resource, units, and comparison measure determine which equality is physically meaningful.

The response to removing an absorber is also determined. For sphere-averaged u, with $e = 4\pi r^2 u = P/c$, the transport equation is $\partial_t e + c\partial_r e = -c\kappa(r,t)e$. Suppose the source remains constant, the initial field is stationary under $\kappa_{\mathrm{old}}(r)$, and absorption is set to zero throughout the modeled domain at $t = 0$. For $t \geq 0$, characteristics give

$$P(r,t) = L\exp\left[-\int_{r_0}^{\max\{r_0,r-ct\}}\kappa_{\mathrm{old}}(s)\,ds\right].$$

Every radius recovers by $(r - r_0)/c$. Downstream of a layer $[a, b]$, the response starts at $(r - b)/c$ and completes at $(r - a)/c$. This is an idealized change of the absorption coefficient, with no radiation injected by the removal process. The physical derivation states the boundary history and all transport assumptions. It supplies an established physical realization of a declared equality and its constrained departure; it does not equate radial shells with organism size classes or derive abundance laws from an inverse-square flux.

S3.3. Classical equipartition and quantized mode energies

Thermal electromagnetic radiation supplies a second physical realization with a different reference measure: counting field modes, including polarization. For a finite set of classical normal modes in canonical equilibrium at $T > 0$, each has two quadratic terms, $H_j = P_j^2/(2m_j) + m_j\omega_j^2 Q_j^2/2$. Gaussian integration over both coordinates gives $Z_j^{\mathrm{cl}} \propto (\beta\omega_j)^{-1}$, with $\beta = (k_B T)^{-1}$, and therefore

$$\langle H_j\rangle = -\partial_\beta \ln Z_j^{\mathrm{cl}} = k_B T.$$

A class of M_i modes consequently has mean energy $M_i k_B T$: the resource density is equal per mode, with independently defined weight $w_i = M_i$. This is equality of equilibrium means, not identical instantaneous energies. Equilibrium preparation or thermal coupling must be justified; uncoupled harmonic modes do not thermalize each other merely because their canonical distribution can be written down. This argument uses standard equipartition and supplies no relaxation time (Tong, n.d.).

Quantization changes the allowed states. Define the resource as thermal excitation energy above the ground state, excluding the temperature-independent zero-point term. For photon absorption

and emission in thermal equilibrium, photon chemical potential is zero and a mode has excitation levels $\epsilon_n = nh\nu$. The geometric sum gives

$$Z_\nu = \frac{1}{1-e^{-x}}, \qquad \bar{\epsilon}_\nu = \frac{h\nu}{e^x - 1} = k_B T f(x), \qquad x = \frac{h\nu}{k_B T}, \quad f(x) = \frac{x}{e^x - 1}.$$

For $x > 0$, f decreases strictly from 1 toward 0. Its limits are $f(x) = 1 - x/2 + x^2/12 + O(x^4)$ near zero and $f(x) \sim xe^{-x}$ at large x. The quantum level spacing relative to thermal energy specifies the departure before any spectrum is observed. Raising temperature or lowering frequency recovers classical relative equality. This is a restriction of the available energy states; it does not establish the additive decomposition into opposed classical currents proved in Section S3.1.

In a large three-dimensional vacuum cavity, counting two polarizations gives the mode density $dM/(V d\nu) = 8\pi\nu^2/c^3$. Multiplication by $\bar{\epsilon}_\nu$ yields the Planck energy density and, for isotropic radiation, the spectral radiance:

$$u_\nu(T) = \frac{8\pi h\nu^3}{c^3(e^{h\nu/(k_B T)} - 1)}, \qquad B_\nu(T) = \frac{c}{4\pi} u_\nu(T).$$

Thus the classical equality per mode gives $u_\nu^{\mathrm{cl}} \propto \nu^2$ and $dU/(V d\ln\nu) \propto \nu^3$, neither a flat linear-frequency nor a flat log-frequency spectrum. If photons are the counted objects, $q(\nu) = h\nu$ and their mean number per mode approaches $k_B T/(h\nu)$ in the classical regime. Extending classical equipartition to infinitely many modes gives divergent total energy; quantization makes it finite, $U/V = 8\pi^5 k_B^4 T^4/(15h^3c^3)$. These are established results (Planck, 1901; Tong, n.d.; Condon and Ransom, 2016), here specifying the resource, comparison measure, and constraint in the allocation framework.

S3.4. FIRAS data as a retrospective mode-energy reconstruction

FIRAS measured the cosmic microwave background against calibrated blackbody references and separated monopole, dipole, and Galactic components (Fixsen et al., 1996). NASA LAMBDA's distributed monopole table contains a 2.725 K blackbody plus the residuals from that published analysis, with marginal uncertainties and a Galactic model column. The residuals already follow temperature and foreground fitting. The table therefore supports a transparent re-expression of the published result; agreement with its embedded template cannot be counted as a new independent confirmation.

Our retrospective calculation retains all 43 rows and fixes $T_0 = 2.725$ K from the product definition. With wavenumber $\tilde{\nu}$ in inverse centimetres, $\nu = 100c\tilde{\nu}$. Radiance remains per hertz, with $1\,\mathrm{MJy/sr} = 10^{-20}\,\mathrm{W\,m^{-2}\,Hz^{-1}\,sr^{-1}}$. Dividing the inferred energy density by the independent mode density gives

$$\frac{\widehat{\epsilon}_\nu}{k_B T_0} = \frac{c^2 I_\nu}{2\nu^2 k_B T_0} = \frac{I_\nu}{B_\nu^{\mathrm{RJ}}(T_0)}, \qquad B_\nu^{\mathrm{RJ}} = \frac{2\nu^2 k_B T_0}{c^2}.$$

The channels span 68.05–639.46 GHz and $x = 1.199$–11.262. Their transformed mode energies range from 0.518 to 1.32×10^{-4} of $k_B T_0$ (Figure S4). They illustrate the quantum departure, not the $x \ll 1$ classical plateau. The CMB can retain a Planck spectrum through collisionless cosmological redshifting; this example does not require current thermal contact with surrounding matter (Condon and Ransom, 2016, Section 2.6.2).

For residual diagnostics we use the source's original residual column, not the difference between rounded monopole intensities and a newly calculated curve. The source's approximate frequency covariance is $C_{ij} = \sigma_i \sigma_j Q_{|i-j|}$, using its 43 published lag coefficients (Fixsen et al., 1996, Section 3.3). The matrix is positive definite. Applying the linear radiance-to-mode conversion to both

axes propagates this covariance to the mode-energy ratios. The residual quadratic form is $r^T C^{-1} r = 49.774$; it is a descriptive audit of the published approximation, with no new fit or goodness-of-fit probability. It need not reproduce the original full-analysis statistic, 46 for 40 degrees of freedom. Diagonal inverse-variance weighting gives residual RMS 18.90 kJy/sr, or 49.28 parts per million of the tabulated peak radiance. This defined amplitude summary does not assume that the channels are independent.

Using exact SI constants at the rounded source coordinates gives a discrepancy $I_{\mathrm{posted}} - [B_\nu(T_0) + r_{\mathrm{published}}]$ of -0.127 to $+7.598$ kJy/sr, with radiances in common units. This discrepancy is retained without assigning a cause or replacing the original residuals. Absolute calibration, foreground modeling, and the prior fits remain inherited from the published analysis. The thermal-radiation report retains the full derivation, source bytes, covariance transcription, and reproducible calculation. This example connects a physical equality and an independently derived quantum departure to established observational evidence under its documented conditions.

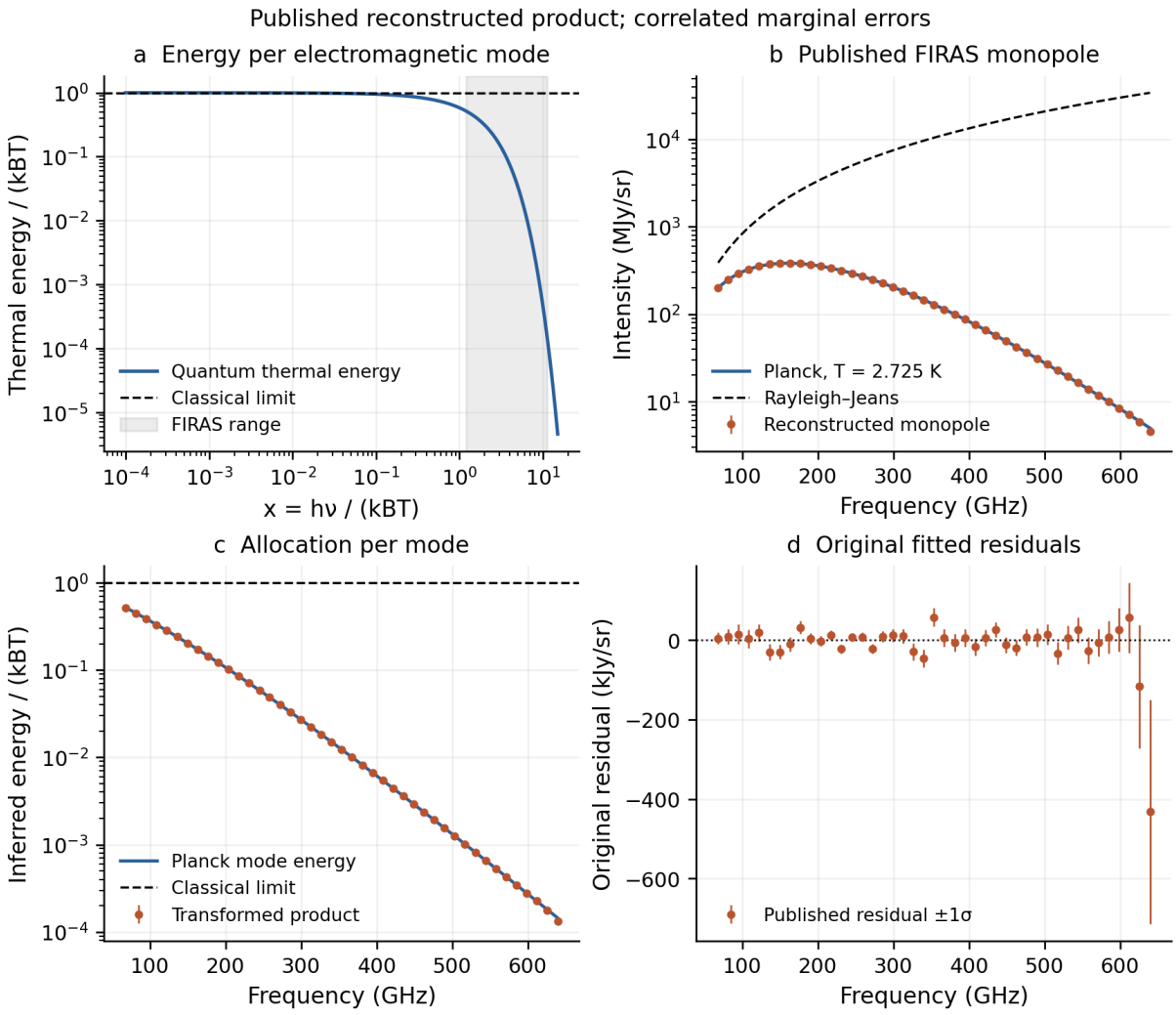


Figure S4. Thermal energy per electromagnetic mode. (A) The theoretical quantum factor $f(x)$ approaches classical equality as $x \to 0$. (B) The NASA FIRAS reconstructed monopole product and Planck and Rayleigh–Jeans radiances at the product's 2.725 K reference temperature. (C) The same product expressed as energy per mode relative to $k_B T_0$. (D) Original published residuals with marginal one-sigma uncertainties; channels are correlated. The Planck template is part of the distributed product, so panels B and C illustrate the published result rather than provide a new independent model test. All 43 channels are retained.

S3.5. Growth, division and resource-weighted observation

Let x be additive cell mass, growing deterministically as $\dot{x} = gx$ with the same $g > 0$ in every cell. Assume conservative symmetric binary division, no death, and a balanced asynchronous population with finite mean mass. Total mass and population number then grow at the same exponential rate g. For division hazard $\beta(x)$, normalized population density p_P satisfies

$$gp_P + (gxp_P)' = -\beta(x)p_P(x) + 4\beta(2x)p_P(2x).$$

Multiplying by x shows that $xp_P(x)$ solves the stationary chronological lineage equation: division has one retained daughter, halved in size. After normalization,

$$p_L(x) = \frac{xp_P(x)}{E_P[x]}.$$

Thus **the mass-weighted population size distribution equals the chronological lineage size distribution**. This size tilt is established prior work, explicitly derived and qualified by Genthon (2022), Section 2.2 and its revised exposition. It is not introduced here as a new theorem of cell biology.

Let B, D be sizes at the beginning and end of completed lineage cycles. Stationarity and symmetric division imply $D \overset{d}{=} 2B$, not necessarily $D = 2B$ within each individual cycle. For monotone growth the fraction of cycles crossing x is

$$H(x) = P(B \le x < D) = F_B(x) - F_D(x) = F_B(x) - F_B(x/2).$$

Each crossing occupies time $dx/(gx)$, while mean cycle time is $\ln 2/g$. The renewal occupation identity and the size tilt therefore give

$$p_L(x) = \frac{H(x)}{x \ln 2}, \qquad p_P(x) = \frac{2H(x)}{E[B^{-1}]x^2}, \qquad E_P[x] = \frac{2 \ln 2}{E[B^{-1}]}.$$

The normalized mass share per logarithmic size is

$$\frac{x^2 p_P(x)}{E_P[x]} = \frac{H(x)}{\ln 2}.$$

This identifies both the measure, $d \ln x$, and the constraint profile from birth/division events. When every cycle crosses an interval, $H = 1$ there and mass is equal across equal logarithmic widths. For fixed birth mass b, $p_P = 2b/x^2$ on $[b, 2b)$: the familiar inverse-square result. Variable birth and division sizes alter the envelope; an exact plateau is not generally present. A lognormal birth law has no strict plateau. The allocation claim is therefore more informative as an envelope prediction than as a search for an inverse-square slope over a selected interval.

For normalized probability densities b_P, d_P of population-tree birth/division event sizes (not event intensities per unit time), population balance also gives

$$\frac{d}{dx}[x^2 p_P(x)] = x[2b_P(x) - d_P(x)].$$

Here the total division-event rate per cell equals g, which cancels the common growth factor. Source-free intervals have a flat resource profile. These population event densities must not be replaced by chronological event densities without the sampling correction above. The resource profile is a stock across simultaneously present cells; lineage duration is used to infer it, not counted as extra mass.

Conditions that matter in nature

The common-growth assumption is stronger than a shared mean rate. Growth noise, division asymmetry, size-dependent loss, and lineage-specific physiology change the relation. With asymmetric partition, the corresponding lineage follows a mass-biased daughter; an ordinary old-pole mother-machine trajectory need not be that lineage. Genthon derives these distinctions. A balanced phase distribution is also an assumption: perfectly deterministic synchronized cycles can oscillate without approaching it. The proof is a statement about a stationary profile, not a general convergence result.

The empirical source measures length L, not mass. Even for constant diameter d, a spherocylinder has volume proportional to $L - d/3$, rather than exactly L. Width, density, polar geometry, and fixation are not independently calibrated here. The numerical study therefore tests a **length-proxy transfer model**. It cannot certify a biomass plateau or explain deviations by selecting an unmeasured correction after seeing the outcomes.

S3.6. Newtonian gravity and complete source accounting

The gravity derivation adds a physical realization of complete source accounting. In the Newtonian regime, the divergence theorem gives

$$\Psi(S) = -\int_S \mathbf{g} \cdot \mathbf{n}\, dA = 4\pi G M_{\mathrm{enc}}.$$

Complete surfaces enclosing the same source have the same inward field flux. For an isolated spherical source, $\mathbf{g} = -GM\hat{\mathbf{r}}/r^2$ and $d\Psi = GM\, d\Omega$: the field flux is equal per solid angle. Acceleration is not an energy current, and nested source representations cannot be added as independent mass stocks. This equality uses the angular measure $d\Omega$; it is not a linear-orthopolic or log-orthopolic law over radius.

In the conventional field representation of gravitational binding energy, $u_g = -|\mathbf{g}|^2/(8\pi G)$. Exterior spherical shells then have $dU/dr = -GM^2/(2r^2)$ and $dU/d\ln r = -GM^2/(2r)$, both nonuniform. That negative energy representation is distinct from an available positive resource budget. The linked derivation retains the source geometry, boundary conditions, attribution and interpretation limits. It also separates static field accounting from the supported-body analogy and from relativistic propagation.

S4. Exploratory sources, methods and results

These convenience sources were inspected before the analyses or governed by locally recorded plans with the exposure qualifications below. They do not constitute a representative sample of natural systems.

S4.1. Sources and scope

Data source	Analysed material	Resource and size coordinate
NOAA GOES XRS flare reports, 2022–2024	10,501 catalogue events; 1,168 paired events in the primary evaluation domain	End fluence in J/m^2; peak irradiance in W/m^2
USGS ComCat, 2010–2024	6,639 events after retaining moment-magnitude types and rounding to 0.1 at threshold 5.5	Magnitude-derived energy proxy as both resource and coordinate
Hatton et al. (2021)	23 one-decade biomass bins, upper 200 m and full water column	Body mass for both
GLOSSAQUA (Ersoy et al., 2025)	1,300 normalized biomass-spectrum estimates from 16 study identifiers	Body mass for both

Table S5. Exploratory data sources, analysed material and resource definitions.

These are four data sources; the two ocean domains are related reconstructions. GLOSSAQUA contains published estimates rather than organism-level observations. Study identifiers do not guarantee independent authors, methods, sites, or measurement errors. The sources are convenience cases for evaluating a hypothesis, not a representative sample of natural systems. Input bytes are fixed by SHA-256 manifests; provenance and source notices accompany the data.

S4.2. Analysis-plan provenance

The pilot configuration explicitly labels itself exploratory. The subsequent aquatic/ocean plan is retained as prereg_2026-09-09.json. In local history, commit 6df364f contains the plan and raw slope data, preceding analysis commit a0c9973. This records an order of commits. It does not establish an immutable public registration, independently verified blinding, or when values were first viewed. Statements of blinding in the historical file are author reports. The current corrections are retrospective, and the later ensemble, stratification, and variance analyses are exploratory. No claim of confirmatory preregistration is needed for the descriptive and mathematical results here.

S4.3. Accounting, domain choice and uncertainty

For bins B_j of logarithmic width w_j, compute $\widehat{\mathcal{O}}_j = \sum_{i \in B_j} q_i / w_j$ and $\widehat{C} = \sum_j \sum_{i \in B_j} q_i / \sum_j w_j$. The normalized profile $\phi_j = \widehat{\mathcal{O}}_j / \widehat{C}$ has width-weighted mean one by construction. This is not evidence of flatness. Empty bins remain in the profile, the final boundary is included, and missing resources require an explicit exclusion policy.

A factor-F endpoint drift across log-domain width L corresponds, for a power-shaped spectrum, to slope tolerance $c = \ln(F)/L$. This differs from requiring every $\phi_j \in [1/F, F]$: the latter permits a maximum-to-minimum ratio of F^2. Both slope and profile deviations matter. A slope equivalence interval combined with a point-estimate profile screen is not full simultaneous equivalence; that would require uncertainty for the whole profile. Failure to establish equivalence is not, by itself, evidence of a nonzero departure.

For flares, the arithmetic mean resource exponent is fitted by Gamma quasi-likelihood on 2022 data, with bounded count-density fitting on 2023–2024 over $[10^{-5}, 10^{-3}]$ W/m^2. Log-resource least squares would target a different conditional quantity unless an appropriate retransformation

were justified. Pilot intervals use 600 calendar-month block resamples for flares and calendar-year blocks for earthquakes. These do not capture all dependence or selection bias. The temporal split separates estimation samples but does not make the exploratory domain choice a prospective prediction.

Power-law diagnostics use maximum likelihood and a refitted parametric-bootstrap KS statistic on declared support, following the approach of Clauset et al. (2009). The rounded earthquake magnitudes are tested on their integer grid: for $K = (M - M_0)/\Delta$, $P(K = k) = (1 - r)r^k$ with $r = 10^{-b\Delta}$. Continuous-reference KS calibration is unsuitable for these ties. Parametric-bootstrap probabilities remain conditional on an independent-event working model; catalogue clustering is a limitation.

S4.4. Aquatic estimates and reconstruction uncertainty

The primary GLOSSAQUA subset requires body mass on the size axis, finite slopes and positive ordered size limits, and the normalized biomass-spectrum convention. For biomass per unit linear mass $B(m) \propto m^t$, equal biomass per log mass requires $t = -1$. We examine $\beta = t + 1$ and each estimate's reported size span. The historical plan uses $F = 1.25$ and $F = 2$, a 2,000-replicate study-block bootstrap for the pooled median, and a joint criterion involving its interval and the fraction of slopes inside their tolerances. That fraction ignores individual slope uncertainty and must not be interpreted as formal individual equivalence. Slope summaries cannot detect within-spectrum curvature. A post-audit sensitivity explicitly excludes StudyID_07 from range-dependent assessment because its reported bounds are physically implausible; no replacement bounds are imputed (S4.8).

For the ocean reconstruction, 20,000 simulated profiles propagate reported multiplicative uncertainty factors using lognormal draws centred on the published log estimates, with log-scale standard deviation $\ln(f)/1.96$. The published bounds match estimate divided and multiplied by f. Independent errors across groups and bins are a working assumption, not a property established by those bounds. The reconstruction is not a sample of independent organisms, and its plateau was selected after inspecting the upper-ocean estimates.

Exploratory random-effects and latent-distribution fits use reported errors where available. Intervals converted to standard errors assume their advertised coverage and approximate normality. Size-spectrum estimators can have biased estimates and miscalibrated intervals (Edwards et al., 2017, 2020). Repeated spectra within studies and estimated error variances further limit inference; independence-based pooled intervals are not primary evidence of a population mean.

S4.5. Earthquakes and the specified energy proxy

Gutenberg–Richter scaling $N(\geq M) \propto 10^{-bM}$ and proxy $E \propto 10^{\gamma M}$ give resource per log energy proportional to $E^{1-b/\gamma}$. Flatness therefore requires $b = \gamma$. At threshold 5.5, the fitted $b = 0.998$ has year-block 95% interval [0.973, 1.024], far below the chosen $\gamma = 1.5$. Thresholds 6.0 and 6.5 give 0.984 and 0.977. Using conversion exponent 1.44 also leaves a substantial discrepancy.

At threshold 5.5 the discrete KS statistic is approximately 0.0077, with a parametric-bootstrap probability about 0.26. This is non-rejection of the fitted geometric model, not proof that the model is exact. A power-like tail can thus coexist with a resource-proxy allocation that disagrees with $O_{\log}$. No discrete lognormal comparison was conducted. Magnitude-derived energy is not independently measured radiated energy; the finding is restricted to the specified proxy and catalogue, without declustering or a regional completeness model.

S4.6. Flares and the joint resource/count prediction

The fitted resource exponent is $d = 0.858$ with 95% interval [0.697, 1.054]. The implied density exponent is 1.858, compared with 2.239 [2.085, 2.399] in the paired evaluation sample. The estimated gap is 0.382 [0.125, 0.620]. A rise-phase fluence sensitivity gives gap 0.411 [0.208, 0.604]. This alternate resource has no missing values in the domain, but does not prove the primary end-fluence analysis free of missingness bias.

For all 1,306 in-domain evaluation events, peak irradiance has fitted exponent 2.274 and KS statistic about 0.032; its bootstrap probability is about 0.02. The power law is therefore inadequate under this working test. Its likelihood cannot clearly distinguish it from the fitted lognormal. Exponent mismatch tests the conjunction of equipartition, power-shaped mean cost, transfer between years, and the count model; it cannot isolate equipartition from all those assumptions.

The gaps at four lower cutoffs are −0.480, 0.191, 0.382, and 0.960. This substantial sensitivity discourages treating one selected range as a universal scaling regime. Fluence is instrument-band irradiance integrated over an event, not total flare energy.

S4.7. The ocean reconstruction and its uneven plateau

The upper-ocean summary reproduces the published abundance slope at approximately −1.039. Its maximum-to-minimum biomass ratio is 38.8 across all 23 bins, and 1.7 within the selected plateau of 16 bins. The corresponding full-water-column plateau ratio is 4.3. The 23 one-decade bins span 23 decades between outer edges; their centres are 22 decades apart. Similarly, the plateau covers 16 decades between edges and 15 between centres.

Under the stated uncertainty propagation, only 2 of 23 upper-ocean bin intervals lie wholly outside the factor-1.25 band; the corresponding full-column count is 3. These are pointwise intervals without multiplicity adjustment. The remaining intervals can include both near-flat and materially non-flat values, so their overlap with the band establishes neither equality nor absence of departures. Correlated reconstruction errors and post hoc range selection further limit the assessment. These results re-express Hatton et al.'s reconstruction; they do not independently replicate it.

The full upper-ocean range has a fitted biomass log-slope of −0.0392 and a propagated 90% interval approximately [−0.058, −0.026], outside the factor-1.25 slope tolerance of 0.00421. Thus the assumed uncertainty model indicates a systematic gradient even though most individual bin intervals overlap the pointwise band. Pointwise overlap does not make the whole-profile question uninformative; its interpretation remains conditional on the reconstruction error model.

S4.8. Aquatic slopes and the equivalence criterion

Among 1,300 normalized biomass spectra from 16 study identifiers, the pooled median slope is −1.015. The study-block 95% interval for its departure from −1 is [−0.100, 0.010]. The median slope tolerance is 0.0317 at $F = 1.25$ and 0.0984 at $F = 2$. This interval is not wholly within either band. Under the historical joint criteria, the result is **not supported**.

The reported-slope fractions inside their individual drift tolerances are 0.07846 ($F = 1.25$) and 0.24231 ($F = 2$), counting boundary cases inclusively. These are descriptive compatibility fractions. Their complements are not estimates of the fraction of true spectra violating equipartition, because measurement error and within-spectrum shape are not accounted for.

These historical fractions also depend on a metadata defect: all 377 StudyID_07 records report bounds of 2×10^{-8} to 2×10^{27} pg C, a 35-decade range with a physically implausible upper body

mass of 2×10^{12} kg C. Excluding those records from this range-dependent calculation leaves 923 slopes from 15 studies. Their point-slope compatibility fractions are 9.32% at $F = 1.25$ and 28.93% at $F = 2$. The source-derived replacement bounds are unknown; the original slopes remain in the location summary. The remaining records also require a methods and units audit. Consequently neither version of the pass fraction should be treated as a validated estimate of ecological prevalence.

Two studies supply 1,016 of the 1,300 estimates, approximately 78%. A central estimate near -1 is worth investigating, but it neither establishes a mean of -1 in a defined population nor establishes equal average biomass occupancy. The stronger ensemble interpretation fails on both statistical and mathematical grounds (S2.3).

S4.9. Further exploratory diagnostics and limits

Latent-shape fits and random-effects summaries describe appreciable variation in the error-reporting subset. Their fitted dispersion includes ecological differences, study methods, and potentially unmodelled dependence. An I^2 estimate is not a measured fraction of variance caused by ecosystem differences. AIC rankings among Gaussian, Laplace, and Student families do not establish adequacy or identify ecological classes. Error-aware, matched-sample pass fractions are internal model checks, not independent predictions of individual failure.

The cross-study comparison of 639 lake-fish estimates with 377 records associated with Gaedke's Lake Constance study does not partition spatial and temporal variance. Different taxa, sampling designs, fitting methods, and measurement uncertainties prevent treating their variance ratio as a bound on one population. Reproducing Arranz et al.'s reported summary statistics from the same underlying observations checks extraction, not independent replication of a dispersion parameter.

Exponent metadata also need source verification. Counts in equal-log bins and abundance divided by linear bin width have slopes differing by one under power scaling. The Perkins et al. (2018) count-bin method illustrates why a database category alone may not identify the estimand. Choosing a mapping because it makes a slope closer to -1 or -2 would bias the test. Secondary conventions remain sensitivity analyses; a complete study-level methods audit is needed before pooling them. Checks of two dominant primary studies do not validate every record in the primary subset.

S5. Frozen forecasts, full comparisons and eligibility

The analyses of Section S4 assess data that had already been inspected. We therefore added tests in which the resource, its cost law, the forecasts and the decision rules are fixed before the evaluated outcomes are decoded, and in which whole units are held out: launches, a calendar year, vessels and selection treatments. Each executed study retains its inputs, frozen protocol, archived algorithms, forecasts and results in an append-only run registry. Three kinds of evidence are kept separate: measurements on engineered systems, an unused observation period of a natural catalogue, and published biological experiments reanalysed retrospectively. Published summaries of those experiments had been read, so their reanalyses are retrospective validations, not blinded tests. A further plant study uses fixed scoring and a development-only geographic split after accidental raw outcome exposure. Its freeze cannot establish that those outcomes were unseen.

S5.1. Controlled allocation measurements

In a controlled workload experiment, separately measured memory and CPU costs of matrix tasks predicted completion profiles under assigned quota pairs within four tolerances frozen before 672 validation tasks (workload pilot). A fresh launch exposed a transfer boundary: the original forecast failed its unchanged tolerance in two of four conditions, and local recalibration recovered one (workload transfer). The quotas were acceptance criteria, so these tests concern the transfer of cost models, not allocation.

An allocation test requires the system, not the experimenter, to divide a resource. In a Python runtime whose global interpreter lock serializes execution, worker threads running matrix kernels of five sizes competed for one execution resource (dimensionality intervention). Separate calibration gave CPU cost degrees 1.870 and 2.686 for the two kernels. The calibrated mean-cost curves predicted complete CPU and job profiles before 16 allocation trials, with maximum CPU-share errors of 0.0035–0.0063 and job-share total-variation errors of 0.0021–0.0124. Replacing the largest-class thread with a second smallest-class thread produced the predicted CPU-share change within 0.0078. A unit-cost alternative had count errors of 0.126–0.273. This success concerns one engineered allocation mechanism; it identifies no geometric dimension and no tendency of natural systems.

S5.2. Calibrated decisions and an unused observation period

Complete-profile equivalence decisions were calibrated on 180,000 independently generated datasets in 90 conditions (profile calibration). Simultaneous bands met the declared gate for dense, bounded, independent blocks at 48 blocks, with minimum coverage 93.95%, but failed at 12 and 24 blocks. With twelve serially dependent blocks, wrong-departure rates reached 43.75%. These failures define observation designs for which no neutrality verdict is available, whatever a nominal interval suggests.

For solar flares, measured rise-phase fluence costs, six logarithmic classes, three forecasts, a tolerance and an application gate were committed before the 2025 NOAA records were acquired (solar validation). Of 3,277 events, 356 met the domain rules. Logarithmic neutrality predicted the class counts much better than linear neutrality, with count total variation 0.125 against 0.573; historical counts did better still, at 0.022. The point profile departed from the $\ln 1.5$ margin, but the calibrated decision remained unresolved, as declared before acquisition, and the instrument mix changed between periods.

S5.3. Published biological measurements

Study	Held out	Resource and cost	Frozen comparison	Outcome
Synechococcus quotas (Harcourt et al., 2024)	Ten cultures at 25 °C	C, N and P per cell against measured diameter	Fixed cube, free power, strain means, temperature trends	Fixed cube best for C; strain means best for N and P
Size-selected *Dunaliella* (Malerba et al., 2018)	Large and small treatments in two folds	Biovolume at carrying capacity in one shared medium	Equal biovolume ($d = 1$), fitted size law, assigned exponents, equal cell number	Equal-biovolume forecast best; $d = 1.02$ describes the post-hoc all-lineage fit
Food-web chemostats (Wojcik et al., 2025)	Twelve polyculture vessels	Algal N stock: biovolume times separately assayed N per volume	Persistence, development response, no-herbivore response, equal stock, two-budget cost ratio	No model beat persistence; the cost-ratio rule failed
Harvested plants (Dillon et al., 2019)	Five Colorado plots; prior raw exposure	Direct aboveground dry mass; $q(m) = m$ by definition	Log and linear neutrality, Ohio histogram, bounded Pareto and Weibull	Pareto best for biomass; trained models better for counts; ecological neutrality unresolved

Table S6. Biological transfer comparisons and their retained scope.

Published elemental quotas of four *Synechococcus* strains tested whether a size-based cost transfers to an unused temperature (cost transfer). Trained on 16–22 °C cultures, a cubic diameter law with one fitted intercept predicted held-out 25 °C carbon quotas within a typical factor of 1.14. For nitrogen and phosphorus, strain means beat both size laws, and a fitted exponent never beat the fixed cube. Extrapolated temperature trends were worst for every element. A cost calibration can therefore transfer for one resource while failing as a size law for another resource in the same cells.

The accounting identity behind orthopolity, $N\,q(V) = R$, predicts how abundance scales with per-object cost when a budget is shared. Thirty *Dunaliella tertiolecta* lineages, artificially selected for about 280 generations into small, control and large classes, were regrown separately in one F/2 medium after replete, N-deprived or P-deprived histories (size budget). The protocol was committed before any small- or large-selected outcome was decoded. Fold A predicted large lineages from small and control lineages; fold B predicted small lineages from control and large lineages. Controls were calibration data in both folds, never a held-out treatment.

The large-selected treatment had 10.4 times the mean cell volume of the small-selected treatment and 0.975 times its attained biovolume: these two treatment means differed by less than 3%. Controls had mean log capacity 11.89, against 11.72 for small and 11.70 for large lineages. The unrounded retained means imply about 18% and 21% higher geometric-mean capacity, respectively. Thus three-percent uniformity does not describe all treatments or all 30 lineages. The all-lineage cost exponent $d = 1.02$ is a post-hoc descriptive fit (Figure S5), separate from the frozen equal-biovolume prediction $d = 1$. The frozen equal-biovolume law ($d = 1$) had held-out errors of 0.124 and 0.179 log units. It outperformed a size law fitted within part of the range,

whose fitted cost dimension was 0.74 in one fold and 1.12 in the other, and an equal-cell-number law, with errors of 1.48–2.08. A carbon cost degree assigned from the *Synechococcus* study ($d = 0.91$) performed comparably; its nitrogen degree ($d = 0.80$) did not. Regrowth after N deprivation nearly restored capacity, whereas P deprivation left increases of about 0.25–0.31 log units in control and large lineages (about 28–36%), but not in small ones. Restoring the medium therefore did not guarantee a return to the replete profile.

In 24 food-web chemostats, a nitrogen pulse redistributed algal resource composition (chemostat response). Pre-pulse N and C per cell volume from separate monocultures converted three algal groups' biovolumes into resource stocks. Forecasts for twelve held-out polyculture vessels were frozen before their post-pulse values were decoded. The pulse moved resource composition by 0.25 total variation on average, about 1.5 times the pre-pulse variation around baseline, yet no forecast improved appreciably on persistence: 0.243 for the development response against 0.247. A two-budget allocation rule, in which a binding secondary carbon budget reweights each group's share by its measured C:N ratio, failed on three counts. Its maximum attainable redistribution, 0.05–0.12, was below every vessel's maximum observed departure over days 0–12, 0.29–0.49. Its ordinal prediction (which group has the smallest time-weighted log share growth) held in 4 of 12 vessels. Its primary midpoint score was 0.251 against persistence at 0.247. No held-out vessel met the declared recovery criterion: return within 0.1 total variation of baseline at two consecutive observed samples, by day 12. These are the original frozen outcomes; the separate follow-up in S5.10 examines their explanation.

Full Dunaliella forecast comparisons

Held-out mean absolute log error of carrying capacity:

Model	Large held out, biovolume	Small held out, biovolume	Large held out, OD	Small held out, OD
Equal biovolume ($d = 1$)	**0.124**	0.179	**0.065**	**0.094**
Assigned carbon exponent ($d = 0.91$)	0.281	**0.175**	0.229	0.121
Fitted size law	0.626	0.259	0.329	0.136
Assigned nitrogen exponent ($d = 0.80$)	0.509	0.274	0.457	0.278
Equal cell number ($d = 0$)	2.078	1.480	2.026	1.513

Table S7. All frozen Dunaliella size-cost forecasts: held-out mean absolute log error in calibrated biovolume and optical density (OD).

The assigned exponents are the pooled carbon and nitrogen cost degrees of the archived *Synechococcus* study, converted to volume ($D/3$). They are labelled cross-taxon hypotheses.

Frozen verdicts, calibrated biovolume:

- equal biovolume outperforms the fitted size law: 0.50 and 0.08 lower; the fitted law is lower in 0 and 3 of 10 lineages;
- equal biovolume outperforms the assigned nitrogen exponent;
- the assigned carbon exponent outperforms the fitted size law;

- the fitted size law outperforms equal cell number;
- the assigned carbon exponent and equal biovolume are not distinguished.

On raw OD, every difference keeps its direction. Two verdicts narrow to "not distinguished": equal biovolume and the assigned carbon exponent, each against the fitted law. Their fold B gaps (0.042 and 0.015) fall under the 0.05 margin.

The fitted law extrapolated badly because the folds disagree.

- Small and control lineages give a slope of $+0.26$ ($d = 0.74$).
- Control and large lineages give -0.12 ($d = 1.12$).

Controls have mean log capacity 11.89, against 11.72 for small and 11.70 for large. Using the unrounded retained means gives about 18–21% higher geometric-mean capacity. Fitting a line within either half of the range then mispredicts the other extreme.

Across all 30 lineages (a post-hoc description), the slope is -0.023 and the implied cost dimension is $d = 1.02$; the OD gives $d = 1.01$. Large lineages have 10.4 times the cell volume of small ones, reach 0.975 times their biovolume and hold 0.093 times as many cells.

Dunaliella restoration comparisons

Mean change in log capacity relative to the replete history (biovolume; OD in parentheses):

Treatment	**After N deprivation**	**After P deprivation**
Small-selected	-0.12 (-0.06)	$+0.01$ ($+0.01$)
Control	-0.04 (-0.02)	$+0.31$ ($+0.16$)
Large-selected	-0.07 (-0.04)	$+0.25$ ($+0.13$)

Table S8. Dunaliella treatment-mean changes in log capacity after deprivation; parentheses give the OD sensitivity.

After N deprivation, the treatment-mean log changes ranged from -0.04 to -0.12, corresponding to geometric-mean capacities about 4–11% below replete. After P deprivation, control and large lineages increased by 0.25–0.31 log units (about 28–36%), while small lineages did not.

Errors of the restoration models in each fold:

Model	**Fold A (large held out)**	**Fold B (small held out)**
Full restoration	0.178	**0.126**
History offset	**0.090**	0.181
Within-history mean	**0.090**	0.262
Within-history size law	0.530	0.624

Table S9. All frozen Dunaliella restoration forecasts: held-out mean absolute log error.

Full restoration and the history offset are not distinguished, because the folds favour opposite models. The development offset, learned partly from lineages that overshoot, transfers to large lineages but not to small ones. The within-history mean outperforms the within-history size law.

The treatment comparison is a retrospective published-experiment validation. Common medium does not independently establish an identical attained limiting-resource budget, and cell number

inferred from biovolume and mean volume is not a separate counted endpoint. These are point errors for ten held-out lineages per fold; no interval or coexisting-class neutrality verdict is supplied.

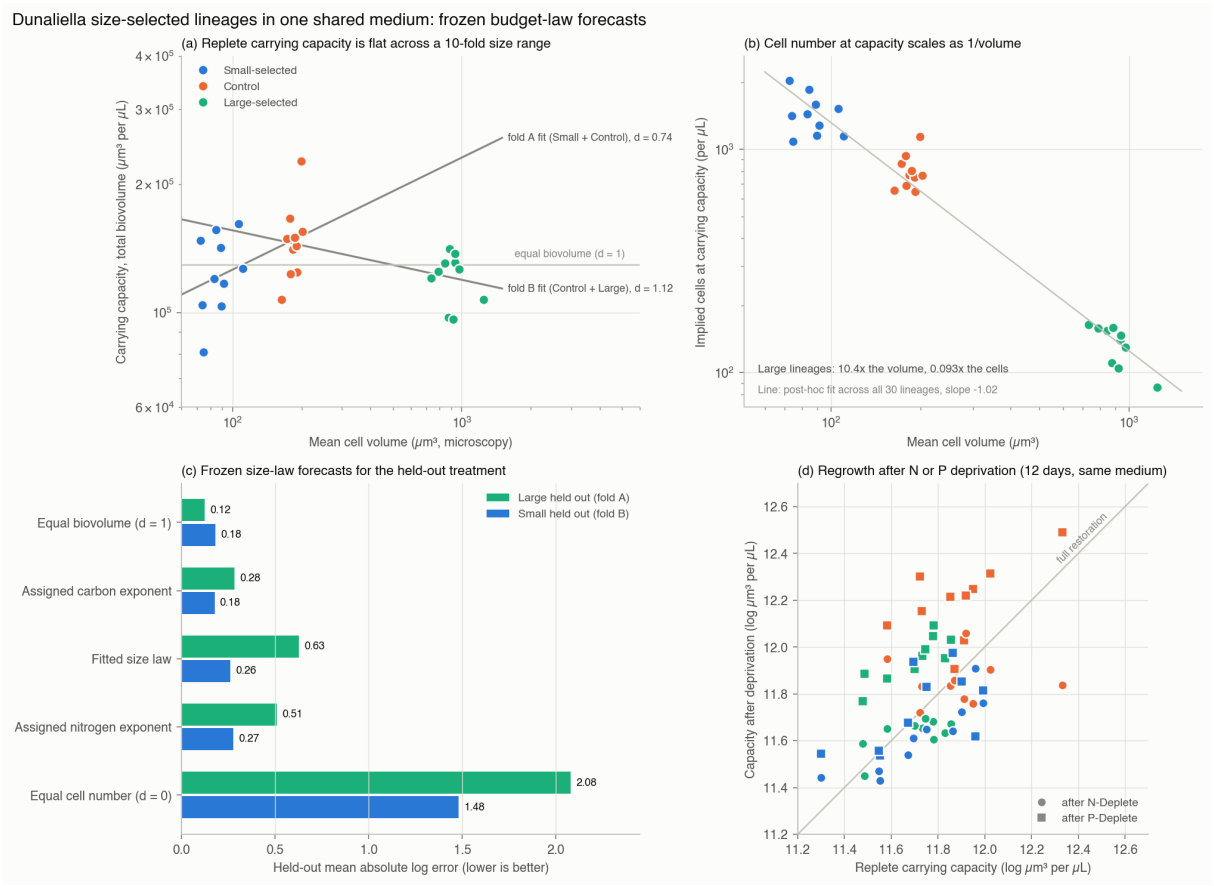


Figure S5. Size-selected Dunaliella lineages regrown in one shared medium, from the data of Malerba et al. (2018). (a) Replete carrying capacity in total biovolume against mean cell volume, with the two cross-fitted size laws and the equal-biovolume law. (b) Implied cell number at capacity. (c) Held-out errors of the frozen forecasts. (d) Capacity after N or P deprivation against replete capacity. Generated from retained outputs by report_dunaliella_size_budget.py.

S5.4. Directly weighed coexisting plant stocks

Ten harvested herbaceous plots provide direct aboveground dry masses for coexisting ramets or stem clusters (Dillon et al., 2019; plant study). Forest and desert allometric masses are excluded. Five Ohio plots supply development fits; five Colorado plots supply evaluation. The protocol, algorithms and forecasts were committed before formal scoring, but raw rows had already been exposed during source inspection. This is a retrospective transfer comparison. It identifies neither a limiting nutrient nor an opportunity budget; $q(m) = m$ defines the measured stock rather than testing an independent cost law.

The frozen domain is 0.01–100 g, split into eight half-decade bins with empty classes retained. The upper bound depends only on Ohio masses. Thirty missing mass records, 144 subthreshold ramets and one above-domain ramet remain in the evaluation membership ledger. The latter weighs 112.55 g and accounts for 24.50% of one plot's known biomass. In-domain mass coverage is 75.48% there and 99.89–99.97% in the other four plots. Scores therefore describe the declared domain; the excluded mass and unknown missing mass cannot be silently absorbed.

The primary score is equal-plot mean total variation between realized normalized biomass and each expected-stock or empirical template. Bounded Pareto scores 0.470, logarithmic neutrality 0.493, linear neutrality 0.507, the development histogram 0.510 and bounded Weibull 0.531 (Figure S6). Pareto beats logarithmic neutrality in three of five plots; large opposing plot differences leave a mean improvement of only 0.023. Count profiles provide a different comparison: the development histogram, Pareto and Weibull score 0.260–0.271, versus 0.412 for linear and 0.495 for logarithmic neutrality. Their binned count log losses are about 1.83, versus 2.08 and 2.42. These are point discrepancies, without a calibrated ecological superiority or equivalence verdict. They also do not contradict the original study's superior within-site Weibull fits: the models here transfer across locales and share a development-fixed domain.

Synthetic calibration fixes census size and draws masses from m^{-2} on the same domain, so each bin has equal expected biomass before measurement rounding. At 160 independent ramets, mean realized stock TV is 0.441 and every simulated census has an empty bin. The largest bin's average normalized share is 3.13%, although its normalized expected stock is 12.50%. At 1,280 ramets these values are 0.244, 87.2% and 9.12%. Thus $E[R_i/\sum_j R_j]$ differs from $E[R_i]/\sum_j E[R_j]$. A separately generated iid 95% TV envelope has exceedance rates 6.2% and 5.2%, but repeating masses in blocks of twenty raises these to 97.8% and 99.9%. This stress test is not an ecological dependence model. Field dependence, inclusion and independent regional replication remain unidentified, so the ecological neutrality decision is unresolved. An uneven small census alone cannot settle a claim about expected allocation.

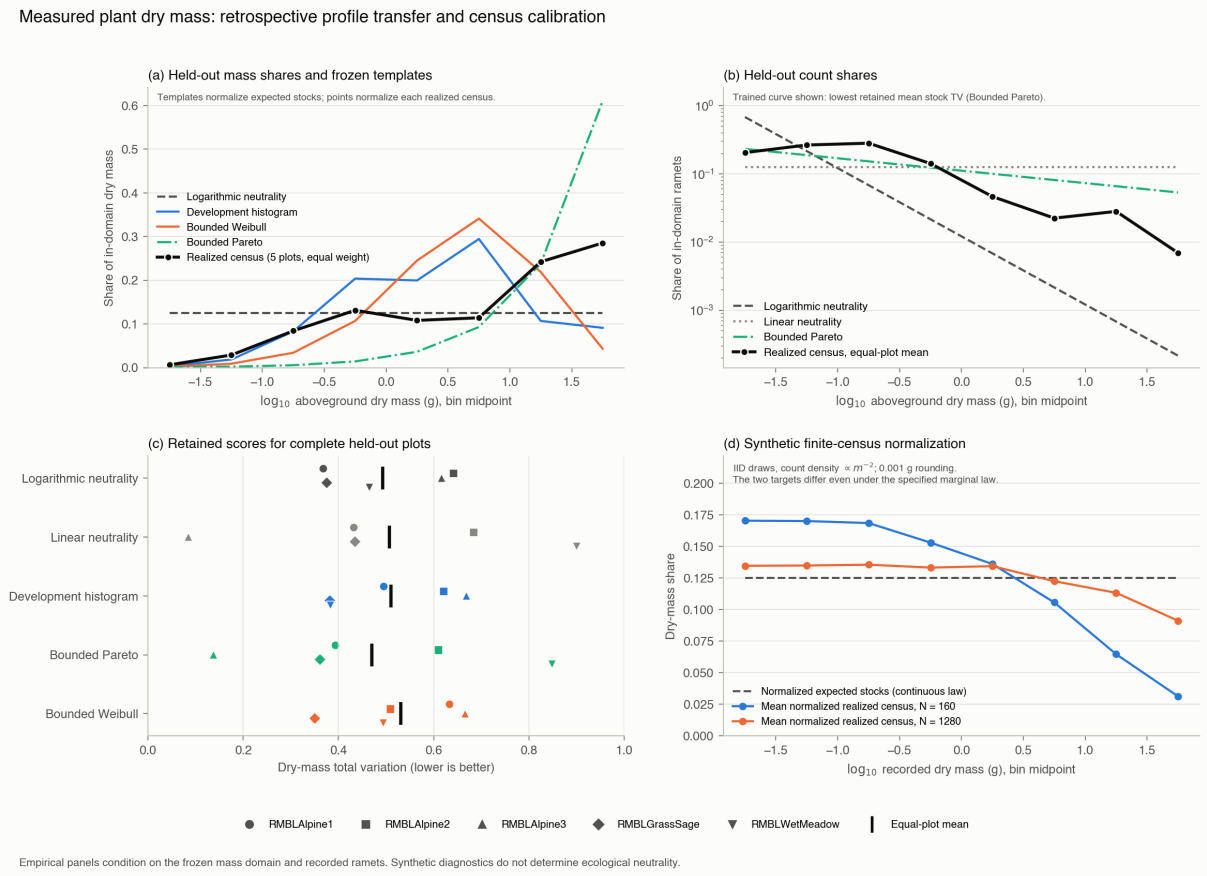


Figure S6. Directly weighed plant profiles and synthetic observation limits. (a) Equal-plot mean Colorado biomass shares and transferred stock templates. (b) Count shares and retained count forecasts. (c) Each plot's biomass discrepancy and the five means. (d) Mean normalized census shares under iid logarithmic neutrality, compared with normalized expected stocks. All panels read retained outputs; report_plant_biomass_profile.py performs presentation only. Prior raw exposure and excluded biomass are retained.

S5.5. Linguistic resources and withheld genres

The linguistic study turns the inverse proposal into a measured transfer comparison (protocol and report). Word occurrences come from the pinned Universal Dependencies English EWT corpus, release 2.18; letter counts L come from original sentence text and phoneme counts P from an independently pinned CMU pronunciation dictionary. The model uses a counting measure on word types, with no fitted rank transformation or post-hoc constraint factor.

Training uses only email, weblog and newsgroup texts: 83 documents and 83,635 retained tokens determine a vocabulary of 2,623 types with at least five training occurrences and a dictionary pronunciation. All six forecasts share this vocabulary. Conditional multinomial likelihood with nonnegative exponents gives

$$\widehat{Q}_w \propto L_w^{0.5153} P_w^{1.8275}, \qquad \widehat{p}(w) \propto \widehat{Q}_w^{-1}.$$

Both exponents are interior. The log-feature correlation is 0.871 and the model information condition number is 11.65; these diagnostics describe the fitted model, without treating dependent tokens as independent measurements of parameter precision. The protocol, code and fitted forecasts were hashed before the official test file was acquired or decoded.

The primary target comprises withheld review and answer genres: 253 documents, 7,329 retained tokens and 77.65% vocabulary coverage after exact-duplicate exclusion. The secondary target contains 63 new documents from the training genres, with 9,618 retained tokens and 79.35% coverage. Ninety-three sentences matching original training text were excluded by the frozen rule, and no training/test document IDs overlap. Coverage is conditional on the tokenizer and fixed vocabulary; excluded tokens remain in the ledger.

Frozen forecast	**New genres**	**Represented genres**
Uniform word types	7.872	7.872
Inverse letter count	7.405	7.457
Fitted letter power	7.239	7.328
Fitted letter/phoneme product	7.055	7.189
Exponential letter cost	7.195	7.304
Smoothed training frequencies	6.136	6.199

Table S10. Held-out conditional log loss in nats per retained token; lower is better. Every parameter and the vocabulary are fixed from training.

The composite improves on letter power by 0.1843 nats per token in new genres; the paired document-bootstrap 95% interval for composite minus letter-power loss is $[-0.1967, -0.1696]$. The secondary improvement is 0.1387, with interval $[-0.1512, -0.1251]$, and the point difference favors the composite in all five genres. The 2,000-resample intervals condition on the fitted forecasts and exchangeability of documents within each evaluation group. They omit training uncertainty and dependence between authors, threads or websites.

The gain demonstrates transferable information in the combined symbolic features (Figure S7). Training word frequencies nevertheless predict much better. Only 83 distinct (L, P) pairs occur among the 2,623 vocabulary types, so the resource model must give many different words identical probabilities. Observed primary letter-resource and inferred-cost shares remain far from uniform over word types, with descriptive total variations 0.693 and 0.689. Within every individual genre the composite distance is larger than the letter-resource distance, despite the small decrease after pooling. These finite-corpus discrepancies are not calibrated neutrality tests. The fitted combination is a predictive candidate; neither equal resource allocation nor a unique physical

linguistic cost has been established. Sonority, duration, context and semantic requirements remain unmeasured here.

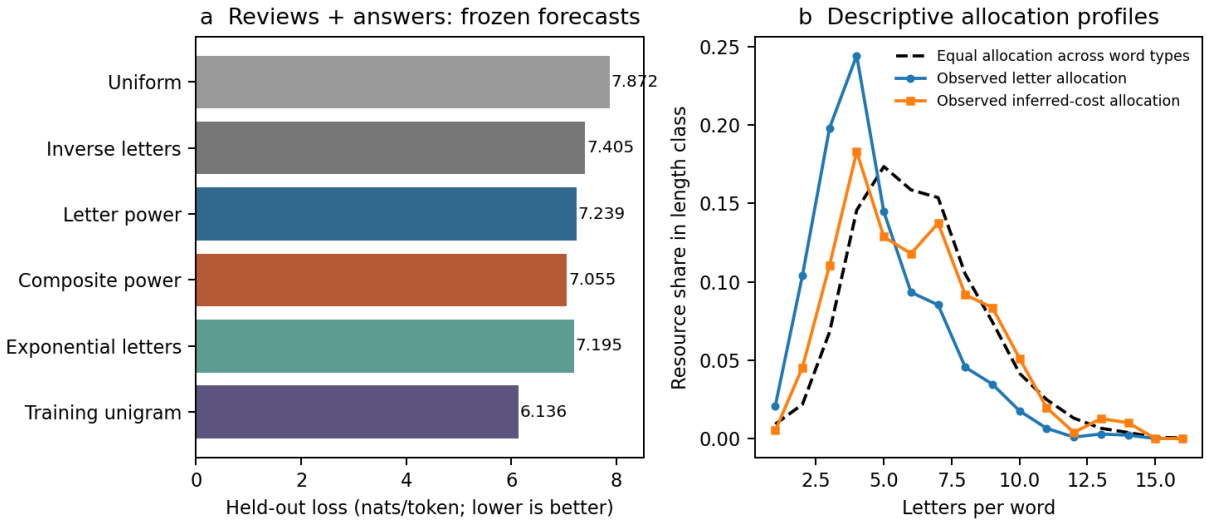


Figure S7. Frozen linguistic predictions. (a) Loss on withheld review and answer genres, including all six declared alternatives. (b) Observed letter and inferred-cost allocation aggregated by word length. Equal allocation over word types gives the dashed type-multiplicity profile, not a flat profile over length. Aggregation conceals variation among words of the same length. The fitted composite improves prediction over letter-only costs but trails training word frequencies; its resource profile remains descriptive.

S5.6. Birth-calibrated bacterial population summaries

Gangan and Athale (2017) measured *E. coli* MG1655 in mother-machine and batch settings at 37°C. We transcribe the printed arithmetic means and variances of their lognormal fits in the source article's Figure 2 (birth/division) and Figure 1 (batch snapshot), using official NCBI images. The source authors retain credit for the experiments and original fits.

Medium	Birth mean	Birth variance	Division mean	Division variance
LB	2.7391	0.9419	5.2833	3.2094
M9 + succinate	1.4913	0.0540	3.2358	0.2084

Table S11. Published bacterial birth/division calibration moments; lengths are in micrometres and variances in square micrometres.

Means are in micrometres; variances in square micrometres. These are fitted moments, not log-space parameters or the slightly different sample statistics quoted in the article prose. Calibration sample counts are 153/110 (LB birth/division) and 133/68 (M9). The source summaries do not establish independent replicates, complete uncensored cycles, or common individual growth rates.

The division mean divided by twice the birth mean is 0.9644 in LB and 1.0849 in M9; division/birth CV ratios are 0.9570 and 0.9054. Exact symmetric stationary compatibility is therefore not satisfied by the separately fitted laws. We do not subtract those incompatible lognormal CDFs and clip negative tails. Instead, the primary conditional model uses the birth fit and imposes $D \overset{d}{=} 2B$; a separate, declared sensitivity uses the division fit divided by two.

The protocol, implementation, calibration sources, and all four forecasts were hashed in the freeze before acquisition of the source article's Figure 1. The source's qualitative findings and calibration

measurements had already been read. This separation protects the numerical comparison from retuning; it is not external preregistration or global blinding. Dryad's raw workbook returned HTTP 403, so raw-cell uncertainty and distribution fit are not evaluated. The archived published summaries remain reproducible offline.

For birth mean m and squared CV c^2, the lognormal model gives

$$E_P[x] = \frac{2m \ln 2}{1 + c^2}, \qquad \mathrm{CV}_P^2 = \frac{1 + c^2}{2(\ln 2)^2} - 1.$$

The chronological comparator gives $E_L[x] = m/\ln 2$ and $\mathrm{CV}_L^2 = \frac{3}{2}\ln 2(1 + c^2) - 1$. All forecasts retain the calibration length scale; no target-culture multiplier is fitted. The fixed-birth comparator sets birth variance to zero.

Results, including the unsuccessful comparison

Medium and quantity	Published target	Population	Lineage	Fixed birth	Division / 2
LB mean	3.2842	3.3737	3.9517	3.7972	3.2845
LB CV	0.3957	0.4139	0.4126	0.2017	0.4004
M9 mean	2.0094	2.0184	2.1515	2.0674	2.1991
M9 CV	0.2146	0.2568	0.2549	0.2017	0.2478

Table S12. Every frozen bacterial endpoint forecast; CV is dimensionless and mean length is in micrometres.

Targets come from 1,217 and 3,665 reported cells respectively. Their printed fit variances are 1.6888 and 0.1859. No inferential interval is manufactured from those counts and rounded fitted moments.

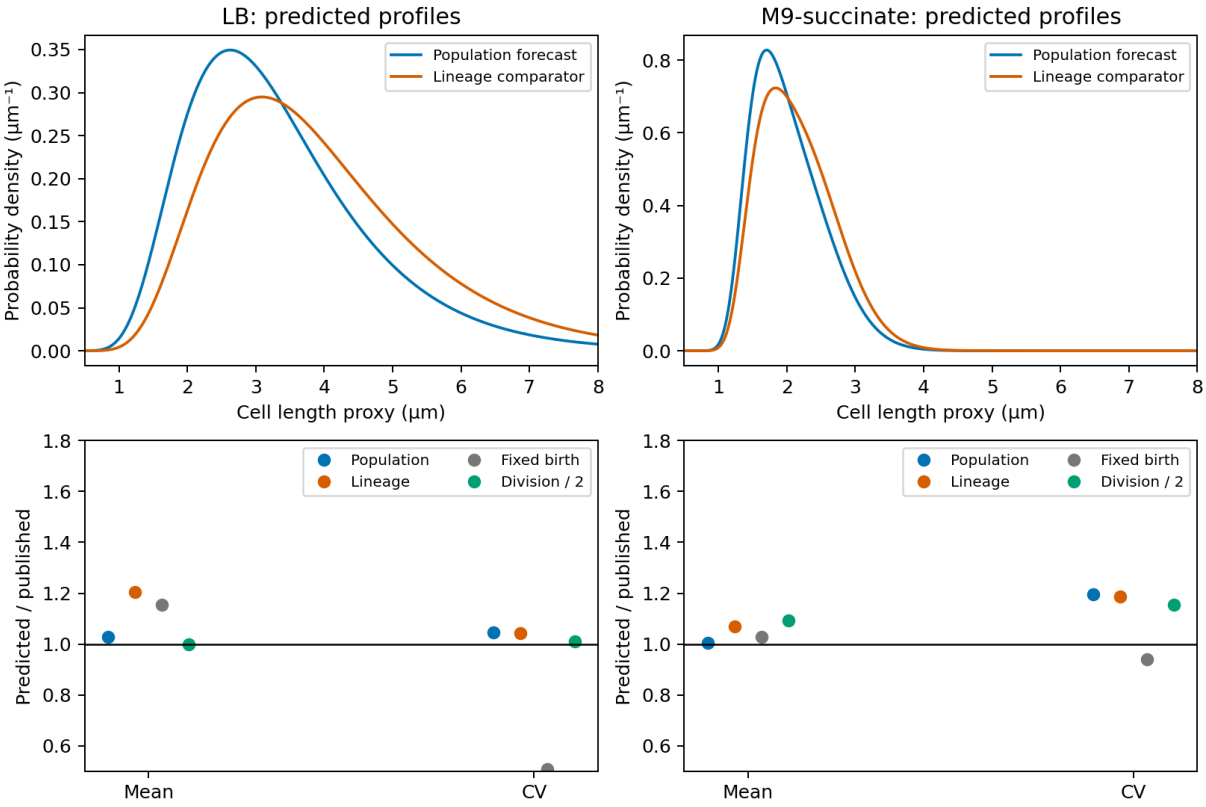
The primary mean forecast is 2.72% high in LB and 0.447% high in M9, compared with 20.32% and 7.07% for the lineage comparator. Absolute log errors are 0.02688/0.004456 for the population and 0.18502/0.06832 for the lineage. Population CV is 4.61% and 19.69% high. The lineage CV errors are slightly smaller, so the declared requirement of improvement on **both endpoints** is not met in either medium. No pooled success verdict replaces that result. The CV advantage is small and structural: the multipliers of $1 + c^2$ in the two squared-CV formulas are 1.04068449 and 1.03972077. This endpoint weakly distinguishes the sampling schemes.

Interpretation correction to the frozen protocol: its statement that failure "rejects this complete length-proxy transfer model" is too strong. The declared descriptive criterion establishes no joint predictive superiority; it supplies no statistical model rejection. The original protocol is preserved, with this clarification also retained in the registry.

The fixed-birth model also predicts M9 variability more closely than the primary model. Division-based calibration happens to match LB very closely but raises the M9 mean error to 9.44%; it remains a sensitivity, not a selected winner.

Figure S8. Top panels are model predictions, not observed histograms. Bottom panels show predicted/published endpoint ratios for every declared model; unity is exact agreement. The machine-readable results retain all errors and calibration diagnostics.

This adds a concrete natural-system mechanism and an independent-experiment mean prediction to the framework. It does not establish accurate full-profile transfer, independently verified neutral eligibility, or a general biological allocation law. The variability discrepancy and calibration mismatch delimit the model's explanatory scope. Resolving them requires growth, partition,



width and sampling information; naming them possible constraints is not a measurement of their effects.

S5.7. Geometric predictions and seismic eligibility

Separate repository registrations compare physical resource readings using earthquake aftershocks, flare ribbons and lake geometry. They concern log-orthopolic allocation per logarithmic size coordinate. Inputs, estimator amendments, prior exposure and decision rules are retained in the test ledger. These studies supply negative or qualified comparisons; the literal registered decisions remain distinct from whether an interval excludes an individual predicted value.

Comparison	**Result**	**Scope of the decision**
Aftershock productivity: rupture area predicts $\alpha = 1$, moment predicts 1.5	Primary $\alpha = 0.63$, 95% interval $[0.51, 0.71]$; rupture-length-window sensitivity 0.90, $[0.76, 0.97]$	Both predictions excluded under these counting schemes; registered verdict inconclusive; strong window dependence
Flare ribbon area: density exponents 2 and 7/3	$\alpha = 3.163$, $[2.962, 3.457]$	Both excluded; registered verdict inconclusive
Global lake radius: survival exponent $\zeta_R = 2$	Original geometry 1.930, iid $[1.919, 1.941]$; local-geometry follow-up 1.859, basin $[1.785, 1.924]$	Original iid equivalence disappears with physical local geometry; follow-up inconclusive against the $[1.9, 2.1]$ margin

Table S13. Retained geometric predictions. Intervals condition on each study's declared sampling and measurement assumptions; they do not establish a universal resource law.

The lake follow-up independently changes projection and quantifies basin dependence. Two local projections agree closely, identifying the original global projection as a material source of the change. Exact 2 lies outside the basin interval, while practical equivalence and practical departure are both unestablished under the frozen margin. Invalid source topology, unmatched basin outlets and an iid goodness-of-fit diagnostic remain qualifications. The separately frozen UK result likewise does not establish a robust spatial rejection: its iid interval excludes 2, but its spatial interval includes it.

The seismic comparison predicts $b = D_2/2$ for rupture territory, using hypocentre correlation dimension, against $b = 1$ independent of that dimension. The Southern California qualification uses published sequence memberships. Forty sequences have at least 200 raw events; quality, completeness and location-resolution gates reduce this to six, below the registered minimum ten. No b or D_2 was fitted.

A subsequent CLASS/HORUS source audit establishes 295,193 unambiguous Italian location/provenance links, of which only 699 have direct moment-tensor magnitudes. The calibrated proxies require an observation model. A separately frozen synthetic check, using the published conversions of Lolli et al. (2020), fails its necessary validation gate in 45 of 56 required scenarios, including 30 of 32 noiseless proxy cases. For example, true $b = 1.5$ gives mean 1.158 under one earlier-ML conversion and lower-support choice; exact source-bin integration gives 1.158 as well. Conversion leaves gaps in rounded magnitude classes which the uniform-bin correction does not represent. This is a controlled counterexample to the estimator's adequacy, not an observed Italian exponent or a test of either physical relation. The Italian attempt therefore stops before real sequence construction or response fitting. A revised likelihood would need to represent original magnitude bins, detection and calibration uncertainty before another physical comparison.

Complete lake geometry and spatial comparisons

The original global and UK outcomes were known before this follow-up was designed. Its local Git freeze protects the declared follow-up against subsequent retuning; it is not a new independent confirmation. The resource reading is territory per $d \ln R$, and all intervals below retain their stated resampling assumptions.

All **1,427,688** original HydroLAKES features were processed. The inclusive 1–20 km range and equivalence band were held fixed.

Geometry / population	**Selected lakes**	ζ	**Iid 95% interval**	**Spatial 95% interval**
Original EPSG:6933, largest ring	119,683	1.9295	[1.9185, 1.9408]	Country: [1.5945, 2.0296]
Local LAEA, largest exterior ring	70,199	1.8614	[1.8477, 1.8760]	Country: [1.6105, 1.9600]
Local LAEA, full lake geometry — primary	70,486	1.8588	[1.8443, 1.8739]	Country: [1.6071, 1.9539]
Local LAEA, catchment-matched lakes	70,455	1.8585	[1.8429, 1.8726]	Basin: [1.7851, 1.9244]
Local geodesic AEQD, full geometry	70,486	1.8588	[1.8444, 1.8726]	Country: [1.6073, 1.9453]
Separate UK regional study, full geometry in EPSG:27700	148	1.4413	[1.1913, 1.7144]	100-km grid: [1.1657, 2.0087]

Table S14. Global geometry and spatial sensitivities, plus the separate UK regional comparison. Survival exponent is ζ; interval assumptions differ by row.

The original result and its literal decision remain preserved. The UK row comes from its separate regional protocol and report, with 29 occupied spatial cells. Its frozen either-interval departure rule triggers on the iid interval, while its grid interval includes 2; robust spatial rejection is therefore not established. The table compares completed calculations, not a menu from which to select a favourable method. All follow-up choices were frozen before its fitted outcomes.

The local-largest-ring comparison attributes most of the exponent change to the projection and consequent radius selection; accounting for holes has a smaller effect. Relative to the original selection, 3,676 lakes enter and 52,873 leave. The full source contains 216,330 hole rings and exactly one exterior ring per feature; 29,112 selected lakes contain holes.

LAEA and AEQD select **exactly the same lakes**. Their exponent estimates differ by about 0.00000021, and their maximum selected radius discrepancy is only 9 parts per million. The longest centre-to-vertex distance in a selected lake is 85.31 km, below the declared 200 km limit. Local projected areas differ from independent geodesic polygon areas by at most about 1.8 parts per million on the selected sample. These checks make residual local projection distortion a poor explanation for the new exponent. They do not establish mapping completeness or intrinsic geodesic exactness.

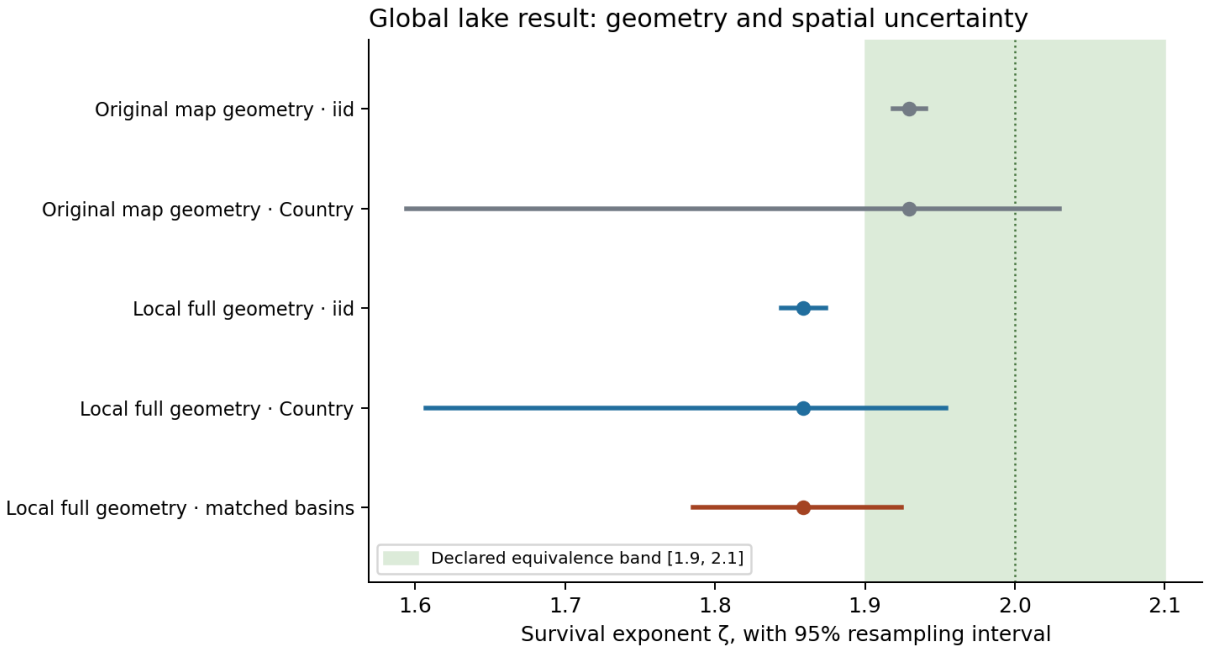


Figure S9. Retained global lake geometry comparisons and their conditional uncertainty intervals.

Spatial and distribution qualifications

The official **HydroBASINS level-3 v1.c** files were reachable. All nine regional archives supply 292 major drainage regions. The deterministic outlet join matches **70,455 of 70,486 lakes (99.956%)**, occupying **246 basins**. There are 31 unmatched outlets and no ambiguous multiple matches. Unmatched lakes remain in the global estimate; the basin interval explicitly describes the matched population. Its point estimate differs from the all-lake estimate by only 0.00023, but the declared complete-coverage gate still fails.

The largest occupied basin contains 7.17% of matched lakes. By contrast, the largest Country block contains 53.67% of selected lakes. Different grouping scales give different intervals; neither grouping guarantees independence between blocks or accounts for mapping error.

379 selected lake geometries (0.54%) are topologically invalid under GEOS. None triggers the corrected orphan-hole conversion check. Their signed moments are retained, as declared; no lake is silently repaired or excluded. The all-valid-topology gate therefore fails. The basin overlay's separate, predeclared linework repair was needed for 209 of 292 basin features; source archives were unchanged. A pre-fit audit found no basin conversion errors or raw longitude jumps over 180°.

The bounded, refitted **iid** KS diagnostic gives D=0.004618 and p=0.01399 (13 of 1,000 simulated statistics at least as large). This is a small absolute CDF discrepancy but fails the declared diagnostic threshold. Spatial dependence prevents interpreting its p-value as a calibrated spatial rejection. It is another reason to describe a fitted exponent without claiming positive power-law support.

The declared `Lake_type == 1` point-only sensitivity gives ζ=1.9752 for 66,074 lakes. This means **not flagged as reservoir or regulated** in this source, not independently verified natural lakes. No interval was declared for this sensitivity, and its near-2 estimate does not replace the primary outcome. Any further classification analysis must disclose that this result is known.

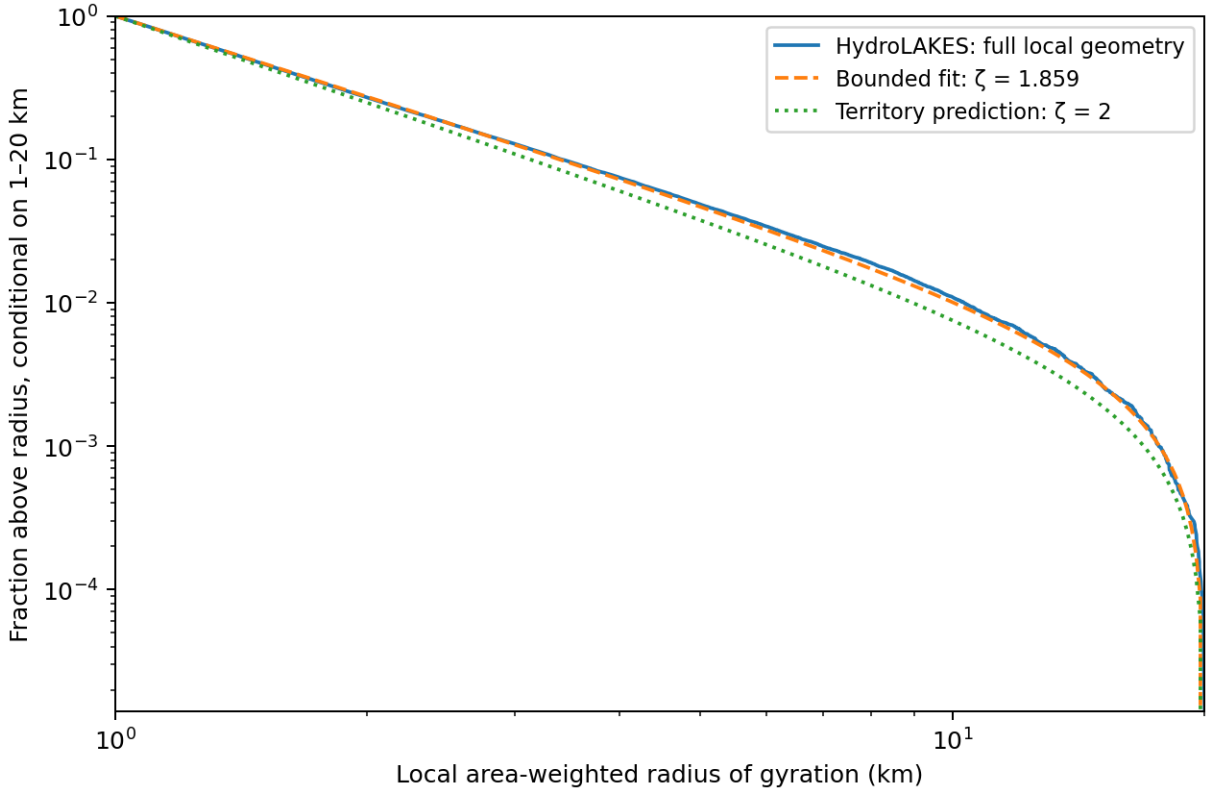


Figure S10. Observed lake-radius survival and bounded fitted model; the iid adequacy diagnostic is qualified by spatial dependence.

Seismic source and resolution eligibility

Necessary condition, in order	Sequences remaining
At least 200 published events	40
At least 200 after location/event-quality filtering	28
At least 200 above the frozen completeness rule	16
Distance interval spans a factor of at least 3 after the location-error floor	**6**
Required before any b/D2 comparison	**10**

Table S15. Southern California sequence eligibility gates, in their frozen order.

The published 1981–2018 catalogue contains 123,275 events in 26,137 sequences. Membership follows the published largest-event index without reclustering, merging or splitting. The source is Zaliapin and Ben-Zion (2020). All event IDs join uniquely to the matching 634,252-event relocated catalogue from SCEDC, `hs_1981_all_comb_K4_A.cat_so_SCSN_v01`, associated with Hauksson, Yang and Shearer (2012). Published latitude, longitude, depth and magnitude agree exactly with that source.

Three full downloads are pinned in the acquisition manifest: the published labels (13.31 MB), their matching source (98.94 MB), and the newer 1981–2025 regional catalogue (152.81 MB). The last is retained for future work and contributes no observations to this calculation. All byte counts and SHA256 digests pass. The raw files remain local and can be restored from publisher URLs.

The execution supplement, config, parsers, qualification code and 22 checks were committed as **`6ae7a8c` before real completeness or distance-range calculations**. Structural source counts and the original synthetic validation were already known and disclosed. This is a local Git freeze, not an externally timestamped registration. The original hypotheses, 200-complete-event threshold, factor-three distance span and ten-sequence minimum remain unchanged. The approximately 1,000/3,000 events needed in the earlier planar/volumetric simulations were not imposed as extra hard thresholds.

What limits the calculation

Quality filtering retains 96,441 events. It requires cross-correlation locations, earthquake types `le` or `eq`, and finite positive absolute horizontal and vertical errors below the declared placeholder cutoff. The ledger retains every event, including exclusions and overlapping reasons. One published clock record has minute=60; it remains flagged and retained because this calculation uses the published memberships and does not use event times.

For each candidate sequence, completeness is rounded MAXC+0.2, bounded by the source's magnitude floor. Sixteen have enough complete events. Their median quadrature absolute-error proxies range from **0.447 to 4.757 km**. With the registered three-error lower radius and upper radius of one tenth of the 90th-percentile pair distance, only six have a factor-three span:

Published sequence ID	Complete events	Mc	Available span
2112	365	2.4	3.415
15732	1,154	2.3	7.032
39006	1,095	2.4	7.943
41355	6,048	2.4	6.498
44940	1,147	2.4	7.603
76958	2,537	2.4	8.754

Table S16. The six published sequences that pass the complete-event and spatial-resolution requirements.

The hypocentre cap is 3,000; the location-error median uses all complete events. Every sampled ID is retained. The ten failed ranges remain in the ledger. Ignoring errors would admit all sixteen, but this was frozen as a diagnostic only and cannot justify a response fit.

This error convention is conservative rather than a complete uncertainty model. Absolute standard-error estimates lack covariance information; shared translation errors can cancel within a relocation component. All sixteen complete sequences span multiple waveform/processing components, so the within-component relative errors cannot simply replace their absolute errors. The finding is about this specified observation model, not proof that these earthquakes can never support a dimension estimate.

Two other limits would remain even if more sequences qualified. Published membership depends on magnitude as well as time and distance, so fixed external labels do not provide magnitude-independent geometry. Also, preferred catalogue magnitudes are not homogeneous Mw, and this source has no magnitude-type column. The SCSN catalogue history describes these magnitude distinctions. A physical test of the moment-based conversion would need independently justified calibration. No D2 contrast, regression slope, or hypothesis interval is claimed here.

Italian observation-model prerequisite and complete required scenarios

There are 400 independent repetitions of 7,000 events in each of 72 cells: two true b-values, four lower magnitude supports and nine observation models. Each eligible sample receives the original MAXC+0.2 and Aki–Utsu calculation, with 500 conditional bootstrap resamples. All 28,800 samples pass the minimum 200-event population gate; the failures concern bias and uncertainty calibration.

The true b-values 1 and 1.5 distinguish the magnitude marginals of the proposed **log-orthopolic** rupture-territory relation b=D2/2 and the b=1 alternative on ideal planar/volumetric supports. This checks the magnitude component, not the full geometry, sequence selection or Deming-regression decision.

The HORUS paper, Tables 7 and 9, supplies the ML/Md conversion coefficients and ML scatter scales. Single-source scenarios isolate their effects: simulate true exponential magnitudes, convert to the source scale, optionally add Gaussian error, round the source to 0.1, convert to Mw proxy and round to 0.01, then apply the registered 0.1 rounding. The normal distribution, hard lower support and homogeneous source populations are explicit simulation assumptions. We have not fitted their frequencies or detection law to the Italian catalogue, reconstructed all HORUS averaging/switching rules, or established that every real mixture has the simulated biases.

Observation model	**Cells passing / 8**	**Mean-bias range**	**Coverage range**
Direct rounded Mw control	7	−0.017 to −0.003	88.3–95.5%
ML before 2011, no scatter	1	−0.342 to +0.067	0–94.5%
ML from 2011, no scatter	1	−0.216 to +0.035	0–90.8%
Md before April 2013, no scatter	0	−0.304 to +0.377	0–32.8%
Md from April 2013, no scatter	0	+0.035 to +0.347	0–45.8%
Earlier ML with source scatter 0.18	0	−0.114 to +0.028	1.3–89.0%
Later ML with source scatter 0.16	2	−0.067 to +0.037	22.5–95.0%

Table S17. All required Italian observation-model scenarios, grouped into eight cells per model; the two shared-slope sensitivities remain outside the necessary gate.

The two additional shared-calibration-slope sensitivities are outside the gate; their 16 cells remain in the output. The full cell table and summary include Monte Carlo errors, Wilson coverage intervals, every failure and the independent population calculation. A single near-threshold control failure could be sensitive to Monte Carlo variation; the large proxy failures are not an inference from that borderline control.

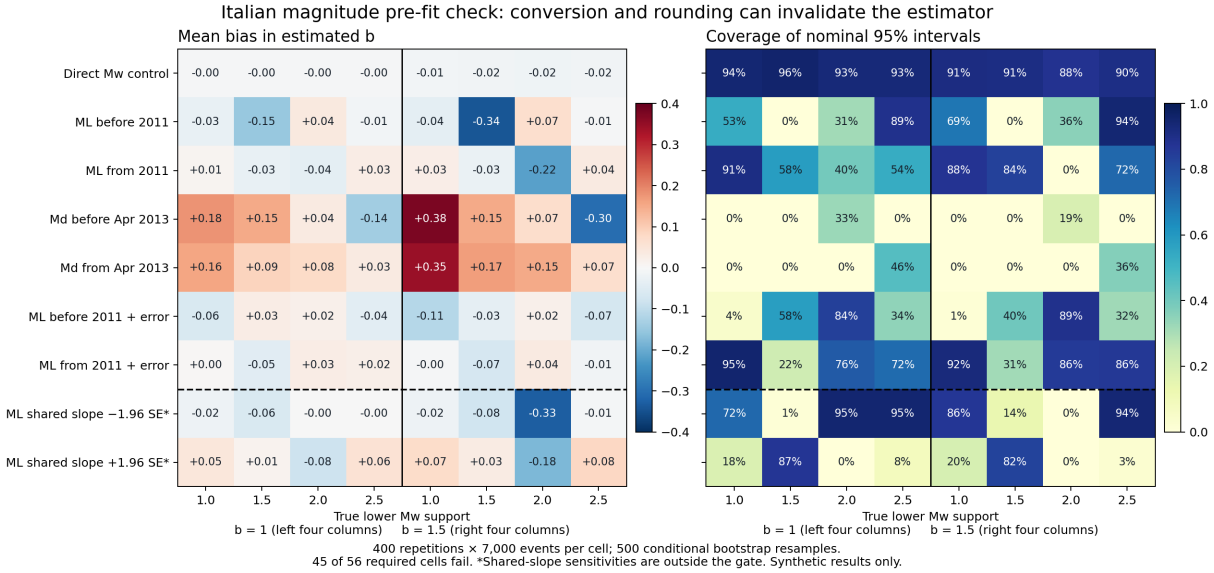


Figure S11. Retained synthetic magnitude-validation cells. Bias and coverage refer to simulated observation models; no real Italian b or dimension was fitted.

Why rounding matters

In the earlier ML conversion, a source grid 1.6, 1.7, 1.8 produces proxy hundredths 1.54, 1.65, 1.75. Under the declared NumPy rounding, the registered tenths are 1.5, 1.6, 1.8: **the 1.7 class is missing**. With true b=1.5 and lower Mw support 1.5, the population MAXC+0.2 is 1.7. Events first appear in the 1.8 bin, while the estimator subtracts `1.7−0.05` from their mean.

The resulting average b is **1.15785**, with Monte Carlo SE 0.00071; an independent integration over source-bin probabilities gives **1.15791**. None of its 400 diagnostic intervals covers 1.5. For the later ML conversion, true b=1.5 above support 2.0 gives mean **1.28430**, compared with the population limit **1.28507**, again with zero coverage. The largest positive nominal bias is an earlier-Md case: mean **1.87694** for true b=1.5.

Even the direct control has the small known finite-bin approximation error: for a complete 0.1 lattice, the population limit of the implemented correction is $2\tanh[0.1b\ln(10)/2]/[0.1\ln(10)]$, or **1.48526** when b=1.5. This remains a magnitude-estimator issue, not an observed earthquake result.

Some population-mode bins nearly tie. The sampled MAXC can then change between repetitions, so an ensemble mean need not equal the single population-mode limit. Mode gaps are retained explicitly; no mismatching cells are discarded.

What a repair would need

For noiseless, singly converted complete tails, an exact geometric likelihood on the **original source bins** recovers b in the independent population calculation. Its simulated point estimates also recover the nominal value. This diagnoses the source of the problem; it is not a registered replacement. With measurement error near a lower detection boundary, even that diagnostic can remain biased (one earlier-ML b=1.5 scenario gives about 1.390).

A future attempt would need a likelihood that represents source-specific bins, calibration, detection and source switching/averaging, plus uncertainty shared by all events using a calibration. A deliberately narrower homogeneous cohort is another design option, but would need fresh

validation and unchanged population/resolution gates. Neither option has been tested on real responses.

HORUS equation 4 also illustrates within-bin jitter using b=1. The retained `_o` input files avoid this extra assumption. Table 9's post-2011 ML covariance, as printed, violates the covariance bound for its two printed standard errors: `0.0011 > 0.065*0.016`. We have neither diagnosed the cause nor repaired it. The two marginal slope perturbations are sensitivities, not a joint calibration uncertainty calculation. These issues strengthen the need for an observation model; the noiseless failures already establish the present stop decision.

S5.8. Scope of the retained comparisons

The positive results concern conditional budget closure: a separately measured or geometric cost, declared shared resource conditions, and abundance or resource shares predicted from them. The runtime experiment and the algal lineages are the clearest cases. In the lineages, volume scaling supplies the cost exponent; the empirical content is the similar attained biovolume at the small and large treatment extremes, together with higher intermediate-control capacities and the retained held-out forecast scores. The capacity theorem makes inverse-volume abundance conditional on that common attained budget. The biological observation concerns whether the lineages actually reach it. A rule that goes beyond closure, by predicting how a restriction redistributes resource among coexisting classes, failed in the grazed, dynamically changing food webs tested. The plant study now observes coexisting class stocks directly, but it does not establish equal expected allocation: the flat template is not best for biomass or counts, and the ecological observation law is uncalibrated. No natural size-class study in S5 has independently identified neutral eligibility and a design sufficient for the expected-allocation verdict. Identifying such a regime, or independently characterizing its constrained allocation, remains a central empirical task. The linguistic study separately shows a gain from an inferred composite in a frozen genre transfer. Its lexical baseline and uneven resource shares retain the limits of that candidate. The bacterial comparison adds a mechanism-derived prediction between separate experimental settings: mean length transfers closely, while variability and the joint comparison fail. Its resource proxy and unverified growth assumptions leave the natural-allocation qualification task open.

S5.9. Data, code and provenance

Code, frozen inputs, configurations, and generated results are in this repository. The make all target verifies input checksums, runs tests, and regenerates the empirical analyses; see the README for setup. Pilot estimates originate in results.json; distribution diagnostics in gof.json; aquatic median, compatibility fractions, and ocean intervals in independent.json. Remaining JSON files contain explicitly exploratory diagnostics. The studies of S5 are registered in an append-only run registry with their inputs, frozen protocols, archived algorithms and outputs. The class-exchange demonstration and retrospective FIRAS calculation are also registered; their reports give offline reproduction commands and interpretation limits. The inverse-resource simulation and linguistic transfer likewise retain configurations, algorithms, predictions, all scored outcomes and source hashes. The linguistic protocol and training fit were frozen before test-file acquisition; this is a retrospective public-corpus study, not external preregistration. The bacterial summary transfer retains official figure images, manual numerical transcriptions, calibration diagnostics and four frozen forecasts. Its numerical target figure was retrieved after the freeze; qualitative outcomes were already known. The geometric tests, seismic eligibility assessment and Italian synthetic magnitude validation retain separate configurations, source receipts and reproduction targets. Their outputs preserve failures and stopping decisions; no real Italian b/D2 result is implied by the simulation record. Plant sources are pinned to author commit

defccc3dcbbbf3ba57ff1572377de88fba83ff7f, with acquisition checksums and complete inclusion ledgers. The registry-verify target audits the registry offline, and each study report gives its replay command; source errata found during those audits are recorded there. Source-specific conditions are recorded in data/NOTICE.md; there is no blanket licence for all inputs. The supplied 2017 manuscript was inspected privately and is not redistributed.

The reports cited in this supplement, rather than this editorial compilation, retain the exact historical protocols and outputs. Numerical corrections to interpretation here do not amend a frozen forecast or relabel a failed comparison. See the status and handoff for subsequent work and current verification counts.

S5.10. Chemostat capacity diagnosis with changing quotas

The retrospective follow-up completed on 11 October 2026 asks whether the earlier chemostat failure can be explained by a different path of resource prices. Its design and tested algorithms were frozen at `13374c5` before this calculation. Outcomes were already known; this is a conditional diagnosis, not a new blinded test. The full report records every comparison, assumption and reproduction command.

For group biovolume $V_j(t)$, fixed nonnegative opportunities w_j and positive elemental densities ρ_{Nj}, ρ_{Cj}, consider

$$V_j(t) = c_t \frac{w_j}{a_t \rho_{Nj}(t) + b_t \rho_{Cj}(t)}, \qquad a_t, b_t \geq 0, \quad a_t + b_t > 0.$$

Writing $r_j = \rho_{Cj}/\rho_{Nj}$, the normalized nitrogen shares satisfy $p_j(t) \propto p_j(0) f_j$, with $f_j = [a_0 + b_0 r_j(0)]/[a_t + b_t r_j(t)]$. For every distinct pair,

$$\frac{f_i}{f_j} \leq \max\left(1, \frac{r_i(0)}{r_j(0)}\right) \max\left(1, \frac{r_j(t)}{r_i(t)}\right).$$

Let K bound all such ratios, allowing all pooled-species ratio endpoints. An event with initial probability x can gain at most $Kx/[1 + (K-1)x] - x$. Maximizing over x proves

$$\mathrm{TV}(p(t), p(0)) \leq U = \frac{\sqrt{K} - 1}{\sqrt{K} + 1}.$$

The bound allows arbitrary prices at both dates and all starting shares. It can be conservative for a particular baseline. All Monoraphidium/Chlorella mixtures are admitted independently at both observations. Possible observed nitrogen shares form line segments; minimizing their TV gives the smallest departure compatible with any mixture. A feasible dual witness certifies a lower bound on that minimum. An exceedance is declared only when this lower bound exceeds U by more than 10^{-8}.

Twenty preliminary quota records cover four species at days −4, 0, 1, 2 and 3. Day-0 main-vessel composition is compared with exact days 1–3, without quota extrapolation. Changing quotas give $U = 0.190065, 0.174899, 0.154853$; keeping pre-pulse quotas fixed gives $U = 0.154853$ throughout. All 144 planned entries are usable. The primary group comprises twelve rotifer-polyculture vessels; the secondary group has twelve single-rotifer-species vessels. Both contain multiple algal species.

Quotas and cohort	Exceeding comparisons	Vessels exceeding at least once
Measured changes, polycultures (primary)	17/36	8/12
Fixed pre-pulse, polycultures	15/36	8/12
Measured changes, monocultures	9/36	8/12
Fixed pre-pulse, monocultures	6/36	6/12

Table S18. Complete descriptive comparisons on days 1–3. Repeated days in a vessel are dependent; these are not independent statistical tests.

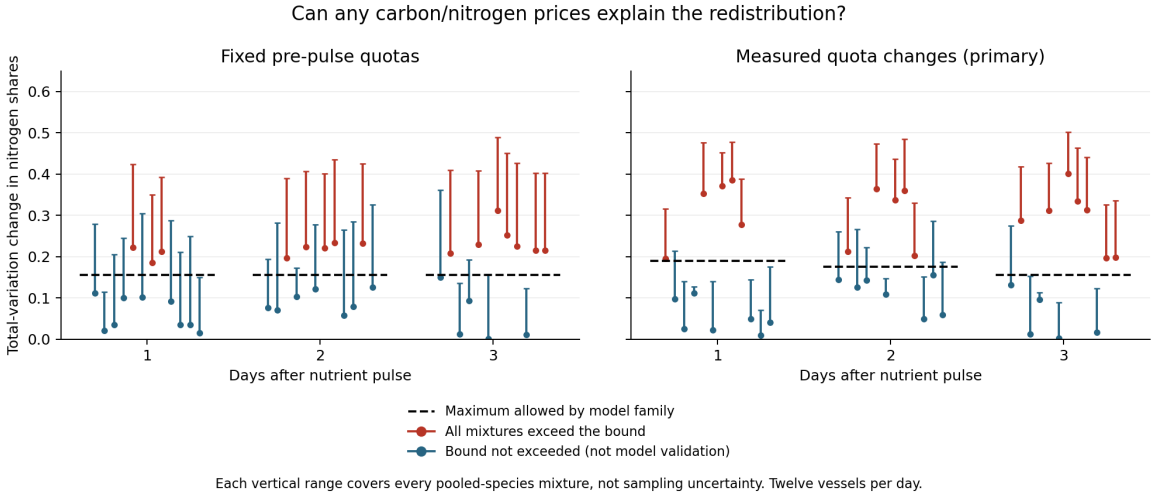


Figure S12. Each vertical range spans all possible observed nitrogen-share changes over pooled-species mixtures, with the minimum marked by a dot. Within each day, vessels run 1–12 from left to right. Dashed segments bound all model price choices. These are mixture envelopes, not confidence intervals. Every red interval lies beyond the corresponding bound.

Price adjustment alone cannot accommodate those eight primary vessels under the transferred quotas and fixed opportunities. Preliminary semi-continuous cultures do not measure the mixed vessels' physiology, however, and assay uncertainty is unavailable. The finding excludes that conjunction at the recorded values, not orthopolity generally, and does not identify which assumption failed. Non-exceedance is a necessary-condition result, not model validation. This finite-group calculation makes no size-spectrum neutrality claim. Its next discriminating study requires main-vessel quotas and a model of changing opportunities or dynamics, followed by an independent prediction.

Ten synthetic tests preceded execution. A separate verifier recomputes all minima by enumerating vertices of the zero-residual line arrangement rather than using the LP solver. It reproduces quota conversions, checks extreme prices, source identities and every verdict. The maximum numerical difference in the minima is 5.6×10^{-17}. Frozen prior results remain intact.

S5.11. Independent phosphorus assays and a failed count transfer

The paired-phosphorus pilot uses Filella and colleagues' same-experiment culture counts/bulk phosphorus (BCO-DMO 1005084) and single-cell X-ray microanalysis (BCO-DMO 1005186). The single-cell quota is independent of the Coulter count it predicts; it is not bulk chemistry divided by that response. The finite measure counts cells in one monoculture, so this is a resource-accounting transfer, not an equal- allocation or log-size-spectrum test.

Of 12 culture/day samples, complete phosphorus assays and unambiguous identity checks leave only High-P/day 1/bottle c (35 assayed cells) and Low-P/day 2/bottle b (33). Ten samples are excluded for missing assays or conflicting/nonunique identifiers; none is relabelled using its values. The first culture calibrates and the second tests. Individual assayed cells are not independent population responses, and the completeness-selected pair cannot establish a treatment effect. Metadata and identifiers were inspected before the local design freeze at `83a8c36`. A missing-volume stop led to a documented amendment at `1569c4f` removing the optional size comparator before numerical response exposure. The two retained cost predictions were unchanged.

Let Q be measured bulk particulate P and q the mean XRMA quota. The fixed- quota forecast is $\widehat{N}_1 = N_0 Q_1/Q_0$; the measured-quota forecast is $\widehat{N}_1 = N_0(Q_1/Q_0)(q_0/q_1)$. These condition on current stock, not feed supply or future resource availability. Q changes from 7.677 to 0.782 micromoles/L, while q changes from 1.061 to 1.389 fg/cell.

Prediction	**Million cells/mL**	**Predicted / observed**	**Absolute log error**
Fixed phosphorus quota	5.552	0.1591	1.8384
Independent XRMA quota	4.241	0.1215	2.1077

Table S19. One independent test culture contained 34.9 million cells/mL. Both predictions miss the declared factor-1.25 accuracy target; the independently assayed quota worsens this prediction. This is a descriptive pilot, not an equivalence test or a population-level rejection.

Raw accounting, $N = 10^6(30.973761998)Q/q$ in the reported units, implies 4.112 times the observed calibration count but 0.500 times the test count. Thus a constant cross-method calibration cannot reconcile the point estimates. Unrepresented particles, preparation losses, cell sampling, sample correspondence or other assay effects remain possible; none is identified by this comparison. Bulk-assay uncertainties are unavailable, and dilution rates also differ between treatments. Independent costs must be compatible with the populations and stocks they explain before allocation can be assessed. A separate Decimal-arithmetic verification reproduces 16 numerical quantities to relative error below 1.7×10^{-16} and checks the unchanged qualification ledger. The protocol, amendment, full source audit and original unsuccessful predictions are retained.

S5.12. Independently assayed field quotas and partial-stock compatibility

Lomas and Martiny's sorted-cell quota archive divides elemental chemistry by direct post-sort counts, distinct from ambient seawater abundance. Its combination with Pacific and Atlantic bulk/count archives therefore avoids defining quota as bulk stock divided by the response. These are published observations: Lomas et al. (2021) already analysed their stoichiometry and modelled community contributions. Our new audit asks whether the independently measured inputs satisfy a necessary stock bound. It is not a new collection or a test of equal allocation. The finite measure counts cells, without imposing equal opportunities on named taxa.

The protocol, inputs, matching audit, code and 13 synthetic checks were frozen at `52d2cc2` before numerical response comparison; public findings and metadata were already exposed. Of 96 quota records, nine have a unique exact cruise/station/depth match, the same reported calendar date and coordinate differences each at most 0.02 degrees. Eight additional records pass only with a one-day date offset. These are nominal sampling matches, not proven common aliquots or time-zone corrections. All other rows remain in the exclusion ledger. Only Prochlorococcus and Synechococcus have usable group definitions across the files.

For available groups, $S_e = 10^{-6} \sum_j N_j q_{je}$ converts ambient counts (cells/mL) and quotas (fmol/cell) to partial standing stock (micromol/L). Missing groups are recorded, not imputed as zero. With accurate representative inputs and complete capture, filtered bulk stock B_e must satisfy $S_e \leq B_e$. More generally, if every included group retains at least elemental fraction r, the necessary condition is $rS_e \leq B_e$. Counts describe whole water while bulk chemistry uses GF/F filtration; complete capture is not established.

Records	**Element**	**Scored**	**Partial/bulk range**	**Above bulk**
Primary, same reported date	C	9	0.0669–0.4397	0
Primary, same reported date	N	9	0.0397–0.2717	0
Additional one-day offsets	C	8	0.1356–1.0554	1
Additional one-day offsets	N	7	0.1099–0.8114	0

Table S20. Carbon/nitrogen compatibility at nine primary sampling records (eight stations), plus eight date-offset records (12 stations combined). Only three primary records contain both cyanobacterial groups; the others contain Synechococcus alone. One additional N record has no usable quota. Repeated elements and depths are not independent inferential replicates.

The sole C/N excess is Pacific station 7, 5 m, with quota and bulk dates differing by one day: Prochlorococcus alone implies 1.80481 micromol C/L against bulk 1.71. Its ratio 1.05544 allows a common minimum elemental retention no greater than $\min(1, B/S) = 0.94747$. The minimum symmetric relative adjustment to the two aggregate quantities is $\max(0, (S - B)/(S + B)) = 0.02697$. Neither quantity is a measured recovery or confidence interval. Other particulate material makes the bound conservative; the comparison identifies no unique cause of the excess.

Phosphorus remains unit-unqualified. Atlantic P was excluded before execution because the metadata use micromol/kg. The frozen Pacific calculation follows the printed micromolar label and records nine ratios below one, but inspection then exposed bulk C:P near 0.1, far below the original study's scale. An nM interpretation would change those ratios by 1,000; it is not an independently confirmed correction. The literal calculation and this post-run qualification are both retained. No P result is counted as physically qualified evidence.

The primary C/N comparisons clear a necessary observation check without establishing full-community closure, assay accuracy or a binding resource. Incomplete taxon coverage makes the check permissive. Independently specified opportunities and constraints, followed by another abundance prediction, remain necessary for a natural-community allocation test. Sorting and shared assay effects, nominal sample identity, missingness and unresolved units cannot be replaced by a target-fitted resource or calibration.

S5.13. Damping-calibrated forecasts of cantilever mode energy

The resource is mean harmonic energy, with potential energy per mode coordinate $k_B T_{\text{fluc}}/2$; the reference measure counts specified mechanical modes. Fontana and Bellon (2023) provide reduced damping and fluctuation measurements for heated cantilevers. Their extended fluctuation–dissipation relation is

$$T_{\text{fluc}} = \int_0^1 T(z) w(z)\, dz, \qquad w(z) = \frac{d(z)\rho(z)}{\int_0^1 d(u)\rho(u)\, du},$$

where ρ is normalized squared mechanical strain and d is local loss. The known physics supplies the prediction; this is not a consequence of neutral allocation alone. It identifies how a thermal gradient and spatial coupling change mode energies before those energies are fitted.

Our protocol was committed at `0428b64` before numerical damping fits; predictions were sealed at `6733362` before scoring numerical fluctuation responses. Published qualitative results were already known, including problematic modes. No publisher theoretical curve was used as input or validation oracle. The reduced archive inherits equilibrium normalization under equipartition and common spectral fits, so this is a retrospective transfer conditional on calibration, not statistically independent measurements or a new equilibrium test.

For a linear temperature profile $T = T_0 + \Delta z$ and local loss $d = A + B\Delta z + C\Delta^2 z^2$, define $\mu_r = \int_0^1 z^r \rho(z)\, dz$. Global damping has the quadratic form $\Phi = a + b\Delta + c\Delta^2$, with $A = a$, $B = b/\mu_1$, $C = c/\mu_2$. Thus

$$T_{\text{pred}} = T_0 + \Delta \frac{a\mu_1 + b\Delta\mu_2/\mu_1 + c\Delta^2\mu_3/\mu_2}{a + b\Delta + c\Delta^2}.$$

Use clamped-free beam curvature for flexure and $\rho(z) = 2\cos^2((m - 1/2)\pi z)$ for torsion. The denominator is fitted damping, never observed heldout damping. Exact quadratic extrema must give nonnegative local loss throughout the complete temperature domain, not merely positive global damping. Undefined or physically invalid predictions remain failures.

The lower eight of eleven C30 conditions and five of seven C30C conditions calibrate inverse-variance weighted quadratic fits. All eleven published mode cases remain: C30 flexural 1–5 plus torsional 1; C30C flexural 1–4 plus torsional 1. The remaining three/two conditions yield 28 targets. The source identifies C30C as the coated C30 sample; these are two preparations, not two independent specimens. Missing raw ramp identifiers and covariance prevent independent-replicate significance claims.

A predictor-only audit found the article's 295 K base conflicts with the archive's 300 K plotting and temperature-column convention. Primary uses $T_0 = 300$ K, $\Delta = 2(T_{\text{avg}} - T_0)$; a fixed 295 K sensitivity leaves observations unchanged. It is not a recalibration of their unknown raw normalization. The publisher-mask sensitivity removes the first C30C flexural-4 calibration loss, matching an unexplained plotting-script mask. No sensitivity replaces the primary result.

Frozen analysis	**Admissible modes**	**Scored targets**	**Thermal MAE**	**Uniform-loss MAE**
Primary, 300 K	10/11	26/28	74.00 K	78.94 K
295 K baseline sensitivity	10/11	26/28	74.47 K	79.77 K
Publisher calibration-row mask	11/11	28/28	67.85 K	75.92 K
All damping rows, secondary	11/11	28/28	70.61 K	75.92 K

Table S21. Cantilever thermal forecasts on the same targets within each analysis. The primary score is conditional on physically admissible modes.

Errors average conditions within mode, modes within preparation, then the two preparations equally. Comparators share each analysis's exact scored rows. Primary spatial-average and cold-clamp errors are 78.77 K and 163.39 K. The primary damping mean absolute relative error, over all eleven fits, is 11.78%. The all-damping secondary has no heldout damping claim. No nominal p-values or full prediction intervals are justified by the reduced archive.

C30 flexural modes 4 and 5 forecast damping within 1.76% and 2.86%, with thermal MAEs 11.39 K and 6.51 K. C30 flexural 3 instead has 34.98% damping error and 255.57 K thermal error. C30C flexural 3 predicts damping within 3.26% yet misses thermal motion by 180.85 K; C30C torsional 1 gives 2.56% and 66.11 K. Global damping agreement therefore cannot by itself justify the assumed spatial weighting. C30C flexural 4 requires negative local loss in the primary, despite positive global damping; the masked sensitivity repairs that case only.

A post-response source check qualifies the C30C flexural-3 discrepancy. Its final archived temperature is 161.12 ± 81.79 K, against a 466.39 K forecast. The plotting script correctly maps this column to mode 3, but sets the panel's range to 250–550 K; the entire displayed error bar falls below it. No response mask or justified correction was found. Retaining this point is necessary for an archive audit, but its origin needs raw data or author clarification before claiming a new physical violation. Neither it nor the other finite observation below the model's base was excluded from scoring.

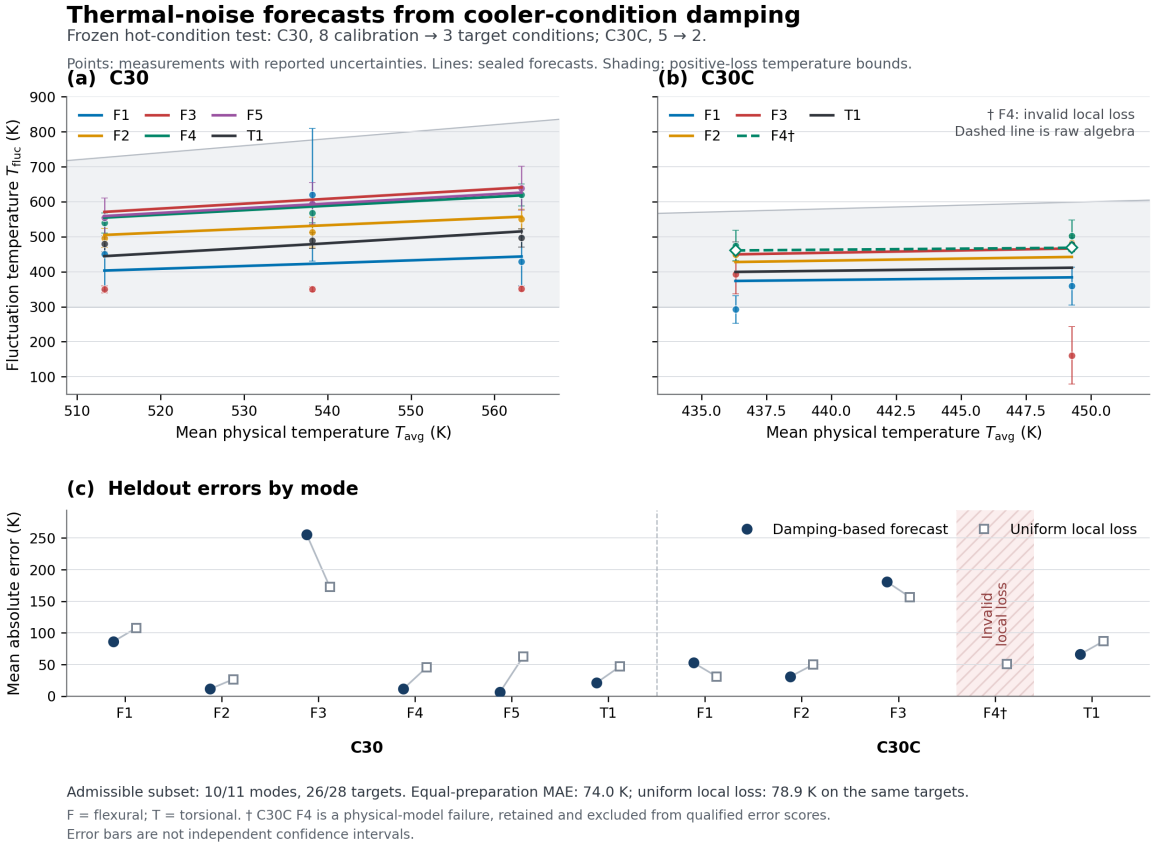


Figure S13. Primary forecasts from colder-condition damping. Points show archived fluctuation temperatures and reported errors; lines show sealed forecasts. Shading marks the allowed temperature envelope for nonnegative local loss. The dashed C30C flexural-4 curve is retained algebra with invalid local loss. Its failed case remains visible in the error comparison. Shared errors are not independent confidence intervals.

The full report includes all mode errors, three comparators, four frozen analyses and the raw plotting discrepancy. Independent QR fitting and adaptive spatial integration verify all results, physical gates and source/forecast hashes. This study adds an executable natural-physics transfer and failure diagnosis. Its modest conditional gain is not evidence of uniform predictive success or universal equal allocation.

S5.14. Pure-to-mixture chemical competition

The public RM8850 zeolite archive (Wedler et al., 2024; Zenodo 17604326) provides pure CO2/CH4 calibration and separate binary equilibrium measurements. The source-qualified intersection at 298, 333 and 353 K supplies 72 pure observations and 45 mixture conditions, or 90 component targets. Absolute uptake, equilibrium gas composition and nominal temperature are used. Every eligible pressure/composition row is retained. The design was committed at `08efaa1` before pure fitting, and predictions at `158a22c` before numerical mixture evaluation. Published qualitative results were known; this is a retrospective transfer audit, not a blind experiment.

The physical approximation is a finite capacity of interchangeable one-molecule sites, with counting measure over species and vacancies. The known competitive Langmuir model (Langmuir, 1918) has

$$q_i = Q\frac{b_i f_i}{1+\sum_j b_j f_j}, \qquad q_0 = \frac{Q}{1+\sum_j b_j f_j}.$$

Here Q is an effective capacity fitted separately at each temperature, b_i is a pure-calibrated affinity, and f_i is reservoir fugacity. Vacancies and occupied sites sum to Q; equal occupied shares require equal $b_i f_i$. This states the resource and the independently calibrated constraint factor. It does not derive allocation from conservation or relabel affinity as a physical additive resource. Similar molecular covolumes motivate testing common capacity, but the source establishes cage volume rather than identical binding sites.

Primary fugacity uses Peng–Robinson (1976), independent critical constants and fixed zero binary interaction. Pure calibration uses pure fugacity; mixtures use mixture fugacity coefficients. The ideal-gas sensitivity recalibrates and predicts throughout using pressure. Fits use positive logarithmic parameters, fixed multistart weighted least squares and U/2 weights. All three primary common-capacity fits pass the declared local identifiability gate and pure RMSE $\leq 5\%$ of the relevant maximum pure loading. The gate measures numerical sensitivity; unknown covariance prevents a confidence-interval interpretation.

The no-competition ablation applies the same fitted pure curves independently at component fugacity. IAST (Myers and Prausnitz, 1965) instead uses separately fitted pure capacities, one additional pure-fit parameter per temperature. For Langmuir curves, spreading pressure is $\Pi_i = Q_i \ln(1 + b_i f_i^0)$. Equal spreading pressure and $\sum_i f_i/f_i^0 = 1$ determine adsorbed fractions; reciprocal total uptake is $\sum_i x_i/q_i^0$. Equal-capacity IAST coincides with competitive Langmuir and supplies a numerical check, not an additional independent hypothesis. No mixture target calibrates any of these models.

Model	Balanced component MAE	Total-loading MAE	Adsorbed CO2 fraction MAE	Within reported U
Shared capacity	0.24746	0.25543	0.04975	39/90
No competition	0.61997	1.20808	0.09402	20/90
Separate-capacity IAST	0.24280	0.22859	0.04983	38/90

Table S22. All primary mixture targets. Loading errors are mmol/g; composition error is dimensionless. Component MAE weights the six temperature-by-gas cells equally. The other means weight temperatures equally. Reported U does not include the full prediction uncertainty or covariance.

Shared component MAEs for CO2/CH4 are 0.465/0.238, 0.301/0.163 and 0.181/0.137 mmol/g at the three temperatures. These are 2.56–5.69% of the corresponding pure maxima. All meet the

frozen 10% engineering criterion; the 60.1% improvement over omitting competition exceeds its required 20%. This is not a 10%-relative-error guarantee for individual mixture points. Only 39/90 predictions fall inside reported measurement error bars. Positive CO2 and negative CH4 biases partially cancel in total loading, reinforcing the need for the declared component score. The pure residuals also exceed reported errors despite meeting the engineering tolerance.

Common effective capacities are 7.891, 7.417 and 7.129 mmol/g; separate CH4 capacities are 5.969, 6.208 and 6.338. Their variation precludes claiming a single temperature-invariant site count. The frozen IAST flags retain 29 upper standard-state extrapolations, reaching 115.7 times measured pure fugacity coverage. A post-response audit of sealed predictors additionally finds 12 mixture CO2 conditions below pure calibration coverage and 11 lower IAST standard-state extrapolations. At 298 K the first pure CO2 measurement already corresponds to about 0.56 fitted occupancy. The dilute response is therefore partly extrapolated. No affected row is removed and no forecast is changed by this coverage diagnosis.

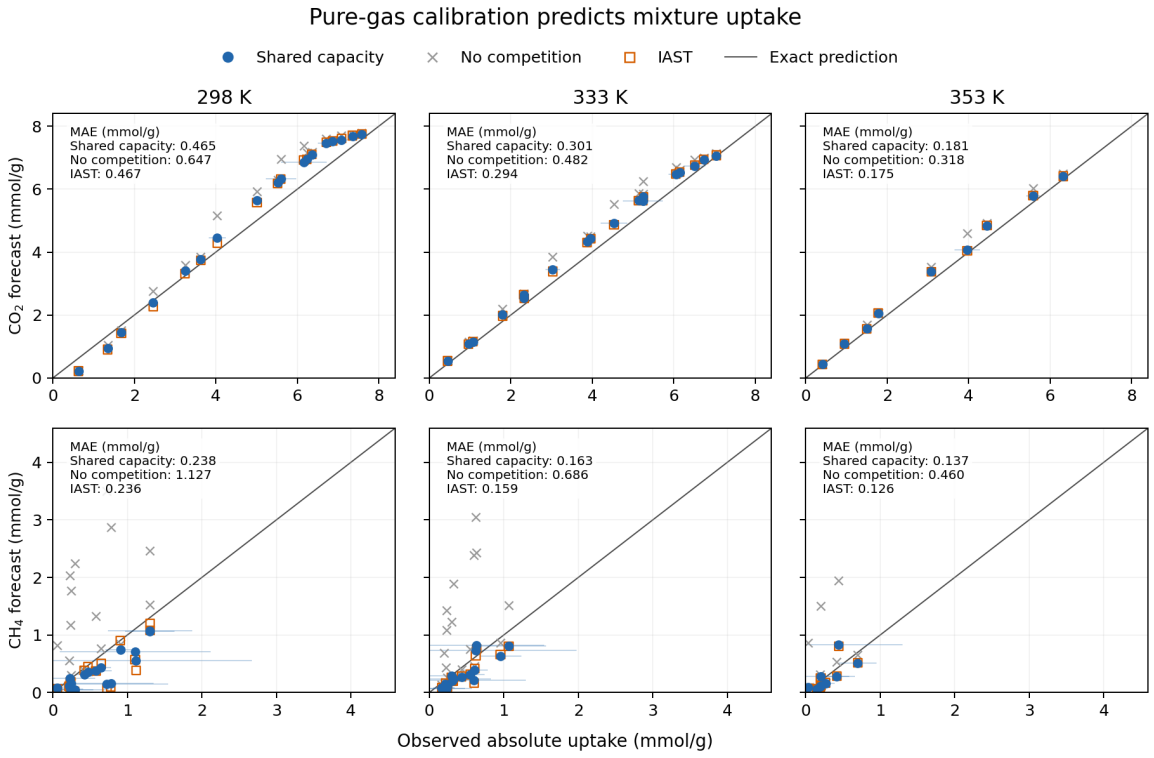


Figure S14. All primary component forecasts, with the same observations for all models. Blue circles: shared capacity; orange squares: IAST; gray crosses: no competition. Horizontal bars show reported expanded observation uncertainty, once per observation at the shared forecast. The diagonal is exact prediction. Gas composition and uptake errors are related, so these are not independent uncertainty intervals.

Ideal-gas sensitivity gives shared, ablation and IAST component MAEs of 0.26401, 0.63130 and 0.25912 mmol/g, respectively, also meeting the pragmatic transfer rule. Independent Gibbs differentiation, variable-projection pure fitting and an adsorbed-composition IAST solver verify 3,294 entries. The full report preserves all source conventions, fit diagnostics, per-sweep errors, coverage limitations and reproduction steps. The gain establishes a useful chemistry instance of constrained allocation at finite accuracy. Established IAST performs similarly and slightly better on loading error; this comparison demonstrates no additional predictive advantage over adsorption theory and does not establish universal neutrality.

S5.15. Grain properties and a narrow-to-wide channel forecast

The new natural-system comparison uses Deal et al. (2026) and its public original archive, including the older Deal et al. (2023) experiments. Of the published narrow sphere/gravel cohort, three rows lack water discharge. Predictor-only qualification leaves 19 spheres for primary calibration and ten gravel runs for a pooled sensitivity and material-specific benchmark. All four new sphere and five new gravel runs are retained. The width changes from 0.0105 to 0.065 m. Both original measurement programs confirm the latter value; README descriptions of a 5-cm channel are stale.

The protocol and implementation were committed at `290acd9` before old-response fitting, and all forecasts at `e86cffc` before numerical target evaluation. Published qualitative outcomes and fitted wide-channel constants were already known, but those constants were not used. A metadata inspection accidentally exposed one numerical target; the other eight remained withheld. This is retrospective local freezing, not blind confirmation. All 30 retained source files have checked local digests; 29 also match publisher checksums, excluding the checksum file itself.

For mass flux $\dot{m}$, the fixed constitutive prediction is

$$q^* = \frac{\dot{m}}{\rho_s W \sqrt{g r_f D^3}} = A \left[F_{\text{shape}} F_{\text{wall}} \frac{HS}{r_f D} - t \right]_+^{3/2}, \qquad r_f = \rho_s/\rho_f - 1.$$

Water discharge, width, depth, water-surface slope and grain size determine the Guo wall correction. Separate grain assays supply settling speed, Corey shape factor and repose angle; archived theoretical equivalent-sphere settling speed and per-run measured bed slope complete the prescribed shape correction. The actual source pipeline uses volume-equivalent diameter D rather than an independently measured D50. That convention is retained, and measured mass flux is normalized consistently instead of importing precomputed flux or flux-inferred stress. All included Guo factors are physically admissible; all primary target stresses fall within calibration support.

Each candidate independently fits nonnegative amplitude A and threshold t on the same permitted old observations; exponent 3/2 is fixed. The reference measure gives equal weight to each grain cohort and then to runs within that cohort. It is an observation measure, **neither log-orthopolic nor linear-orthopolic size allocation**. No grain-size abundance distribution is tested, and boundary momentum conservation alone does not determine transport.

Prediction	Primary balanced MAE	Pooled-calibration MAE
Guo + measured shape	0.04751	0.03478
Guo + unit shape	0.08283	0.04326
Hydraulic radius + measured shape	0.20859	0.18475
No wall correction + measured shape	0.06888	0.06888

Errors are in dimensionless flux. The primary gain is 42.64% versus unit shape and 77.22% versus hydraulic radius, satisfying separate frozen 20% improvement rules. Shape slightly worsens sphere error and improves gravel error. A benchmark with separate Guo/unit-shape fits to old sphere and gravel data gives MAE 0.04297, 9.56% better than the primary full model, with four fitted parameters instead of two. The pooled shape gain is 19.61%; calibration choice therefore matters. The no-wall model extrapolates and predicts zero throughout.

The accuracy rule requires calibration RMSE and each target-cohort MAE to be at most 20% of the largest old sphere flux, with at least three active calibration points. The primary fit has

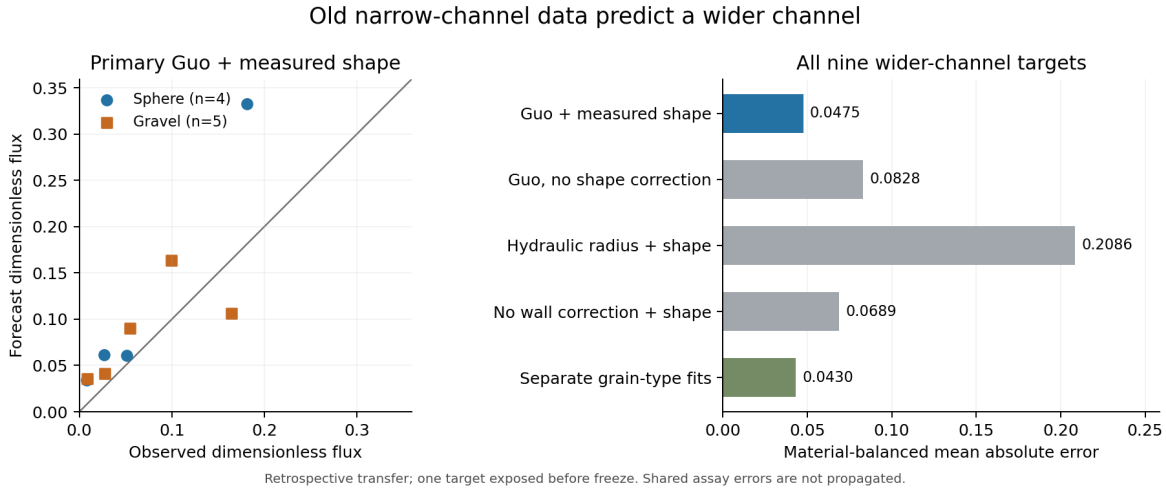


Figure S15. All nine primary forecasts against observations, and balanced errors for models fitted without the wider-channel responses. The diagonal denotes exact prediction. The material-specific benchmark has additional calibration parameters. Shared input uncertainties are not propagated.

19 active points and passes. However, the resulting tolerance, **0.23155**, exceeds the largest wide observation, **0.18134**; even zero forecasts satisfy the target-error portion of that rule. The nominal accuracy pass is consequently weak. Primary MAE is 83.11% of the wide sphere mean and 55.69% of the gravel mean. All sphere runs are overpredicted; the lowest sphere and gravel fluxes are overpredicted by factors 4.43 and 4.09. These descriptive diagnostics qualify the frozen pass rather than replacing it.

Independent real-algebra hydraulics, force-balance drag coefficients, multi-start two-parameter fitting and separate scoring verify 2,467 quantities with maximum relative difference 1.93e-8. Forecasts, results, verification and figure replay byte-for-byte. Grain-batch and pump-calibration errors are shared; there is no population-level significance claim. Existing criticism of shape identification (Chen, Durán and Pähtz, 2026) remains relevant. The full report retains every prediction, source convention, tolerance limitation and benchmark. This is a useful example of a constrained prediction improving under changed geometry, with documented failures. It establishes neither a new transport law nor universal neutrality.

S5.16. Independent protein assays and a failed workload transfer

Borkowski et al. (2018) measured normalized reporter production capacity in cell extracts and living bacteria. We calibrate on twelve initiation-sequence variants expressing mKate, then predict ten different protein/mKate constructs from their separate extract assays. All new constructs share the BCD2 initiation design. The empty insert is excluded in advance. Four Viob constructs form one family, and six other proteins each form one family. Primary errors give equal weight to these seven families, then to constructs within a family. This counting measure tests neither log-orthopolic nor linear-orthopolic allocation across sizes.

If reporter production tracks a free pool and bound demand scales linearly with that pool, extract capacity x defines relative workload $L = (1 - x)/x$. The specified transfer assumes one common environment factor $k \geq 0$:

$$\widehat{y} = \frac{1}{1 + k(1/x - 1)}.$$

This is conditional on a mass-action demand model; conservation alone does not imply it. Both workload and k are inferred quantities, not independent absolute ribosome or energy

measurements. Equal-weight least squares on the twelve calibration means gives $k = 0.78346$ and RMSE 0.05543. Each alternative has its own calibration: linear burden $y = 1 - k(1 - x)$, affine $y = a + bx$, a constant equal to mean calibration capacity, and direct $y = x$ without parameters. Direct transfer is the reciprocal model at $k = 1$, so this comparison tests whether the fitted environment correction transfers. No forecast is clipped or refitted on targets.

The design and synthetic checks were committed at `b1dcb08` before calibration; forecasts and their verification at `bef1f23` preceded all ten numerical target responses. The source's qualitative outcomes and correlations were known. This is retrospective archival validation. Original publisher workbook and supplement copies match Europe PMC byte-for-byte. Exact cells, source hashes and exposure chronology are in the protocol and source audit.

Fixed model	**Family-balanced MAE**	**Construct MAE**	**Maximum error**
Reciprocal resource pool	0.10184	0.09322	0.18446
Direct assay	0.09464	0.08867	0.19494
Linear burden	0.10950	0.09870	0.20908
Affine transfer	0.10603	0.09635	0.20017
Constant capacity	0.20210	0.16837	0.44332

Table S23. Errors on all ten new protein constructs. Primary weighting treats four related Viob constructs as one family. The frozen absolute accuracy limit is 0.10 capacity units; comparative advantage requires 20% lower error against each alternative.

The reciprocal model narrowly misses absolute accuracy and improves only 3.95% over affine and 7.00% over linear transfer. Its 49.61% improvement over constant capacity shows useful assay information, while direct assay transfer predicts better by the primary error comparison. The reciprocal model is 7.60% worse than direct transfer. Ordinary construct weighting brings its error below 0.10 but does not improve on direct transfer; it cannot replace the frozen family measure. The joint rule fails. Its signed bias is +0.03589, and the largest miss is an overprediction of atf1 by 0.18446 capacity units.

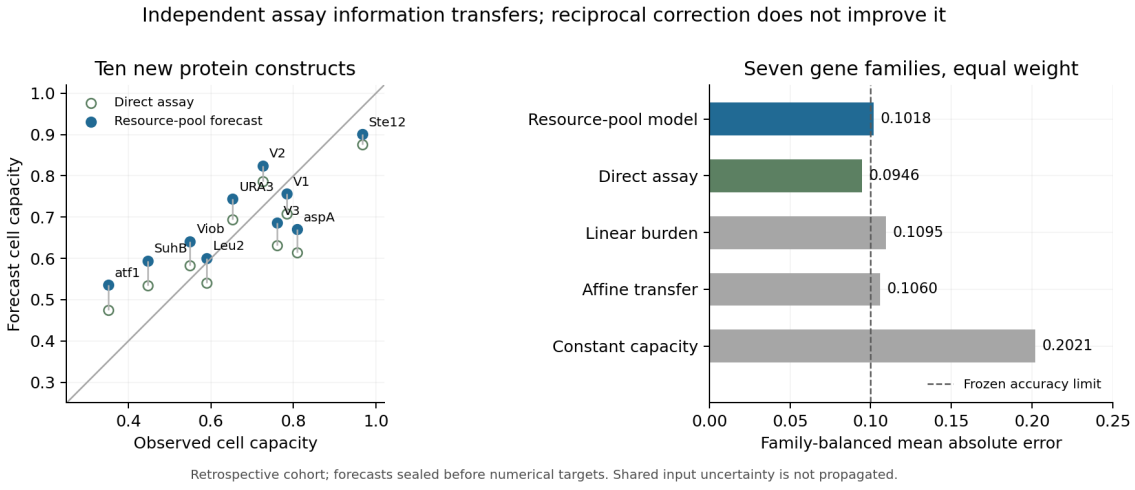


Figure S16. All ten forecasts from the reciprocal model and direct extract assays, compared with observed cell capacity. Connecting segments show the effect of the fitted correction. The right panel compares family-balanced errors; the dashed line is the frozen accuracy limit. Shared input errors are not propagated.

Five target assays extrapolate beyond calibration support, which spans 0.59855–0.84955; target assays span 0.47471–0.87667. All were flagged before evaluation and retained. Reported target

means match triplicate means exactly. Reported spreads match sample SD, although article captions say standard error. Three normalized measurements share controls and batches; no population confidence interval is inferred. Extract capacity uses a maximum production rate, while cell capacity uses a mid-exponential rate per optical density. The source's 29/37-degree incubation descriptions remain unresolved.

Independent workbook extraction, 41-start least squares, centered affine regression and explicit family averaging check 195 quantities before outcomes and 334 after evaluation, with maximum absolute difference 6.25e-9. Forecasts, results, verification and figure reproduce byte-for-byte. The complete report preserves every error and source-summary check. The result limits the specified fitted workload transfer: the independent assay is useful, but its fitted environment correction adds no demonstrated predictive value. It does not reject the reciprocal functional family or establish an absolute resource cost or neutral biological allocation.

S5.17. Independent information costs predict speech across families

Coupé et al. (2019) estimated syllable entropy from written corpora and measured recorded speech duration separately. Their ID is conditional entropy given the preceding within-word syllable or a word-onset null; ShE is unigram entropy. Neither uses speech duration. Canonical syllable count NS describes each translated text, rather than each speaker's realized articulation. Recorded duration D excludes manually identified pauses longer than 150 ms; it is neither elapsed reading time nor pure phonation time.

The locally frozen hypothesis is a common information capacity:

$$\widehat{D} = \frac{NS\,ID}{C}.$$

Counting measure gives equal source-defined families, languages within family, speakers within language, then available texts. This is neither log-orthopolic nor linear-orthopolic size allocation. Equal capacity is tested, not imposed independently; C is inferred, and these measurements do not identify metabolic cost. The source's constant-information-rate account already proposes this relation, so the comparison is a transfer test, not a distinct mechanism for orthopolity.

Seven Indo-European languages provide 789 calibration readings from 70 speakers. Ten languages in eight other families provide 1,499 target readings from 100 speakers. The publisher and repository files match byte-for-byte at 2,288 rows, although the article states 2,265. No corresponding exclusion was identified, so every row is retained. The source has 15 shared translated text identities, but 261 calibration and one target speaker/text cells are absent. Declared weighting handles their contribution to the estimand, not possible selection bias.

Design `eeab4d3` preceded numerical duration calibration; forecasts `26b78d8` preceded all target-duration decoding. Published aggregate rates, correlations and the equal-information-rate conclusion were known. Fits minimize weighted squared log-duration error. The information model gives $C = 40.3967$ bits/s. All alternatives use calibration only: constant syllabic rate; constant unigram-information rate; an unconstrained fitted power of ID; and geometric duration for the corresponding translated text. The power exponent is 0.44019, estimated from only seven language-level predictor values. There is no target refitting, clipping or post hoc language exclusion.

Let e_l be the mean of $\ln \widehat{D} - \ln D$ within language l, first averaging each speaker's readings, then speakers. The primary error E averages $|e_l|$ across languages and families. Thus opposite individual errors can cancel. The secondary reading error instead averages the absolute log residual before grouping.

Model	Primary language log MAE	Reading log MAE	Largest language discrepancy factor
Conditional information	**0.07411**	**0.11005**	1.20144
Constant syllabic rate	0.15440	0.17611	1.46453
Unigram information	0.14489	0.16766	1.38937
Fitted density power	0.09438	0.13295	1.34228
Same-text duration	0.10205	0.14968	1.20805

Table S24. All frozen forecasts on ten target languages. Primary and reading errors use hierarchical counting measure. The final column is $\exp(\max_l |e_l|)$*, which is distinct from the average-accuracy criterion.*

The conditional model reduces primary error by 52.00% against constant syllabic rate, 48.85% against unigram information, 21.47% against fitted power and 27.38% against same-text duration. All frozen requirements pass: E is below ln(1.20), the syllabic gain exceeds 20%, and error is no more than 1.10 times the best unigram, fitted-power or same-text benchmark. This is a tolerance, not a formal noninferiority test; here the model actually has lower descriptive error than every comparator. The separate 10% contextual-information gain criterion also passes. The same-text comparison shows that the gain is not fully reproduced by transferring the shared message's average duration.

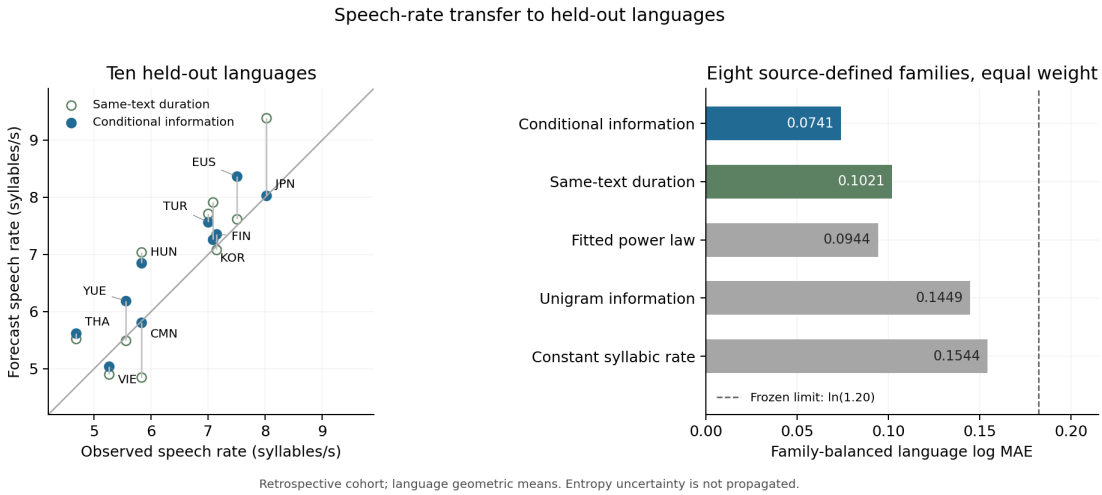


Figure S17. Observed and forecast geometric speech rates for every target language, and the five primary error scores. The reference line denotes exact prediction. Error bars are omitted because entropy/input uncertainty is not propagated; the right-panel dashed line is the frozen average-accuracy limit.

The average discrepancy factor $\exp(E)$ is 1.07693. It does not bound every language or reading: Thai's forecast/observed duration ratio is 0.83233, a factor of 1.20144 in the opposite direction. Hungarian is 0.85096, Basque 0.89663 and Cantonese 0.89840; these departures remain. The complete report retains every language ratio. Basque, Japanese, Thai and Vietnamese lie outside calibration ID support [5.29,7.09]; they were flagged before outcomes and retained. Shared texts, related languages, rehearsed reading, selected speakers and rounded entropy estimates limit generalization. No population confidence interval is inferred.

Eight synthetic and access-control tests precede calibration. Independent tab-field decoding and 50-digit Decimal arithmetic verify 17,658 quantities before outcomes and 25,472 afterward, with maximum difference 1.45×10^{-14}. The result demonstrates useful cross-family prediction from independent contextual-information inputs in this cohort. It establishes neither a new language law nor universal neutral allocation; it supports a capacity relation whose independent entropy input predicts better than the declared alternatives and whose departures remain visible.

S5.18. Membrane exchange predicts a later cohort

Benstead et al. (2026) provide raw absorbance in both chambers of membrane diffusion cells and separate optical standards. We retained the original public ZIP and fixed four automated acid-orange/PEG25-on-PES pairs from 5 May 2025 as calibration (96 chamber observations), with four manual pairs from 3 June as targets. Their eight baseline readings are inputs; all 56 later readings were numerically withheld until predictions were verified and committed. Published aggregate diffusivities and qualitative method comparisons were known. The protocol was frozen at 55b3970 before numerical standards or selected measurements, the remaining implementation at 8c7f60a, and forecasts at 68694fd before target release. This is a local retrospective freeze, not external preregistration or global blinding. The reference measure is compartment volume, neither log-orthopolic nor linear-orthopolic size allocation.

Each chamber holds 11 mL; sampling and water replacement remove 0.02 mL per collection. With $f = 549/550$ and j previous collections, raw contrasts obey $d_j = \Delta f^j \exp(-2gt_j/11)$ under constant effective conductance g. Each run's five independent nominal standards fit the full affine optical model $A = aC + b$; source-derived concentrations, fitted diffusivities and post-experiment thickness are unused. Source thickness units conflict, and conductance therefore transfers without a thickness-based permeability claim. Unknown contrast amplitude absorbs the optical slope; subtraction cancels the intercept. Raw contrast fits require no concentration conversion.

All optical and baseline qualification gates pass. The four fitted g values are 0.18052, 0.16103, 0.15908 and 0.13129 mL/h. Their whole 99% confidence sets, obtained by inverting the absolute residual-sum-of-squares $F_{11,3}$ test, are respectively [0.09848,0.29711], [0.08697,0.26136], [0.08572,0.25822] and [0.06772,0.21193]. Each excludes persistence and instantaneous mixing. The separate 44-residual calibration stock test gives $F = 0.26688$, p=0.98244. It retains shared baseline and optical-intercept covariance. These statements assume independent Gaussian absorbance errors with the same variance for standards and samples within each run. Five standards do not establish that assumption, temporal stability, matrix equivalence or accurate nominal dilution. Only three error degrees of freedom produce broad uncertainty.

For measured target baseline total T_0 and contrast D_0, the fixed forecast is

$$\widehat{C}_{D,A}(t_j) = \frac{f^j}{2}[T_0 \pm K(t_j)D_0], \qquad K(t) = \frac{1}{4}\sum_{k=1}^{4} \exp(-2\hat{g}_k t/11).$$

The primary error is mean absolute contrast error divided by each pair's initial total, equally averaged over seven later times and four pairs. Persistence uses $K = 1$; instant mixing uses $K = 0$ at positive times. The empirical alternative interpolates the mean earlier sampling-adjusted contrast and holds its last value beyond 66 h. It is not clipped or refitted to targets.

Forecast	**Primary error**	**Exchange improvement relative to alternative**
Exchange kinetics	0.042285	–
Persistence	0.547178	92.27%
Instant mixing	0.462221	90.85%
Empirical trajectory	0.041383	-2.18%

Exchange is better than both limiting models for all four pairs, and better than empirical interpolation for only one. Its superiority over every benchmark fails; no population equivalence or significance is inferred from four membranes. Pairwise exchange errors are 0.049057, 0.064851, 0.044069 and 0.011163. The largest individual contrast error is 0.090677. Average joint chamber absolute error is 0.023254 of initial total; average total-stock absolute error is 0.021982, shared by all models' fixed total forecasts.

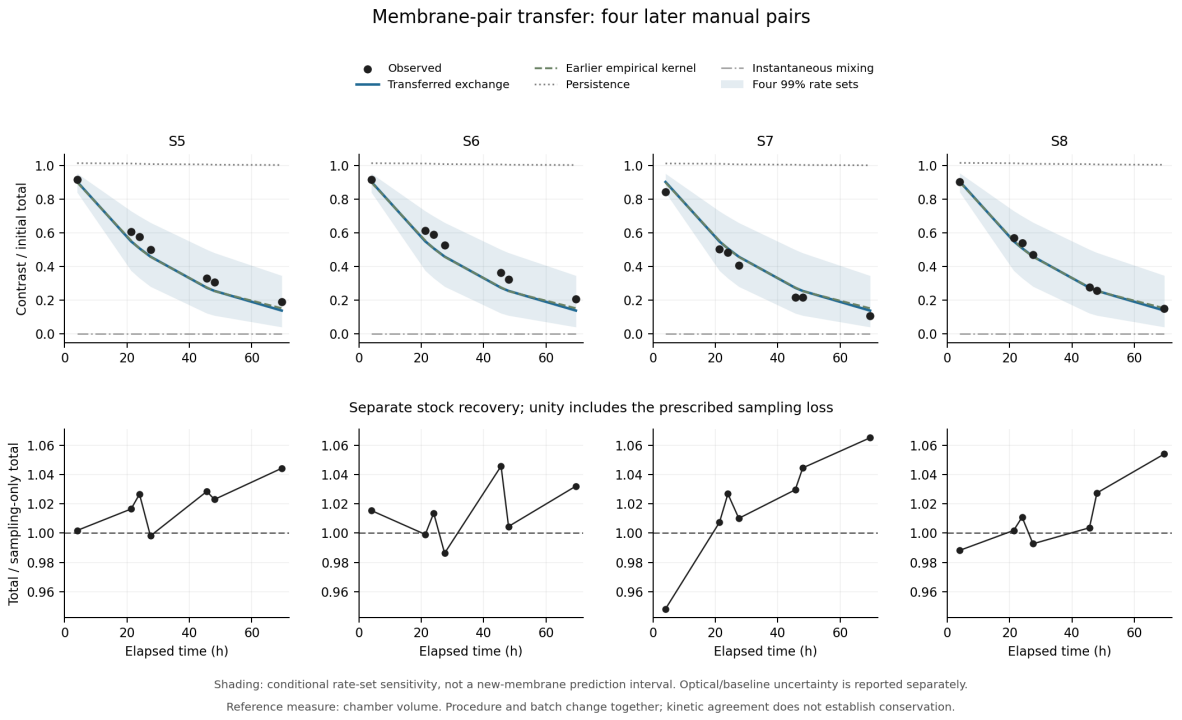


Figure S18. Later-pair normalized contrasts, the four fixed forecasts and the conditional envelope over earlier 99% rate sets; lower panels show measured total divided by sampling-adjusted initial total. The envelope is not a prediction interval for a new membrane. Optical/baseline covariance is retained separately in the forecast artifact.

The independent target stock test gives $F_{28,3} = 2.03717$, p=0.30855. Observed recovery ratios range from 0.94825 to 1.06511. Non-rejection under the declared observation model does not prove conservation. Kinetic comparison does not decide this question: all four models share the same total forecast. Target recipient baseline estimates are slightly negative (-6.16 to -7.66 micromolar) and remain unclipped. All eight target baselines extrapolate the 0–80 micromolar aliquot calibration support slightly, while all 56 later measurements lie within it. The prediction artifact retains the full 56 by 56 shared optical/baseline covariance; conditional standard deviations span 8.09–11.78 micromolar. A declared elapsed-time shift of ten minutes changes point forecasts by at most 1.803 micromolar. These sensitivities exclude unmeasured stirring, actual sequential sampling times and temporary volumes. Method, date and membrane batch change together, so equivalence of automated and manual collection is not identified.

Independent polynomial root isolation and observation-error covariance maps verify qualification. Event-based ODE integration, complex-step uncertainty derivatives and separately assembled target stock covariance verify transfer: 5,184 quantities before release and 7,496 afterward, maximum absolute difference 4.37×10^{-11}. The full report retains raw values, all benchmarks and reproduction commands. The result shows useful transfer of finite exchange kinetics, while failing to improve on the empirical trajectory alternative. Ordinary diffusion gives the same model; this does not distinguish a unique orthopolity mechanism or test the proposed pair-to-network topology intervention.

References

Arranz, I., Fournier, B., Lester, N. P., Shuter, B. J., and Peres-Neto, P. R. (2022). Species compositions mediate biomass conservation: The case of lake fish communities. *Ecology*, 103, e3608. https://doi.org/10.1002/ecy.3608.

Benstead, C., Politi, M., Bergsman, D., and Pozzo, L. (2026). An automated sampling workflow for parallel long-term membrane diffusion cell testing. *Digital Discovery*. https://doi.org/10.1039/D5DD00523J.

Bishop, C. J., and Peres, Y. (2017). *Fractals in Probability and Analysis*. Cambridge University Press. Section 1.1. https://www.math.stonybrook.edu/~bishop/fractalbook.pdf.

Borkowski, O., Bricio, C., Murgiano, M., Rothschild-Mancinelli, B., Stan, G.-B., and Ellis, T. (2018). Cell-free prediction of protein expression costs for growing cells. *Nature Communications*, 9, 1457. https://doi.org/10.1038/s41467-018-03970-x.

Boyd, S., and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. Sections 4.7 and 5.6. https://web.stanford.edu/~boyd/cvxbook/.

CMU Sphinx contributors (accessed 7 October 2026). *CMU Pronouncing Dictionary*. Pinned source.

Chen, Y., Durán, O., and Pähtz, T. (2026). On the testing of grain shape corrections to bedload transport equations with grain-resolved numerical simulations. *Earth Surface Dynamics*, 14, 75–83. https://doi.org/10.5194/esurf-14-75-2026.

Clauset, A., Shalizi, C. R., and Newman, M. E. J. (2009). Power-law distributions in empirical data. *SIAM Review*, 51, 661–703. https://doi.org/10.1137/070710111.

Condon, J. J., and Ransom, S. M. (2016). *Essential Radio Astronomy*. Princeton University Press. Chapter 2, radiation and absorption. https://www.cv.nrao.edu/~sransom/web/Ch2.html.

Coupé, C., Oh, Y. M., Dediu, D., and Pellegrino, F. (2019). Different languages, similar encoding efficiency: Comparable information rates across the human communicative niche. *Science Advances*, 5, eaaw2594. https://doi.org/10.1126/sciadv.aaw2594.

Cuesta, J. A., Delius, G. W., and Law, R. (2018). Sheldon spectrum and the plankton paradox: two sides of the same coin—a trait-based plankton size-spectrum model. *Journal of Mathematical Biology*, 76, 67–96. https://arxiv.org/abs/1607.04158.

Damuth, J. (1981). Population density and body size in mammals. *Nature*, 290, 699–700. https://doi.org/10.1038/290699a0.

Deal, E., et al. (2023). Grain shape effects in bed load sediment transport. *Nature*, 613, 298–302. https://doi.org/10.1038/s41586-022-05564-6.

Deal, E., et al. (2026). Generalizing sediment transport experiments from narrow flumes to wider channels. *Journal of Geophysical Research: Earth Surface*, 131. https://doi.org/10.1029/2025JF008712. Data: https://doi.org/10.20383/103.01325.

Dillon, K. T., et al. (2019). On the relationships between size and abundance in plants: beyond forest communities. *Ecosphere*, 10, e02856. https://doi.org/10.1002/ecs2.2856.

Edwards, A. M., Robinson, J. P. W., Plank, M. J., Baum, J. K., and Blanchard, J. L. (2017). Testing and recommending methods for fitting size spectra to data. *Methods in Ecology and Evolution*, 8, 57–67. https://doi.org/10.1111/2041-210X.12641.

Edwards, A. M., Robinson, J. P. W., Blanchard, J. L., Baum, J. K., and Plank, M. J. (2020). Accounting for the bin structure of data removes bias when fitting size spectra. *Marine Ecology Progress Series*, 636, 19–33. https://doi.org/10.3354/meps13230.

Eeckhout, J. (2004). Gibrat's law for (all) cities. *American Economic Review*, 94(5), 1429–1451. https://doi.org/10.1257/0002828043052303.

Ersoy, Z., et al. (2025). GLOSSAQUA: A global dataset of size spectra across aquatic ecosystems. *Ecology*, 106, e70050. https://doi.org/10.1002/ecy.70050.

Fabbri, R., and Oliveira Jr., O. N. (2017). *A simple model that explains why inequality is ubiquitous*. Supplied manuscript dated 17 March 2017, 13 pp.; publication status unverified.

Fabbri, R. (2024). The Orthopolity cosmological principle and the Natural distribution law. Author's essay, 14 August. https://ttm.github.io/2024/08/14/power.html.

Fixsen, D. J., Cheng, E. S., Gales, J. M., Mather, J. C., Shafer, R. A., and Wright, E. L. (1996). The cosmic microwave background spectrum from the full COBE FIRAS data set. *The Astrophysical Journal*, 473, 576. https://doi.org/10.1086/178173. Source article: https://arxiv.org/abs/astro-ph/9605054.

Gabaix, X. (1999). Zipf's law for cities: An explanation. *The Quarterly Journal of Economics*, 114(3), 739–767. https://doi.org/10.1162/003355399556133.

Gaedke, U. (1993). Ecosystem analysis based on biomass size distributions: A case study of a plankton community in a large lake. *Limnology and Oceanography*, 38, 112–127. https://doi.org/10.4319/lo.1993.38.1.0112.

Gangan, M. S., and Athale, C. A. (2017). Threshold effect of growth rate on population variability of *Escherichia coli* cell lengths. *Royal Society Open Science*, 4, 160417. https://doi.org/10.1098/rsos.160417.

Fontana, A., and Bellon, L. (2023). Linking fluctuation and dissipation in spatially extended out-of-equilibrium systems. *Physical Review E*, **107**, 034118. https://doi.org/10.1103/PhysRevE.107.034118. Data: https://doi.org/10.5281/zenodo.7640708.

Genthon, A. (2022). Analytical cell size distribution: lineage-population bias and parameter inference. *Journal of the Royal Society Interface*, 19, 20220405. https://doi.org/10.1098/rsif.2022.0405. Revised exposition: https://arxiv.org/abs/2206.06146v2.

Harcourt, R., Garcia, N. S., and Martiny, A. C. (2024). *Synechococcus* batch culture data (cell quotas and ratios (C, N, P), size, and diameter) from laboratory experiments in 2021 to 2022 with related isolates cultured across a range of temperatures. BCO-DMO dataset 926311, version 1. https://doi.org/10.26008/1912/bco-dmo.926311.1.

Hatton, I. A., Heneghan, R. F., Bar-On, Y. M., and Galbraith, E. D. (2021). The global ocean size spectrum from bacteria to whales. *Science Advances*, 7, eabh3732. https://doi.org/10.1126/sciadv.abh3732.

Habeck, M. (2014). Bayesian approach to inverse statistical mechanics. *Physical Review E*, 89, 052113. https://doi.org/10.1103/PhysRevE.89.052113.

Haus, H. A., and Melcher, J. R. (1989). *Electromagnetic Fields and Energy*. Prentice-Hall. Section 11.2, Poynting's theorem. https://web.mit.edu/6.013_book/www/chapter11/11.2.html.

Ioannides, Y. M., and Skouras, S. (2013). US city size distribution: Robustly Pareto, but only in the tail. *Journal of Urban Economics*, 73(1), 18–29. https://doi.org/10.1016/j.jue.2012.06.005.

Jaynes, E. T. (1957). Information theory and statistical mechanics. *Physical Review*, 106, 620–630. https://doi.org/10.1103/PhysRev.106.620.

Jaynes, E. T. (1968). Prior probabilities. *IEEE Transactions on Systems Science and Cybernetics*, 4(3), 227–241. https://bayes.wustl.edu/etj/articles/prior.pdf.

Keshavarz, A., Wang, Y., and Boyd, S. (2011). Imputing a Convex Objective Function. *Proceedings IEEE Multi-Conference on Systems and Control*, 613–619. https://stanford.edu/~boyd/papers/imputed_objective.html.

Langmuir, I. (1918). The adsorption of gases on plane surfaces of glass, mica and platinum. *Journal of the American Chemical Society*, 40, 1361–1403. https://doi.org/10.1021/ja02242a004.

Levin, D. A., and Peres, Y. (2017). *Markov Chains and Mixing Times*, second edition, with contributions by E. L. Wilmer. American Mathematical Society. https://pages.uoregon.edu/dlevin/MARKOV/mcmt2e.pdf.

Lolli, B., Randazzo, D., Vannucci, G., and Gasperini, P. (2020). The Homogenized Instrumental Seismic Catalog (HORUS) of Italy from 1960 to Present. *Seismological Research Letters*. https://doi.org/10.1785/0220200148.

Lomas, M. W., Baer, S. E., Mouginot, C., Terpis, K. X., Lomas, D. A., Altabet, M. A., and Martiny, A. C. (2021). Varying influence of phytoplankton biodiversity and stoichiometric plasticity on bulk

particulate stoichiometry across ocean basins. *Communications Earth & Environment*, 2, 143. https://doi.org/10.1038/s43247-021-00212-9.

Malerba, M. E., Palacios, M. M., and Marshall, D. J. (2018). Do larger individuals cope with resource fluctuations better? An artificial selection approach. *Proceedings of the Royal Society B*, 285, 20181347. https://doi.org/10.1098/rspb.2018.1347. Data: https://doi.org/10.5061/dryad.4mh47r7.

Marshall, D. J., et al. (2022). Long-term experimental evolution decouples size and production costs in *Escherichia coli*. *Proceedings of the National Academy of Sciences*, 119, e2200713119. https://doi.org/10.1073/pnas.2200713119.

Myers, A. L., and Prausnitz, J. M. (1965). Thermodynamics of mixed-gas adsorption. *AIChE Journal*, 11, 121–127. https://doi.org/10.1002/aic.690110125.

Newman, M. E. J. (2005). Power laws, Pareto distributions and Zipf's law. *Contemporary Physics*, 46, 323–351. https://arxiv.org/abs/cond-mat/0412004.

Piantadosi, S. T., Tily, H., and Gibson, E. (2011). Word lengths are optimized for efficient communication. *Proceedings of the National Academy of Sciences*, 108(9), 3526–3529. https://doi.org/10.1073/pnas.1012551108.

NASA LAMBDA (accessed 6 October 2026). COBE FIRAS CMB monopole spectrum, version 1. Product description and reconstruction provenance. https://lambda.gsfc.nasa.gov/product/cobe/firas_monopole_spect.html.

Planck, M. (1901). Ueber das Gesetz der Energieverteilung im Normalspectrum. *Annalen der Physik*, 309, 553–563. https://doi.org/10.1002/andp.19013090310.

Peng, D.-Y., and Robinson, D. B. (1976). A new two-constant equation of state. *Industrial & Engineering Chemistry Fundamentals*, 15, 59–64. https://doi.org/10.1021/i160057a011.

Perkins, D. M., et al. (2018). Bending the rules: exploitation of allochthonous resources by a top-predator modifies size-abundance scaling in stream food webs. *Ecology Letters*, 21, 1771–1780. https://doi.org/10.1111/ele.13147.

Poynting, J. H. (1884). On the transfer of energy in the electromagnetic field. *Philosophical Transactions of the Royal Society of London*, 175, 343–361. https://doi.org/10.1098/rstl.1884.0016.

Sheldon, R. W., Prakash, A., and Sutcliffe, W. H., Jr. (1972). The size distribution of particles in the ocean. *Limnology and Oceanography*, 17, 327–340. https://doi.org/10.4319/lo.1972.17.3.0327.

Tao, T. (2011). *An Introduction to Measure Theory*. Graduate Studies in Mathematics, 126. American Mathematical Society. Exercise 1.4.34. https://terrytao.wordpress.com/wp-content/uploads/2012/12/gsm-126-tao5-measure-book.pdf.

Tong, D. (n.d.). *Statistical Physics*. University of Cambridge Part II Mathematical Tripos lecture notes, Sections 1.3, 2.2.1 and 3.2. Accessed 6 October 2026. https://davidtong.org/pdfs/teaching/statistical-physics/statphys.pdf.

Visser, M. (2013). Zipf's law, power laws and maximum entropy. *New Journal of Physics*, 15, 043021. https://arxiv.org/abs/1212.5567.

Wedler, C., Ferre, A., Azzan, H., Danaci, D., Petit, C., and Pini, R. (2024). Binary adsorption equilibria of three CO2+CH4 mixtures on NIST reference zeolite Y (RM 8850) at temperatures from 298 to 353 K and pressures up to 3 MPa. *Journal of Chemical & Engineering Data*, 69, 4216–4229. https://doi.org/10.1021/acs.jced.4c00358. Data: https://doi.org/10.5281/zenodo.17604326.

Wojcik, L. A. M., Pfennig, A., Flamm, S., Klauschies, T., Rosenbaum, B., Weithoff, G., and Gaedke, U. (2025). Top-down control and species composition influence nonlinearly the short-term response of experimental food webs to a nutrient pulse perturbation. *Proceedings of the Royal Society B*. https://doi.org/10.1098/rspb.2025.1969. Data: https://doi.org/10.5061/dryad.51c59zwj5.

Universal Dependencies contributors (2026). *English EWT*, release 2.18. Corpus description and pinned source.